Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 **May/June 2017**

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Section A

Question	Answer	Marks
1(a)(i)	*any three of:* low; flat; valley / valley floor; foot of slope / bottom of hill; grass / shrubs / bushes / trees; (semi) desert / bare soil;	**3**
1(a)(ii)	*any two of:* they are on low(er) ground / in a valley /are usually on high(est) ground / hill / not on high(est) ground / hill; winds stronger on higher ground; mountains / hills block wind;	**2**
1(a)(iii)	*any two advantages plus one disadvantage OR any two disadvantages plus one advantage:* *advantages:* away from population / remote / isolated / empty area; poor quality land / cheap land / large area; railway to bring raw materials; *disadvantages:* no nearby, market / industries / settlements / lengthy transmission lines / remote; no water for cooling; no nearby labour force;	**3**
1(b)	*any two of:* high / good rainfall; rain, throughout the year / well distributed; temperatures above freezing point / warm / mild winters; little evaporation;	**2**

Question	Answer	Marks
2(a)(i)	86 000;	**1**
2(a)(ii)	snow melt;	**1**
2(a)(iii)	onshore winds drive sea water onto the, land / coast / inland; low air pressure causes sea (surface) to rise;	**2**
2(a)(iv)	*any three of:* sewers overwhelmed; water supplies become contaminated / lack of safe water; <u>spread</u> of named disease, i.e. diarrhoea / cholera / typhoid; mosquitoes breed in (stagnant) water; malaria; chest infections; contaminated food supplies (in fields / stores / shops / homes); disease spreads easily in crowded relief camps;	**3**
2(b)	*any three of:* farmers lose income as, crops / animals, destroyed / washed away / killed; loss of power reduces, output / sales; industries that use water lose, productivity / sales; loss of output as, factories / offices, damaged; workers unemployed so no income; floods take long time to subside so long term effects on economy; transport routes damaged so transport costs increase / cannot ship goods; high restoration / rebuilding costs; exports reduced, reducing income; tourism industry in dam area reduced; high cost of, rescue / relief / treatment;	**3**

Question	Answer	Marks
3(a)(i)	September;	**1**
3(a)(ii)	*any two of:* shorter duration / 2 months difference; 4 months versus 6 months; August to November versus July to December;	**2**
3(b)(i)	*any three of:* ozone protects from / absorbs UV rays from, the Sun; (protects from) skin cancer / sunburn; (protects from) cataracts / eye damage; (protects from) (DNA) mutations; (protects from) damage to the immune system; reduction in ozone / more UV, decreases photosynthesis reducing rates of crop growth;	**3**
3(b)(ii)	chlorofluorocarbon / CFC / halon / methyl bromide / methyl chloroform / HCFC / tetrachloroethane / trichloroethane / NO / NO_2 / N_2O / NO_X;	**1**
3(c)	*any three of:* not all countries, may enforce the ban / agree the ban; CFCs / halogens / chemicals, that destroy ozone are, stable / stay in the atmosphere a long time; last 40–120 years (accept any figure in between); some products made before the ban are still in use; HCFCs only slowly being phased out;	**3**

Question	Answer	Marks
4(a)(i)	4–4.1 million;	**1**
4(a)(ii)	20;	**1**
4(a)(iii)	the birth rate was increasing rapidly / the population was growing rapidly / many females of child-bearing age;	**1**
4(b)	*any four of:* one / two-child policies; financial deterrents / incentives to limit the number of children; family planning / birth control clinics / programmes; free / cheap / available, contraception; media advertising / information or awareness programmes / education on family size; providing education for females; encouraging females to have careers; making abortion / sterilisation available; encouraging later marriage;	**4**
4(c)	*max two of:* better health care / more doctors; better diets; better sanitation / water supply; *plus any of, up to max three:* high numbers at present in the 20+ age group; (leading to) lower death rate/higher life expectancy;	**3**

Section B

Question	Answer	Marks
5(a)	metamorphic; igneous; sedimentary;	**3**
5(b)(i)	*any three of:* vegetation and soils removed; overburden removed; waste rock / soil stored, out of the way; ore loosened using explosives; diggers / mechanical shovels; load ore onto trucks;	**3**
5(b)(ii)	*any three of:* pit infilled using waste rock; use as landfill site; soil replaced; fertilisers added; vegetation planted / agriculture; *allow* to flood; and use as, reservoir / leisure / nature reserve / aquaculture;	**3**
5(c)(i)	Australia;	**1**
5(c)(ii)	China; 977 (million tonnes);	**2**
5(c)(iii)	little / no, iron ore mined in those countries; many industries that use, iron / steel;	**2**

Question	Answer	Marks
5(d)(i)	*any three of:* formed over <u>millions</u> of years or in carboniferous (era); from dead, organisms / trees / plants; *reference to* lack of oxygen / anaerobic; on sea beds / in swamps; formed a thick layer of peat; covered with, sediments / mud / sand; pressure turns, vegetation / peat, to coal OWTTE;	**3**
5(d)(ii)	*any three of:* shafts, bored / dug (to access coal seams); coal cutter cuts coal; coal transported to shaft by train; coal raised to surface; tunnel roof supported by pit props;	**3**
5(e)(i)	*6 correct [3]* *4 to 5 correct [2]* *2 to 3 correct [1]* sulfur dioxide, nitrogen dioxide, limestone, iron ore, carbon dioxide, slag;;;	**3**
5(e)(ii)	*any four of:* sulfur dioxide and/or nitrogen oxides, produced; dissolves in / mixes with / reacts with, water (in the atmosphere); forming weak acids / lowering pH; such as, sulfurous / sulfuric / nitric; rain falls;	**4**
5(f)(i)	*allow any answer in range of:* 535–540 (million tonnes);; *(if answer incorrect, allow one mark for correct figures for China (55–60 mt) and RoW (480 mt) [1])*	**2**
5(f)(ii)	2007;	**1**

Question	Answer	Marks
5(f)(iii)	*any three of:* RoW little changed whereas China has increased (greatly); 1988 China (only) 55–60 million tonnes, RoW 480 million tonnes; China increase greatly, RoW, little change / slight decline / fluctuates; 2013 China 710 million tonnes, RoW 460 million tonnes; China highest in 2013 whereas RoW highest in 1988; China increased by 645–650 million tonnes, RoW decreased by 20 million tonnes; China lowest 55–60 million tonnes, RoW lowest 410 million tonnes; China increased by 660 million tonnes, RoW decreased by 20 million tonnes;	**3**
5(f)(iv)	got worse / declined;	**1**

Question	Answer	Marks
5(f)(v)	*Level of response marked question:* Level 3 [5–6 marks] Answers must look at both sides of the argument and reach a conclusion. Detail of at least three aspects required for max marks. Two aspects covered well and a balance will be sufficient for five marks. Level 2 [3–4 marks] Answers may look at both sides of the argument but with only limited detail. More likely they will be one-sided. Good answers of this type covering at least two aspects can achieve four marks, whereas those that lack some detail or cover one aspect in detail should be awarded three marks. Level 1 [1–2 marks] Answers will be basic with little detail or explanation. Most will be descriptive or a list of problems or solutions caused by increasing population without really addressing the question. No response or no creditable response [0]. *Level of response marking indicative content:* *Candidates may well refer back to iron production producing CO_2, SO_2, NOx and therefore state that as countries industrialise more air pollution will be caused. More able candidates will look at other ways development leads to air pollution; such as increased car ownership; more flights taken; increased electricity production; other mineral processing leads to air pollution. On the other hand, candidates may quote success in stopping CFC production; increasing use of renewable energy sources; recycling as resources become rare and expensive, flue gas desulfurisation and cleaner transport.*	**6**

Question	Answer	Marks
6(a)(i)	30 (°C);	**1**
6(a)(ii)	**D**; **C**;	**2**
6(a)(iii)	*4 to 5 correct [4]* *3 correct [3]* *2 correct [2]* *1 correct [1]* desert savanna tundra equatorial cool temperate interior;;;;	**4**
6(b)(i)	all three bars correct;; *(if one bar incorrect, allow one mark for two correct bars [1])*	**2**
6(b)(ii)	5 (years);	**1**
6(b)(iii)	drought / shortage of water; *any problem resulting from above such as,* loss of income / food shortages / malnutrition;	**2**

Question	Answer	Marks
6(b)(iv)	*any three of:* *for years of water shortage:* water storage systems; use of groundwater; for irrigation; growing drought resistant crops; *for years of excess water:* drainage channels; building on, higher land / stilts; soil erosion prevention methods (*allow to max of two marks*);;	**3**
6(c)(i)	*any two of:* lots of exposed roots; straight trunks; narrow trunks; long / narrow leaves;	**2**
6(c)(ii)	roots covered in water / swampy;	**1**
6(c)(iii)	*any three of:* monsoon forest lose leaves in dry season, TRF timing of leaf loss throughout year; monsoon forest 3 layers /no emergent, TRF 4 layers / emergent / TRF has more layers; monsoon forest deep roots, TRF shallow roots; greater biodiversity / more species in TRF; greater biomass in TRF; TRF (appears) evergreen, monsoon forest is not; TRF has buttress roots, monsoon forest does not; TRF less groundcover than Monsoon forest;	**3**

Question	Answer	Marks
6(d)(i)	(flows) north to south / southwards / SSE / (bearing) 160–180°;	**1**
6(d)(ii)	5; China;	**2**
6(d)(iii)	*any two of:* from (southern) China to Cambodia / list of countries / all countries except Vietnam / throughout length of the river; cluster / 5 in N Laos; cluster / 5 in S Laos and Cambodia; few in China;	**2**

Question	Answer	Marks
6(d)(iv)	*any three farmers points plus two fishermen points OR any three fishermen points plus two farmers points:* *Farmers:* *negative points:* lack of floods; so no silt deposited; leading to loss of fertility; leading to increased costs of fertilisers; lack of flooding means certain crops (rice) can't be grown; salt penetrates delta; reducing fertility of the soil; farmland lost due to, dam / reservoir; farmers displaced; reduced, yield / food / income; *positive points:* water stored for irrigation; double cropping possible; floods, stopped / reduced; so no, crop / animal, loss; increased, yield / food / income; *Fishermen:* *negative points:* dams stop fish migration; causing decrease in, breeding / fish stocks; may lead to extinction of some species; salt water penetrates delta; reducing number of fresh water fish; reduced, catch / food / income; *positive points:* fish farming possible in reservoirs; leading to increased, catch / income;	**5**

Question	Answer	Marks
6(d)(v)	*any three of:* to aid development; to power industry; to export electricity to neighbouring countries; to provide power for increasingly affluent population; to provide power for increasing population; floods devastate economy;	**3**
6(e)	*Level of response marked question:* Level 3 [5–6 marks] Answers must look at causes of water shortages and strategies that could overcome them in affected countries and reach a conclusion and be aware that not all countries face water shortage. Detail of at least three aspects for max marks. Two aspects covered well and a balance will be sufficient for five marks. Level 2 [3–4 marks] Answers should look at causes and strategies that could overcome water shortages but with only limited detail or concentrate on strategies. Good answers of this type covering at least two aspects can achieve four marks; those that lack some detail, three marks. Level 1 [1–2 marks] Answers will be basic with little detail / explanation. Most will be descriptive or a list of problems caused by lack of rain or increasing demand without really addressing the question. No response or no creditable response [0]. *Level of response marking indicative content:* *Some countries are water poor; e.g. in arid and semi-arid regions; so will struggle to meet demand; especially as populations increase. Water supply may be reduced by upstream countries extracting more water from rivers. Climate change may reduce rainfall in some parts and increase it in others. As Earth warms rainfall more likely to increase. Underground supplies largely being used faster than they are refilled. More pollution of water as increased industry and use of pesticides. Coastal countries may use desalination; but this is expensive and uses a lot of energy. Recycling of water is important for countries facing water poverty. Some countries have sufficient water due to high rainfall and/or low population.*	**6**

Cambridge O Level

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Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/11**

Paper 1 **May/June 2017**

MARK SCHEME

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International Examinations

Section A

Question	Answer	Marks
1(a)(i)	division at either 26 or 80; largest segment shaded as Malaysia / smallest segment shaded as China;	**2**
1(a)(ii)	*allow any answer in range of:* 61–63 (thousand tonnes);	**1**
1(a)(iii)	*any three of:* low grade / many impurities; so low price / could not compete on the world market; difficult to access / mountainous / steep slopes / no communications / distance from port; deep so expensive / difficult to access; thin / small deposits expensive to exploit;	**3**
1(b)(i)	they contain other substances which are not used / need to extract the useful metal from the rest of the rock;	**1**
1(b)(ii)	*any three of:* recycle used products; increase efficiency of use; extract from spoil heaps; substitute other materials; example of a substitute for a mineral; influence of new technology, e.g. miniaturisation;	**3**

Question	Answer	Marks
2(a)(i)	urban population with access to good sanitation;	**1**
2(a)(ii)	6.8;	**1**
2(a)(iii)	all better than the 2015 target except rural access to safe water / target already achieved in 3 out of the 4 / <u>only</u> rural access to safe water had not reached the 2015 target;	**1**
2(b)	*any three of:* economies of scale / cost per property cheaper to provide; easier to access in cities / rural populations often more remote / inaccessible; cities are wealthier authorities / have wealthier people; cities are important economically / industries / factories need water; more pressure can be put on the authorities in cities; better educated population able to campaign more effectively; politicians usually live in urban areas;	**3**
2(c)	*any four of:* contaminated with animal faeces; contaminated with fertilisers / insecticides / pesticides / chemicals used in farming; contaminated by domestic waste; bilharzia snails; fleas deposit (Guinea) worm larvae in water; typhoid / cholera / dysentery / diarrhoea often caught by drinking contaminated water; mosquitoes breed in stagnant water;	**4**

Question	Answer	Marks
3(a)(i)	*any two of:* pipe; sprinkler / sprays; rotates; movable;	**2**
3(a)(ii)	*any one of:* some evaporates; some is falling on bare soil;	**1**
3(b)(i)	trickle drip irrigation;	**1**
3(b)(ii)	it goes to the root / less evaporation;	**1**
3(c)(i)	*any two of:* it helps keep a protective cover of plants on the soil; wet soil particles are heavier and less likely to blow away; water binds soil particles together;	**2**
3(c)(ii)	*any three of:* overgrazing / overstocking / animals exceed the carrying capacity; without rotating the pasture; leaves soil bare; no roots to hold the soil; wind blows / water washes, particles away; movement trails / hoof damage;	**3**

Question	Answer	Marks
4(a)(i)	high / medium;	**1**
4(a)(ii)	*any two of:* houses close together / people in close proximity; lack of clean water supplies; lack of sanitation / sewage disposal;	**2**
4(b)	*any four of:* better / good, education / schools; better health care / more, hospitals / clinics / doctors; better standard of living; better entertainment / 'bright lights'; better food supplies; electricity; better water supply / sanitation;	**4**
4(c)	*any three of:* the settlement grows very quickly / migrants continue to move in; provision of services is expensive; the city has limited funds available; the city does not always own the land; settlement sites are difficult areas to get services to; difficulty of relocating people;	**3**

Section B

Question	Answer	Marks
5(a)	thermometer; anemometer; barometer / barograph; rain gauge;	**4**
5(b)(i)	*allow any answer in range of:* 260–270 (mm); August;	**2**
5(b)(ii)	7 (months);	**1**
5(b)(iii)	*any three of:* increases from January to April; with maximum in April of 34°C; decreases to September / with temperature of 28°C; increases to October / with temperature of 30.5 / 31°C; decreases to December / with minimum in January / December of 25.5°C / 26°C; decrease in wet season; lowest temperature when no rainfall; two maxima / two minima;	**3**
5(c)(i)	*any three of:* dry / dead / brown, grass; grass in tufts / long grass; bush with, no leaves / dry; bush with thorns / thorny bush; scattered / clustered / few, trees; some trees have no leaves / leaves on trees; trees with flat tops; trees of different heights;	**3**
5(c)(ii)	*any two of:* grass, taller / greener; leaves on all, trees / bushes; plants have flowers;	**2**

Question	Answer	Marks
5(c)(iii)	*producer:* organisms that manufacture their own food by the process of photosynthesis; *consumer:* consumers feed on the producers / other animals;	**2**
5(c)(iv)	*any four of:* grass reduced / bare ground / loss of vegetation; less food for grazing wild animals; so decrease in grazing animals; leading to decrease in predators; leading to soil erosion; young trees / bushes eaten; leading to loss of tree cover; loss of habitats for, insects / birds; goats strip leaves from trees so trees die; AVP;	**4**
5(d)(i)	*1950 allow any answer in range of:* 4–5 (million); *2015 allow any answer in range of:* 44–45 (million);	**2**
5(d)(ii)	*allow any answer in range of:* 60–70 (million);	**1**
5(d)(iii)	*any four of:* lowering of death rate / longer life expectancy; reasons for decrease in death rate (*allow to max of two marks*);; birth rate has, remained high / only fallen slowly; reasons why birth rate little changed (*allow to max of two marks*);; more women in fertile age group;	**4**
5(e)(i)	*correctly drawn bars for:* *male 20–24 2.0 million;* *female 5–9 3.1 million;*	**2**

Question	Answer	Marks
5(e)(ii)	*any four of:* schools / education, for large numbers of children; large young dependent population; high unemployment as young reach working age; shortage of housing; shortage of medical supplies; shortage of food; infrastructure problems explained (water, sanitation, etc.) (*allow to max of two marks*);; social problems explained (crime, overcrowding, etc.) (*allow to max of two marks*);; problems of few old people explained;	**4**
5(f)	*Level of response marked question:* Level 3 [5–6 marks] Answers must look at both sides of the argument and reach a conclusion (five marks). For six marks candidate needs to show that they understand reduction in population growth rate only slows, rather than reduces, population. Detail of at least two aspects needed for this level. Level 2 [3–4 marks] Answers may look at both sides of the argument; but with only limited detail. More likely they will be one-sided. Good answers of this type covering at least two aspects can achieve four marks; those that lack some detail, three marks. Level 1 [1–2 marks] Answers will be basic with little detail / explanation. Most will be descriptive or a list of problems caused by increasing population without really addressing the question. No response or no creditable response [0]. *Level of response marking indicative content:* *Some may take the view that increasing population means increasing depletion of resources; food and water shortages and environmental pollution. Others may take the view that changes to renewables will reduce climate change and resource depletion and that recycling and fairer distribution of resources is the answer. Balanced views will take in aspects of both sides and may suggest that stabilising population should be the aim. Some may refer to change in China's one child policy. Slowing rate of population growth will not stop population growing; it will simply grow more slowly.*	**6**

Question	Answer	Marks
6(a)(i)	*any three of:* within the tropics; (mainly) close to coasts; more on east coasts (than west coasts); especially between Asia and Oceania / in the Pacific; some in middle of oceans (around islands); list of the areas;	**3**
6(a)(ii)	too cold / AVP;	**1**
6(b)	positive relationship OWTTE; anomaly at 220 coral species;	**2**
6(c)(i)	*all three correct, in the same order and correct shading [2]* *one correct and correct shading [1]* *all three correct but no shading / incorrect shading [1]*	**2**
6(c)(ii)	Oceania; *allow any answer in range of:* 58–59; low;	**3**
6(c)(iii)	*any three of:* variance in protection methods; enforcement of protection; demand for fish; for example, population size, good infrastructure, eat a fish diet etc.; some reefs are more productive therefore more fishing is done; education of those who catch fish; accessibility of reefs;	**3**
6(c)(iv)	*any one of:* growth of population / growth of demand; technology, such as sonar; increase in size of, boats / nets;	**1**

Question	Answer	Marks
6(c)(v)	*any four of:* quotas; international agreements; banning fishing in certain areas; increase size of net apertures; smaller nets; no fishing during breeding season; increased use of aquaculture; use of sustainable fishing methods, e.g. rod and line;	**4**
6(d)(i)	phytoplankton;	**1**
6(d)(ii)	(*phytoplankton*), zooplankton, corals, fish, (*seals*);; *(if answer incorrect, allow one mark for two correct organisms [1])* *arrows drawn going up food chain*;	**3**
6(d)(iii)	*any three of:* increase in fish; increase in predatory fish; which may cause decline in fish numbers; decrease in crown-of-thorns starfish; so corals increase; but increase in fish numbers may cause decrease in corals; invertebrates decrease; so may be an increase in corals; OR invertebrates increase as fewer predatory fish; so corals decrease; if corals decrease zooplankton increase; and phytoplankton decrease; if corals increase zooplankton decrease; and phytoplankton increase; AVP;	**3**
6(d)(iv)	*any two of:* have few predators so no control of numbers; so large numbers will eat a lot of coral; each one can destroy, $6\,m^2$ / a large area of coral, per year;	**2**

Question	Answer	Marks
6(e)	*any two from each category, up to max six:* *oil refinery:* oil causes birds' feathers to lose ability to regulate body temperature / so they cannot fly / so they lose their waterproofing; birds / fish, may swallow oil which is poisonous / eggs or larvae killed by oil; eggs / larvae, of many fish killed by oil; oil damages fish gills so they die from lack of oxygen; oil on the surface stops photosynthesis so phytoplankton die / reduces dissolved oxygen / suffocates fish; oil clumps onto the sea bed and coats coral / other organisms; *farming:* release of pesticides may poison, fish / other marine life; excess, fertilizer / animal waste, can lead to, algal blooms / causes eutrophication; dead / decaying algae / plants, decrease oxygen levels so marine animals die; soil erosion / sediment, washed into the water kills corals; *lead mining and processing:* lead (which is a heavy metal) is, poisonous / toxic; invertebrates and other sea-bed feeders ingest lead; concentrates in organs such as the liver as little lost through faeces; so such organisms are very toxic to consumers; bioaccumulation of lead can occur; acid from lead processing lowers the pH of seawater; lead affects the nervous system of fish;	**6**

Question	Answer	Marks
6(f)	*Level of response marked question:* Level 3 [5–6 marks] Must include an international dimension for this level. Answer will give details of problems of controlling pollution. Level 2 [3–4 marks] Answers will cover a number of aspects with limited explanation **or** maybe one reason in depth. Level 1 [1–2 marks] Answer may well be a list or descriptive rather than an explanation **or** may provide a basic explanation of one or two points. A list of sources of pollution but no methods of control would achieve max Level 1. No response or no creditable response [0]. *Level of response marking indicative content:* *Marine pollution spreads with ocean currents and so its effects are not limited to the area offshore from the polluter; so if some nations take action there will still be pollution from others. Pollution may also occur outside territorial waters; such as oil spillages; cleaning oil tanks illegally and particularly dumping of plastics from ships. So while local action can reduce marine pollution on a local scale; that environment will still be affected by external sources. Vast amounts of pollutants such as plastics are long lasting; CO_2 increases acidity and atmospheric CO_2 levels are at an all-time high.*	**6**

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Question	Answer	Marks
1(a)(i)	fault / faulting	1
1(a)(ii)	folded / folding (sideways) compression	1
1(a)(iii)	*advantage of A* *any 1 of:* can be accessed, from the surface / by adit / drift / open cast; straight seam / easy access; *disadvantage of A* *any 1 of:* thin seam; need two shafts to access both sides of the fault / tunnel needs to change height; *advantage of B* *any 1 of:* thick seam; continuous / more coal; *disadvantage of B* *any 1 of:* deep(er) / shaft needed to access it / can't be accessed from surface; change in height of tunnel; more subsidence possible;	4
1(b)	coal is a sedimentary rock that formed on land; the main content of coal is carbon;	2
1(c)	*any 2 of:* to prevent depletion of coal stocks; to reduce release of carbon dioxide which causes global warming; to reduce release of sulfur dioxide / oxides of nitrogen, which cause acid rain; belief that potential dangers of nuclear can be effectively controlled; volume of waste generated with coal is greater; reference to relative amount of energy released;	2

Question	Answer	Marks
2(a)(i)	*any 2 of:* low / little (annual amount); decreases westwards / higher towards the east / less near the ocean; varies from less than 50 mm in the west to more than 300 mm in the east; AVP;	**2**
2(a)(ii)	highest – E medium – C lowest – D 3 correct = 2, 1/2 correct = 1	**2**
2(a)(iii)	*any 2 of:* there is not enough to satisfy all the demand; to preserve enough for other users; so river does not dry up;	**2**
2(a)(iv)	*Expect E but allow C or D if justified. Accept any sensible reason, such as:* E – users are of the greatest value to the economy / to ensure supply to hospitals / sanitation / drinking water for the (human) population / AVP C – commercial / wildlife farms may be for food / valuable to the economy / AVP; D – tourism / wildlife parks are valuable to the economy / AVP;	**1**

Question	Answer	Marks
2(b)(i)	*any 2 of:* dams across rivers; storage reservoirs; pipelines (take the water to E); tap underground supplies / wells / boreholes; desalination of sea water; water capture (clarified); emergency water tanker / bowser; construction of new river route / channels;	**2**
2(b)(ii)	*any 1 of:* reuse / recycle, water; use appliances economical with water; mend, dripping taps / bursts quickly; catch / store, rainwater; AVP;	**1**

Question	Answer	Marks
3(a)(i)	wind;	**1**
3(a)(ii)	hill-top / upper part of hill / free from trees or obstructions;	**1**
3(a)(iii)	open / exposed to wind / no or few obstacles to slow the wind / way from houses (so won't cause noise or vibration);	**1**
3(a)(iv)	*any 3 of:* blade / arm; tower / shaft / pillar / pole; turbine / generator (at centre of blades); transformer (at base) nacelle / housing (case around generator) etc.;	**3**

Question	Answer	Marks
3(b)(i)	*any 3 of:* wind-borne pollution; rainfall cannot be prevented; source region a distance away; source region may be in another country; depends on prevailing wind;	**3**
3(b)(ii)	*any 1 of:* damage to leaves of trees / plants; acidification of lakes / rivers / soil; harms fish / aquatic life; damages limestone rocks;	**1**

Question	Answer	Marks
4(a)(i)	denitrifying;	**1**
4(a)(ii)	ammonium;	**1**
4(a)(iii)	*any 1 of:* they eat the plants that contain nitrates; produce urine / faeces / waste; containing nitrogen; their bodies decompose to return ammonium to the soil;	**1**

Question	Answer	Marks
4(b)	food chain disrupted; animals will, lose their preferred habitat / will die / relocate; water loving plants / example of, will die; animals / plants requiring dry habitats, become established; soil will be, drier / contain more air; anaerobic bacteria replaced by aerobic bacteria; plant succession may occur; shrinkage of peat reduces the level of the land;	**3**
4(c)	population pressure / increased population; the need for more land for housing; farming; industry; to rid the area of a source of water-related disease; to remove the habitat of dangerous animals / e.g. alligators / crocodiles etc.;	**4**

Question	Answer	Marks
5(a)(i)	*any 3 of:* close to / around / along / north of, the Arctic Circle; 60–75 degrees north (accept figure within latitude range); in the northern hemisphere; along the northern margins (of continents); in North America / Europe / Asia / named country;	**3**
5(a)(ii)	*any 1 of:* the area of tundra will reduce / shrink / become smaller; the margin / edge of tundra will move further north;	**1**

Question	Answer	Marks
5(a)(iii)	*any 4 of:* carbon dioxide: factories / power stations / burning fossil fuels / burning rainforests / deforestation / vehicle exhausts; methane: intensive cultivation / paddy fields / landfill; nitrous oxide / nitrogen oxides: vehicle exhausts / transport; CFCs: aerosols / fridges; water vapour: power stations;	**4**
5(b)(i)	both points correctly plotted; correct completion of the line:	**2**
5(b)(ii)	<table><tr><td>number of months below freezing</td><td>9</td></tr><tr><td>the month with the highest precipitation</td><td>July</td></tr><tr><td>annual temperature range</td><td>34 (°C)</td></tr></table>	**3**

Question	Answer	Marks
5(b)(iii)	*any 4 of:* low growing / close to the ground; for protection against the wind; short roots; t to avoid the permafrost / frozen ground; small leaves; t to reduce loss of water by transpiration; life cycle is short; to flower and set seed in short growing season; cup / rosette shape; efficient light collection in centre of plant / retains heat; reproduce by budding / division; to make maximum use of short growing season;	**4**
5(c)(i)	an animal that eats green plants / an animal that eats another animal	**1**
5(c)(ii)	*any 4 of:* disrupts current food chain; more food for polar bear and wolves; less food for arctic fox and snowy owl; less territory for arctic fox; numbers of arctic fox may decrease: red fox might out compete arctic fox; numbers of primary consumers would reduce / named example;	**4**

Question	Answer	Marks
5(d)(i)	*any 3 of:* from dead organisms; on sea beds / swamps; covered (in sediments); compressed / pressure; reference to millions of years;	**3**
5(d)(ii)	Prudhoe Bay; Valdez; Yukon;	**3**
5(d)(iii)	*any 1 of:* route too dangerous / risk of oil spill; north part of the sea would be frozen for much of the year; more direct route / faster transport; cheaper once built;	**1**
5(e)(i)	*any 3 of:* vegetation would be damaged / habitat loss; vegetation will take time to recover; oil spills / any impact of oil spills; caribou cannot migrate and look for food / caribou would have less food / forced to migrate; could disturb breeding of the caribou; might frighten caribou away; visual pollution / ruin the view / noise pollution/ air pollution (if qualified); melting permafrost / boggy ground;	**3**
5(e)(ii)	*any 2 of:* to stop the warm oil coming in contact with the ground / prevent melting of the permafrost; so caribou (and other animals) could pass underneath the pipeline ; reduces heat loss from oil in pipeline; easier to build / maintain;	**2**

Question	Answer	Marks
5(f)	Level marked question Indicative content: Fossil fuels will run out / are non-renewable / alternative energy sources are renewable / will last into the future Fossil fuels cause global warming / air pollution / acid rain / alternative energy sources are cleaner / less polluting Fossil fuels are subject to price fluctuations / security of supplies / conflicts Fossil fuels can cause problems during transport, e.g. oil spills Dangers of mining accidents / extracting fossil fuels Scars on the landscape from mining Use of alternative fuels will make fossil fuels last longer Some alternative energy supplies are cheaper to run Sites for alternative energy are limited Some need a lot of land to generate a small amount of energy Costly to set up Some countries may not have the technology – developing Unreliable, e.g. sun does not always shine idea Abundant supplies of fossil fuels locally Fossil fuels are energy rich	**6**

Question	Answer	Marks
6(a)(i)	correct plot	**1**
6(a)(ii)	8500 <u>million</u>	**1**
6(a)(iii)	*any 5 of:* declining infant mortality rate; high / increasing birth rates; lack of available contraception; people (in LEDCs) cannot afford contraception; lack of education about how to use contraception; high infant mortality rate so people have more children in the hope that some will survive; death rates are falling / high life expectancy / people live longer; cures for diseases / better medical facilities; vaccinations; improved sanitation; improved water supply; government incentives; religious beliefs / ban on abortion / contraception; abundance of food / better farming methods;	**5**
6(b)(i)	*any 5 of:* fuel wood / for heating and lighting; subsistence farming / slash and burn / shifting cultivation / to provide food; commercial farming / cash crops / plantations; urbanisation / settlement; timber / commercial logging; mineral extraction; cattle ranching / to sell beef to other countries; roads / for communication;	**5**
6(b)(ii)	20676 (km^2)	**1**

Question	Answer	Marks
6(b)(iii)	*any 2 of:* it has decreased / fallen; approximately halved in 3 years; correct data quote;	**2**
6(b)(iv)	greater concern for environmental issues / concerns over global warning / new rules or laws;	**1**
6(c)(i)	*any 3 of:* fewer trees to photosynthesise; (therefore) less carbon dioxide removed; increased levels of carbon dioxide in the atmosphere; carbon dioxide released from burning rainforests;	**3**
6(c)(ii)	*any 4 of:* soil has no protection from trees / soil left bare; (less interception so) more surface run-off; sun can dry soil; (less humus / decaying vegetation so) soil becomes infertile; rainwater leaches minerals from soil; less humus so soil less cohesive / loses structure; no roots to bind the soil; soil is eroded;	**4**
6(d)	*any 4 of:* draining of wetlands; dam building; (intensive) agriculture / grazing; trawling; tourism or specific example; named type of relevant pollution, e.g. heavy metal, fertilisers, oil spill, acid rain; mining; construction / road building;	**4**

Question	Answer	Marks
6(e)(i)	suitable scale and *y*-axis labelled; correct labelling of countries; correct plots;;	**4**
6(e)(ii)	Kenya;	**1**
6(e)(iii)	*any 1 of:* more land available; more wildlife / more endangered species; government supports / invests in national parks; more concern for the environment; AVP;	**1**
6(e)(iv)	*any 1 of:* sustainable harvesting; wildlife reserves; world biosphere reserves; gene banks; education; laws: zoos; AVP;	**1**

Question	Answer	Marks
6(f)	Level marked question Indicative content: agroforestry community forestry reforestation sustainable harvesting fuel wood planting genetic engineering efficient use of timber recycling alternatives materials to timber education prevention of deforestation laws and permits / licensing monitoring introduction of eco-tourism forest reserves protected areas	**6**

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **October/November 2016**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

Question	Answer	Marks
1(a)	it is the numbers / variety / eq; of different types / species / biological groups of living organisms / living things / plants and animals;	**2**
1(b)(i)	Tanzania;	**1**
1(b)(ii)	Rwanda; Burundi;	**2**
1(c)(i)	water would be contaminated after vehicle washing all day / eq / as a control / to compare results;	**1**
1(c)(ii)	41.6; 8.8;	**2**
1(c)(iii)	*any 3 of:* pH lower in all washing sites; phosphate higher; salinity higher; use of figures to support; ORA	**3**
1(c)(iv)	*any 2 of:* oil; petrol / eq; diesel; brake fluid; battery acid; shampoo / cleaning agents;	**2**
1(c)(v)	table drawn; headings; all data filled in a tally form;	**3**
1(d)(i)	*any 5 of:* fertiliser / phosphates / nitrates encourage algal growth / bloom; light cannot penetrate; so plants do not photosynthesise; algae / plants die; decomposed by microbes / eq; bacteria increase; oxygen used up / high BOD; (some) fish die / fish populations decrease;	**5**
1(d)(ii)	*any 1 of:* good source of, protein / omega oils / eq / vitamin A / vitamin D / iodine; helps children grow; maintains / boosts, immune system;	**1**

Question	Answer	Marks
1(d)(iii)	*any 1 of:* unhealthy: because of levels of, phosphate / sewage / named pollutant e.g. heavy metal; or healthy: because only small change in phosphate / not much sewage contamination / good source of protein / omega oils / eq / vitamin A / vitamin D / iodine;	1
1(e)(i)	*any 2 of:* (plan two is better as) more sites so more representative / better average; numbers counted rather than, presence / absence; same fixed time for counting snails / eq;	2
1(e)(ii)	as there are no sites without washing; no comparison is possible no control sites / eq;	2
1(e)(iii)	6.4;	1
1(e)(iv)	*any 2 of:* defined size of quadrat / eq; same number at each sample site; random sampling;	2
1(e)(v)	bilharzia / schistosomiasis / blood fluke / katayama fever / eq;	1
1(e)(vi)	human waste goes into river / swimming stage can get to water snail / eq; swimming stage burrows into human skin / eq;	2
1(e)(vii)	*any 2 of:* vehicle washing reduces snail numbers / eq; so less snail eggs; not enough food for (young) fish; so less adults to reproduce;	2

Question	Answer	Marks
1(f)(i)	B D A C; (any two correct = one mark) 4 correct = 2, 2 / 3 correct = 1	**2**
1(f)(ii)	*any 2 of:* ploughing down slope; run-off; no vegetation cover; wind erosion; animal wastes enter lake; overgrazing / over-cultivation; allow any suitable examples	**2**

Question	Answer	Marks
2(a)(i)	*any 1 of:* closest to lake / similar distance from lake / eq; two in wetland, two in savannah; so comparison possible / eq;	**1**
2(a)(ii)	*any 1 of:* at random from a list; every third house / eq; other valid sampling method;	**1**
2(a)(iii)	*any 2 of:* to collect data about all household activities / all points of view / division of labour / an example stated or described / eq; males not always head of household / eq; to avoid bias / incomplete data collection / eq;	**2**

Question	Answer	Marks
2(a)(iv)	*any 2 of:* so all the questions asked were the same / eq; to record all the information; so results were reliable; could be processed / summarised;	**2**
2(b)(i)	key completed; both axes fully labelled with appropriate scale (plots to cover at least half of the grid); one plotting error = two marks, two plotting errors = one mark);;	**4**
2(b)(ii)	no large difference between any assets; two examples quoted from, graph / table;;	**3**
2(b)(iii)	*any 2 of:* more people need to buy food; and less food being taken, to market / for sale; reference to supply ad demand; physical factors such as, drought / heavy rain; AVP;	**2**
2(b)(iv)	*any 2 of:* questions such as: which type of, crops / named crops (e.g. maize, beans, sorghum) do you grow?; how big are your fields?; when are your harvesting times?; how much do you harvest?; how valuable is your harvest?; do you sell crops?;	**2**
2(c)	*any 4 of:* any suitable ideas in these four areas: sustainable / environmental / social / financial. agricultural advisors; laws about pollution; subsidies / eq / for, drought / pest resistant / high yield seed; control of local industries; control of land ownership; AVP;	**4**

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 **May/June 2016**

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Cambridge is publishing the mark schemes for the May/June 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

Section A

Question	Answer	Marks
1(a)(i)	line at 76 or 85; shading correct with ref. to key;	**2**
1(a)(ii)	87;	**1**
1(a)(iii)	*any 2 of:* few narrow valleys for dam building; few powerful rivers; few rivers with steep courses; climate too dry / dry season; plenty of other cheaper sources; HEP expensive; developing country / lack of finance; AVP;	**2**
1(b)	*increase in energy demand* industrial development/increase in wealth of population increases demand for electricity in homes / increased population; *increase in CO_2 emissions* reduction in use of fossil fuels/use of alternative energy sources / environmental laws enacted;	**2**
1(c)	*any 3 of:* vegetation cleared / loss of habitats; waste heaps / scenic beauty lost; pit head buildings / scenic beauty lost; (but allow scenic beauty lost only once) subsidence hollows; slimes; water pollution from waste washed into water courses; railways / roads; air pollution from transporting coal away; staff quarters / houses / settlement;	**3**

Question	Answer	Marks
2(a)	A; D; B; E; C; all correct = [4], 3 correct = [3], 2 correct = [2], 1 correct = [1]	**4**
2(b)(i)	*any 2 of:* for agriculture / farming / arable / pastoral / animals; for irrigation; for HEP;	**2**
2(b)(ii)	*any 1 of:* poor quality; polluted / contaminated; acidic / saline; too deep underground; low population such as a mountainous area; low population in very cold area / water in form of ice/snow;	**1**
2(c)(i)	salt removed from water / soil;	**1**
2(c)(ii)	*any 2 of:* expensive; the water has a poor taste; many areas are away from coasts / sea / no access to salty water; needs much energy; often not environmentally friendly process / polluting;	**2**

Question	Answer	Marks
3(a)(i)	line at 190 mm;	**1**
3(a)(ii)	all year/every month; more December to May than June to November / trend of decrease and then increase / owtte;	**2**
3(a)(iii)	*advantages* enough warmth for crops to grow all year; enough rain for crop growth all year / irrigation; crops grow rapidly in hot, wet conditions; double cropping is possible; abundant rain for HEP; rain all year for reliable HEP; sufficient rain for domestic / industrial use; AVP; *disadvantages* heavy rain can damage the crop; risk of flooding; rain leaches nutrients from the soils; heavy rain washes soil away / rain causes soil erosion; heavy rain turns earth roads muddy / disrupts transport; hot wet conditions / high humidity hard to work in / live in; AVP; *at least one advantage and one disadvantage for maximum mark*	**3**
3(b)(i)	*any 2 of:* cold; frozen ground / snow / ice; plants cannot take up water / photosynthesise for many months; short growing season / 3 months or less	**2**
3(b)(ii)	glasshouse / greenhouse / under glass; heated / artificial light / hydroponic / details of hydroponics / watering / irrigation;	**2**

Question	Answer	Marks
4(a)	*any 3 of:* grass; short; clumps / tussocks; scrub; taiga; conical / coniferous trees; dense in background; scattered / sparse / lack of vegetation in the middle of the photograph;	**3**
4(b)(i)	*any 2 of:* gullies / dissected / owtte; development for second mark; steep slopes; development for second mark; soil eroded / removed so crops can't grow; bare / little vegetation for animals to graze;	**2**
4(b)(ii)	*any 1 of:* clearance of forest / lack of vegetation; overgrazing; run-off from heavy rain / snow melt; water run-off down the slope;	**1**
4(c)(i)	processes **changing** land in which plants can grow to land where they cannot / from productive to non-productive land / owtte;	**1**
4(c)(ii)	*any 3 of:* maintain plant / vegetation cover / prevent deforestation / afforestation; limit stock numbers rotational grazing; crop rotation / use fertiliser; controlled irrigation; introduce birth control / keep family sizes small; farmer training for sustainable land use; avoid intensive farming; plant trees as windbreaks; AVP;	**3**

Section B

Question	Answer	Marks
5(a)(i)	660–680; 2011;	**2**
5(a)(ii)	1983;	**1**
5(a)(iii)	18 years;	**1**
5(a)(iv)	flooding;	**1**
5(b)(i)	(fruit / veg need a lot of water) so close to the river for irrigation / fertile soil close to river; extensive cattle farming has less need of water / less need of fertile soil;	**2**
5(b)(ii)	*any 2 of:* (many years of) low rainfall / below average rainfall; so grass / fodder doesn't grow much / dies back; reduced food for cattle / cattle may die; lack of water for cattle to drink; farmers lose money / go bankrupt / forced to sell cattle;	**2**
5(b)(iii)	*any 4 of:* heavy rain; so ground becomes saturated; leading to surface runoff; following (long) dry period; so would be limited vegetation; to hold soil together; when rain falls it washes soil away;	**4**

Question	Answer	Marks
5(b)(iv)	*any 4 of:* nitrates and / or phosphates; washed into water sources; nitrates in drinking water harm human health; in rivers / lakes cause algae to increase rapidly; bacterial decomposition of algae when they die; (leading to) oxygen deficiency; and so eutrophication; leading to death of fish / invertebrates;	**4**
5(b)(v)	*any 3 of:* educating farmers; reducing fertiliser / pesticide use; using natural / organic fertilisers / such as manure / compost; using natural bio-controls for pests; less intensive fruit growing; reduce stocking levels on cattle farms; mixed cropping; crop rotation; store water for irrigation trickle drip irrigation; AVP;	**3**
5(c)(i)	period of abnormally low rainfall;	**1**
5(c)(ii)	Europe;	**1**

Question	Answer	Marks
5(c)(iii)	mainly in the tropics; except for China / India; east side of continents; in a band through north Africa; Oceania; South / south east Asia; eastern South America; *max 2 on named locations*	**3**
5(c)(iv)	removal of vegetation / desertification; climate change / global warming;	**2**
5(d)	*any 3 of:* drought means vegetation dries out; accidental burning by farmers; climate change causing loss of habitats / lack of water; expanding human activities / deforestation; pollution of environments;	**3**
5(e)	*any 4 of:* weakening of trade winds; reversal of warm / equatorial current; to flow eastwards; increase in rainfall; so no upwelling of cold water; so lack of nutrients at surface; causing decline in numbers of marine creatures;	**4**

Question	Answer	Marks
5(f)	*Indicative content* droughts tend to last longer and affect larger areas than floods or cyclones; floods and cyclones cause more damage to property than droughts; all can lead to soil erosion and loss of crops / animals; cyclones and floods are short-term and may require emergency rescue, shelter, food, etc.; effects of drought, being longer term, can be planned for, but can cause far more deaths than the others if no food aid, etc.; most environments recover from such disasters; it will depend on the severity of each disaster as to environmental effects, though flooding and cyclones more likely to result in pollutants being washed into water courses or the sea; droughts less environmentally damaging; Level 3 5–6 marks must communicate the features of the climatic hazards. Answers the question with detailed consideration of climatic hazards. Must look at both sides of the argument, so must be some discussion of floods and/or droughts and their relative impacts. Level 2 3–4 marks considers both sides (i.e. other climatic hazards besides cyclones) with brief details OR detailed consideration of one climatic hazard (probably cyclones) Level 1 1–2 marks basic descriptive points with little or no reasoning. May just be a list of impacts of cyclones and an agreement without mention of other climatic hazards. no response or no creditable response, 0.	**6**

Question	Answer	Marks
6(a)(i)	crust; mantle; core; all 3 correct = [2], 1 or 2 correct = [1]	**2**
6(a)(ii)	*any 3 of:* two plates moving towards each other; both are continental plates; collision cause plates to buckle; creating folds and mountains / fold mountains;	**3**
6(a)(iii)	*any 2 of:* no oceanic plate; so no subduction; so no melting of plate for magma;	**2**
6(b)(i)	constructive (divergent);	**1**
6(b)(ii)	two arrows pointing away from central rift; fault labelled at central rift;	**2**
6(c)(i)	south / southwest of Iceland / on the Atlantic mid-oceanic ridge;	**1**
6(c)(ii)	igneous;	**1**
6(d)(i)	21 / 22;	**1**
6(d)(ii)	*any 4 of:* no plants in 1965; increased up to 1975–8; then remained constant until 1985; constant again 1987 to 1989; then a rapid increase; up to 55 / 56 plant species in 2000; accept other valid descriptions;	**4**

Question	Answer	Marks
6(d)(iii)	*any 3 of:* sudden increase; transported seeds to Surtsey; in droppings / feathers; droppings provide nutrients for plant growth;	**3**
6(d)(iv)	*any 2 of:* change in the types of plant species (that occupy a given area through time); from bare ground to climax vegetation; can use descriptive example such as bare ground e.g. moss/lichen/trees /forest;;	**2**
6(e)(i)	*any 4 of:* cold water pumped down well; passes through cracks / joints / fissures in the rock; which heats it; hot water rises (under pressure); at surface (drop in pressure) turns water to steam; turns turbines / generator;	**4**
6(e)(ii)	has hot rocks close to the surface (or similar) / volcanic;	**1**
6(e)(iii)	yes or clearly implied; available to future generations /lasts forever; Earth's heat / water not used-up or can be reused; OR max 2 if 'no' given as answer AND reasons as 50 year max life for power plant at that location; if no control of cold water pumping;	**3**

Question	Answer	Marks
6(e)(iv)	*any 4 of:* burning fossil fuels produces carbon dioxide / carbon monoxide; carbon dioxide is a greenhouse gas; responsible for enhanced global warming; max 1 on consequences of global warming; burning produces nitrogen / sulfur oxides; which cause acid rain; burning produces soot / particulates; which cause smog / toxic gases / health problems; max 1 for impacts of extraction if well explained;	**4**
6(f)	*Indicative content* advantages such as geothermal, fertile soils, minerals, tourist destination and therefore source of income; disadvantages such as risks from lava, lahars, ash clouds, pyroclastics, etc.; may also discuss little choice as nowhere to move to, tradition, etc., but these can only be peripheral to the argument; Level 3 5–6 marks must communicate the hazardous / beneficial features of volcanic eruptions / regions; must reach a conclusion having covered both advantages and disadvantages with developed arguments / explanations; may be more detailed on one side than the other; Level 2 3–4 marks must communicate the hazardous and/or beneficial features of volcanic eruptions / regions. covers both advantages and disadvantages with brief arguments / explanations. OR one sided looking at either advantages or disadvantages with developed arguments / explanations. Level 1 1–2 marks basic descriptive points with little or no reasoning. May just be a list of for or against. no response or no creditable response, 0.	**6**

Cambridge International Examinations
Cambridge Ordinary Level

CANDIDATE NAME	

CENTRE NUMBER						CANDIDATE NUMBER					

5497282150

ENVIRONMENTAL MANAGEMENT **5014/11**

Paper 1 **October/November 2018**

2 hours 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.
You may lose marks if you do not show your working or if you do not use appropriate units.

Write your answers in the spaces provided on the Question Paper.
All questions in Section A carry 10 marks.
Both questions in Section B carry 40 marks.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **25** printed pages and **3** blank pages.

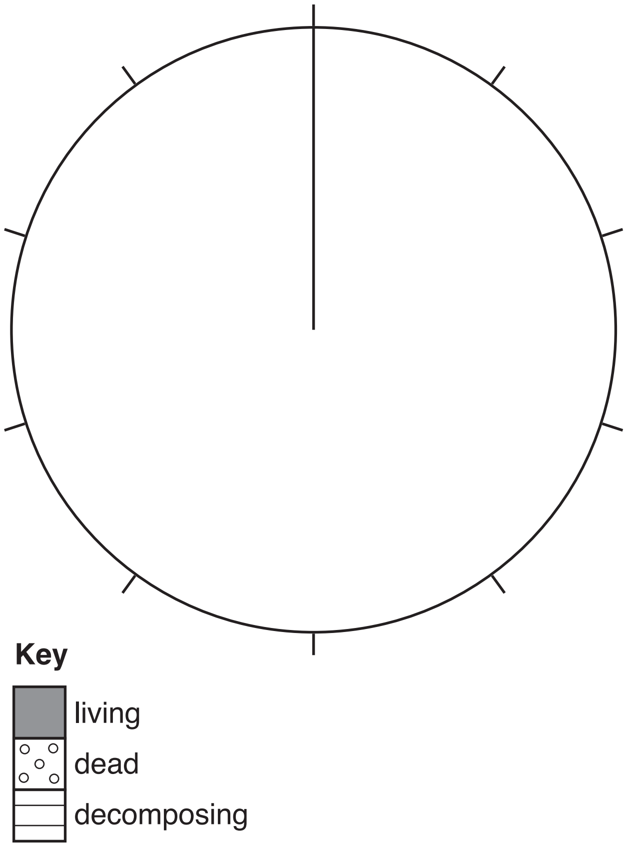
Section A

Answer **all** the questions.

1 **(a)** The table shows the average percentages of types of organic matter in the soil.

(i) Complete the pie graph, using the table and key provided.

type of organic matter	average percentage
living	3
dead	9
decomposing	88



[3]

(ii) State **one** type of living soil organism that consumes dead organic matter.

....[1]

(iii) State **one** type of living soil organism that decomposes organic matter.

....[1]

(iv) Name **two** components of soil other than organic matter.

1 ....

2 ....
[2]

(b) Explain why growing grass improves the soil.

....

....

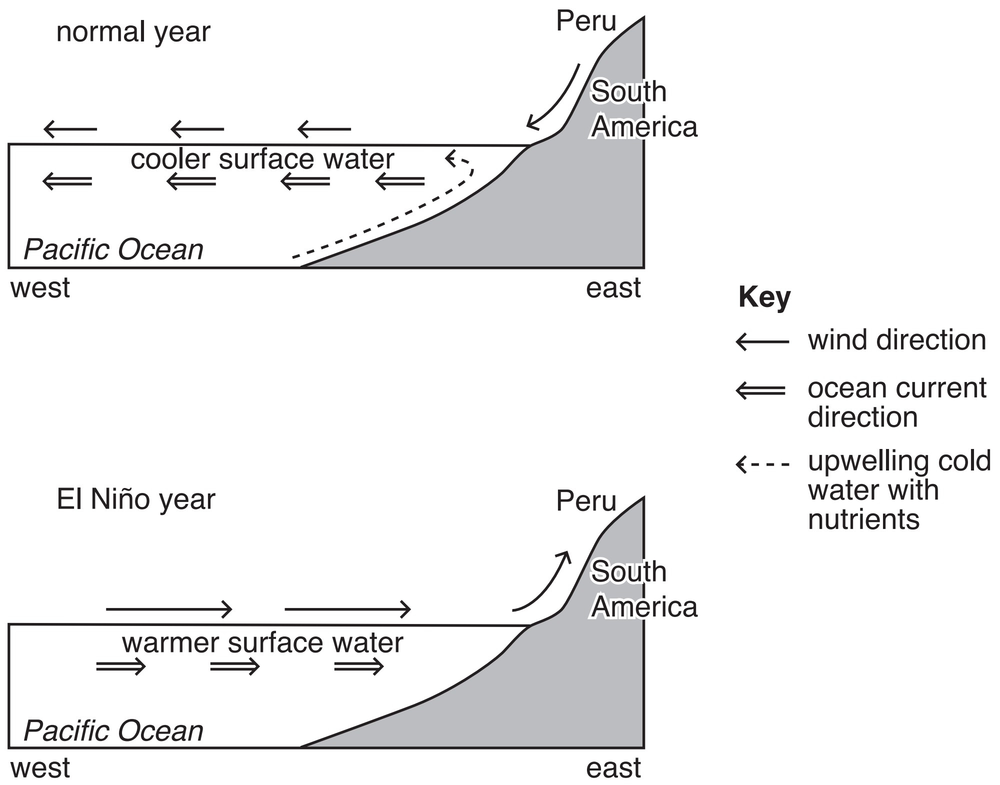
....

....

....

....[3]

2 **(a)** The diagram shows conditions in the eastern Pacific Ocean in a normal year and in an El Niño year.



(i) Use the diagram to give **one** reason why the surface ocean current flows westwards from Peru in normal years.

....

....[1]

(ii) State how the weather in Peru will differ from normal in an El Niño year.

....

....

....

....[2]

(iii) Use the diagram and your own knowledge to explain why Peru has a good fish catch in a normal year.

....

....

....

....

....

....[3]

(iv) Describe why more fish die in an El Niño year compared with a normal year.

....

....[1]

(b) There are disputes between countries in some parts of the world about who owns areas of ocean.
Suggest reasons why.

....

....

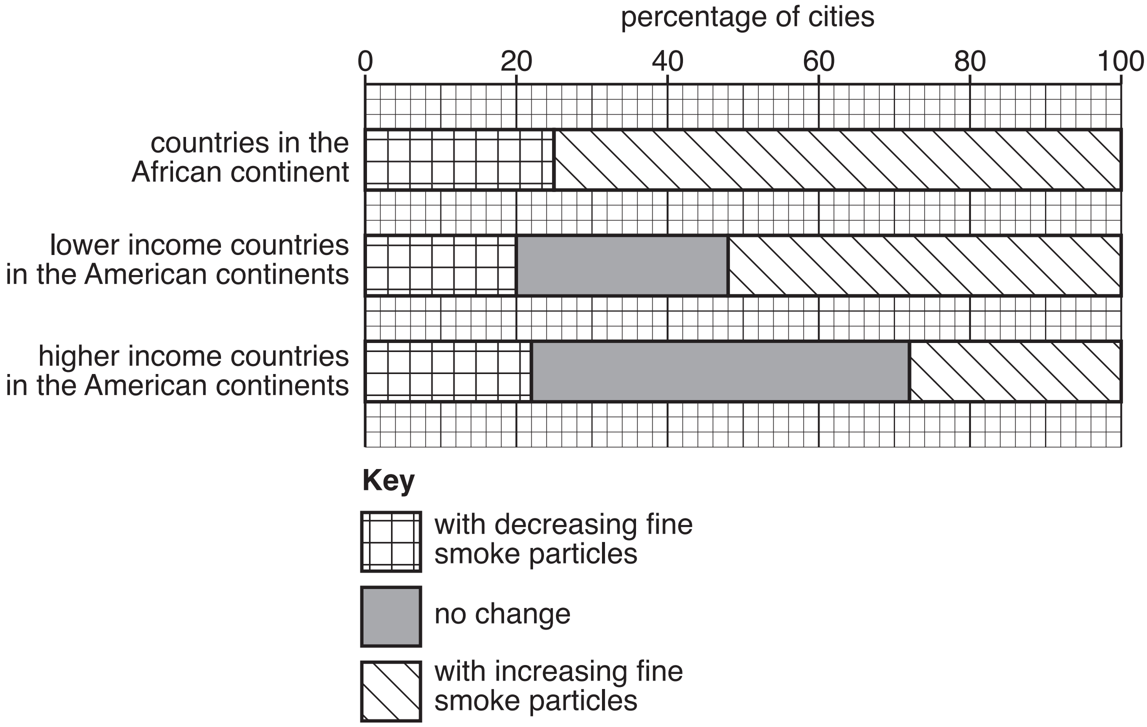
....

....

....

....[3]

3 **(a)** The graph shows the percentage of cities in selected countries that have increasing and decreasing concentrations of fine smoke particles in the atmosphere.



(i) State the percentage of cities from countries in the African continent that have an increasing concentration of fine smoke particles in the atmosphere.

....% [1]

(ii) Compare the levels of increasing fine smoke particles from lower and higher income countries in the American continents.

....

....

....

....[2]

(iii) Describe how fine smoke particles in the atmosphere damage people's health.

....

....[1]

(b) Explain why cities have more air pollution than rural areas.

....

....

....

....

....

....[3]

(c) Suggest why it is difficult to control air pollution in developing countries.

....

....

....

....

....

....[3]

4 **(a)** The photograph shows vegetational succession in an area where vegetation is regrowing.

(i) Complete the table by writing in the letters **A**, **B** and **C** to match the stages in the vegetational succession.

area	description of stage in vegetational succession
....	vegetational succession is being stopped
....	vegetational succession has been happening for a long time
....	vegetational succession has been happening for a shorter time

[2]

(ii) State the type of farming that is taking place in area **C** in the photograph.

....[1]

(b) (i) Describe how the vegetation in the final stage of a vegetational succession differs from earlier stages.

....

....

....

....

....

....[3]

(ii) Describe how seed dispersal can influence a vegetational succession.

....

....[1]

(c) Suggest how allowing natural vegetation to regrow will change the ecosystem.

....

....

....

....

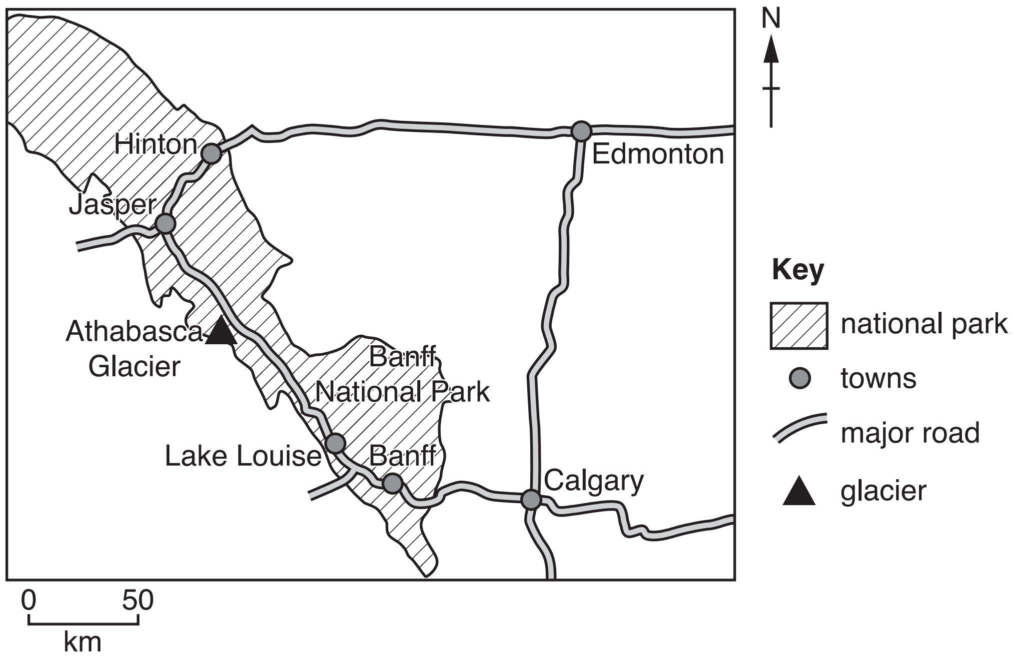
....

....[3]

Section B

Answer **both** questions.

5 The map shows Banff National Park in the Rocky Mountains of Canada. The park contains 6641 km^2 of protected wilderness area, with mountains, glaciers and dense coniferous forest. The park has approximately four million visitors every year.



(a) (i) Suggest possible benefits and negative impacts of tourism on national parks.

benefits ....

....

....

....

negative impacts ....

....

....

.... [4]

(ii) State **two** strategies for reducing the impact of tourism on national parks.

1 ....

....

2 ....

.... [2]

(b) In October 2014, a small earthquake occurred near Banff National Park.

Suggest why people live in earthquake zones even though there is a risk to their safety.

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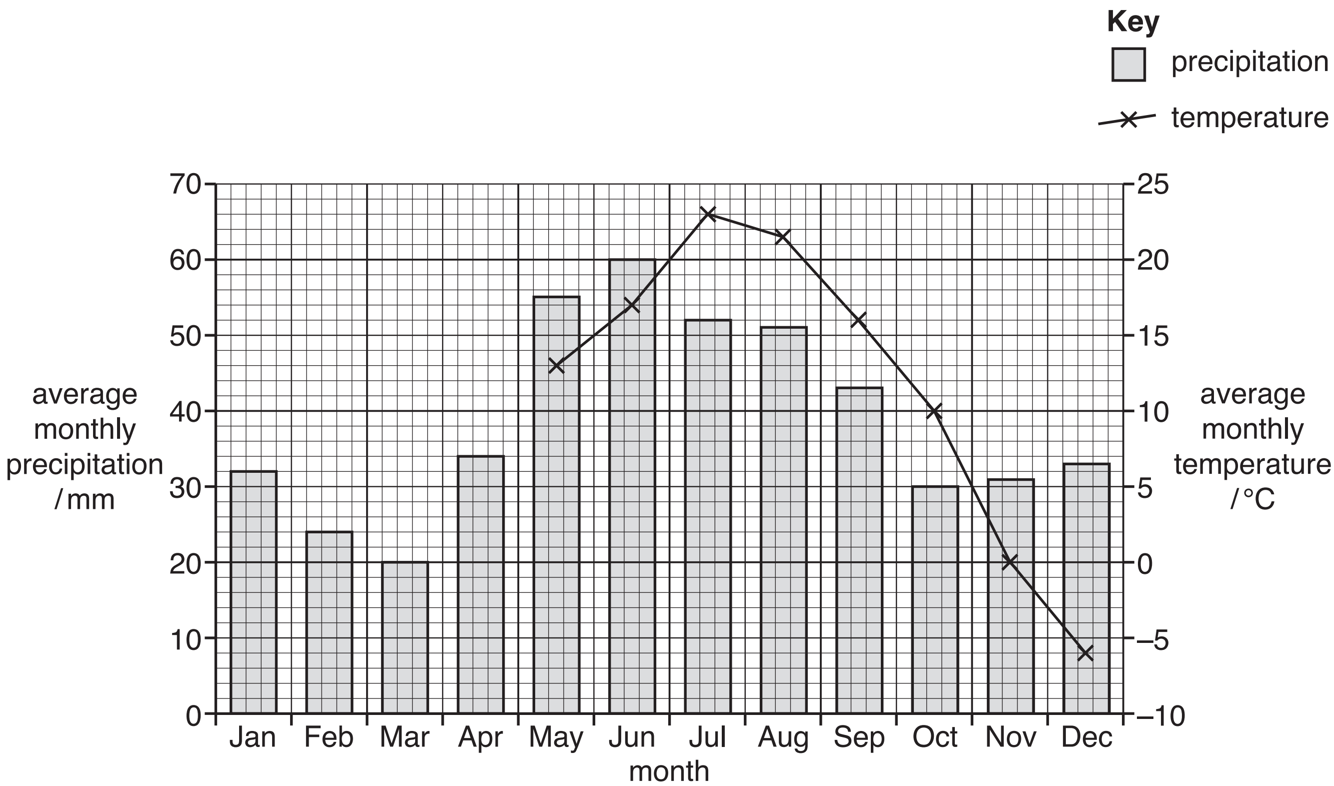
....[4]

(c) The graph shows climate data for Banff National Park.



(i) Use the data in the table to complete the climate graph.

month	Jan	Feb	Mar	Apr
average monthly temperature / °C	–5	0	4	8

[2]

(ii) Calculate the average annual range in temperature.

.... °C [1]

(iii) Calculate the total precipitation in the wettest three months.

.... mm [1]

(iv) Use the graph to describe the pattern of temperature and precipitation in Banff National Park.

....

....

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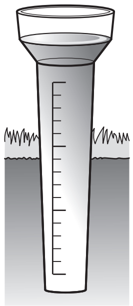
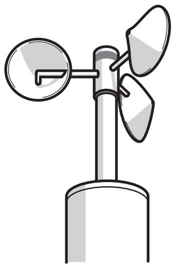
.... [4]

(v) The diagram shows instruments used for measuring elements of weather.

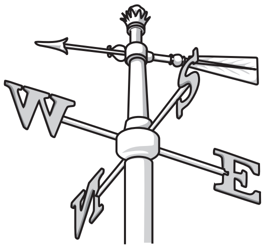


A



B

C



D

Complete the table to identify the names of the instruments and the element of weather each instrument measures.

instrument letter	element of weather that the instrument measures	name of instrument
A	....	....
B	....	....
C	....	Campbell-Stokes recorder
D	wind direction	....

[4]

(vi) In June 2013, 200 mm of rain fell in less than two days in Banff National Park. The rainfall caused major flooding of many rivers.

Describe the problems for people when rivers flood.

....

....

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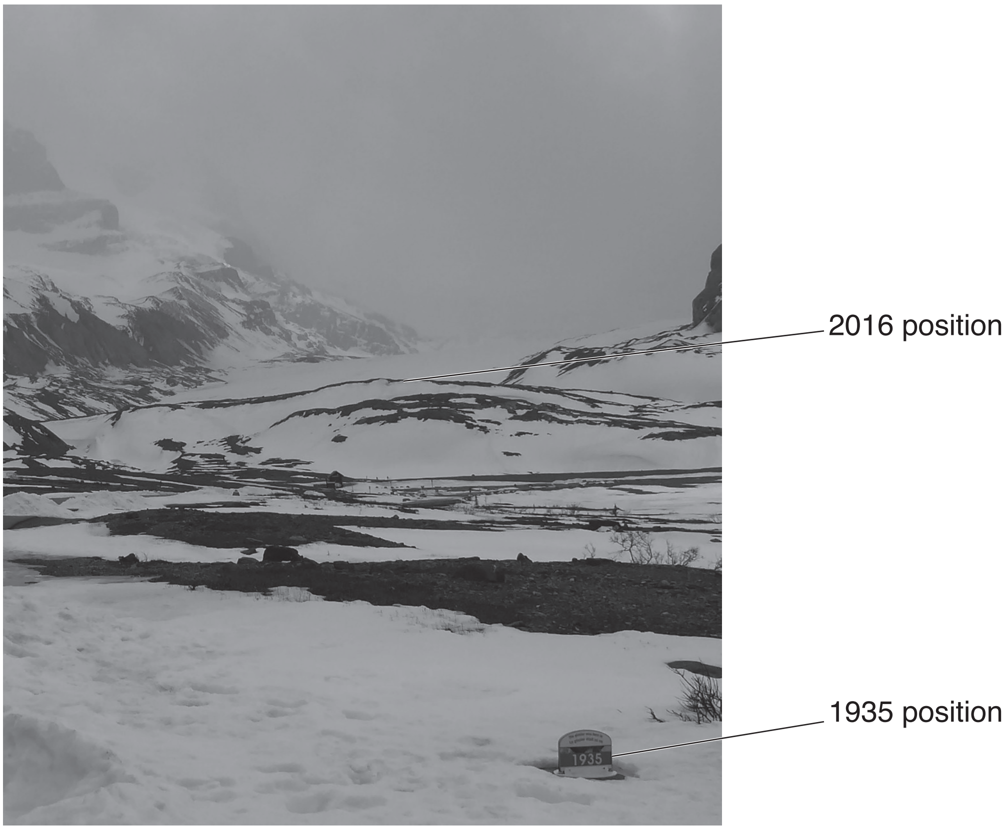
....[4]

(d) The Columbia Icefield is located in the far north-west of Banff National Park. The Athabasca Glacier is a glacier in the icefield.

A glacier is a thick ice-mass that formed thousands of years ago.

The photograph shows the position of the Athabasca Glacier in 1935 and its position in 2016.

The glacier has retreated more than one kilometre since 1935.



Suggest why the glacier has retreated since 1935.

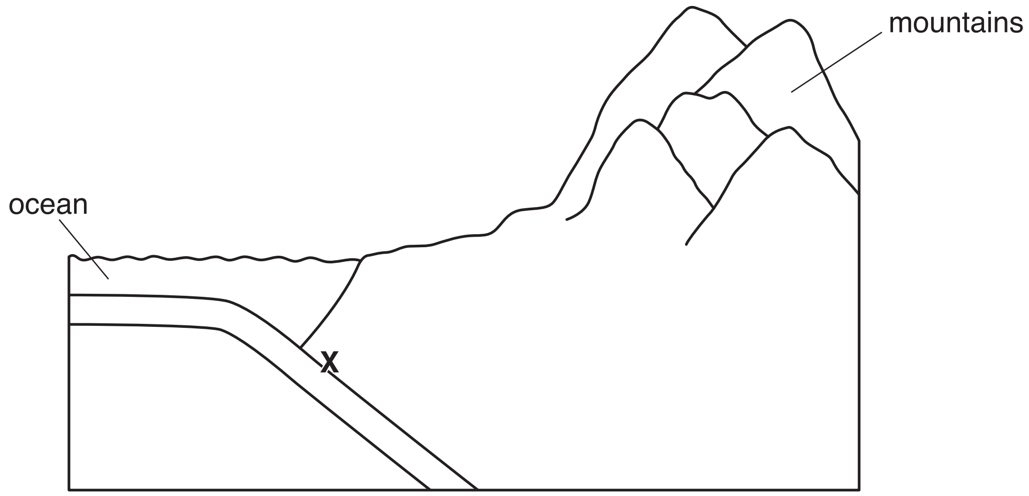
....

....

....

....[2]

(e) The diagram is a cross-section of a type of plate boundary near Banff National Park.



(i) Add labels **A**, **B**, **C** and **D** to the diagram to match the list given.

A oceanic crust
B continental crust
C mantle
D trench

[4]

(ii) State the name of the zone marked with an **X**.

....[1]

(iii) On the diagram, draw arrows to show the direction of movement of the two plates. [1]

(f) Some areas of Banff National Park have geothermally heated groundwater.

'Geothermal energy could be used as a global alternative to fossil fuels.'

To what extent do you agree with this statement? Explain your answer.

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....[6]

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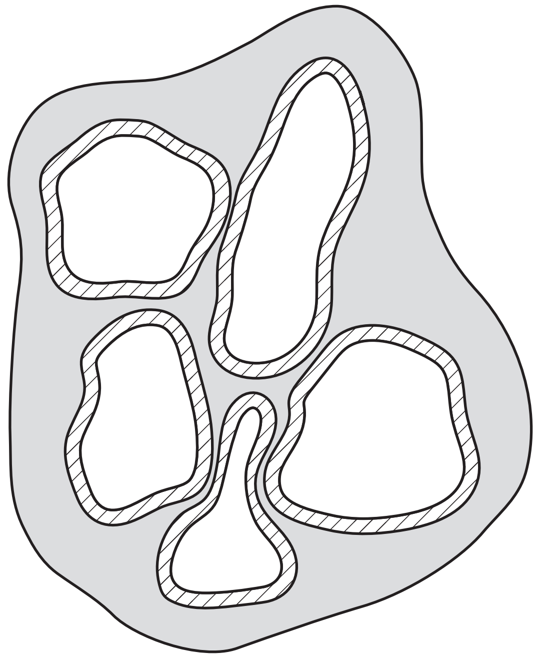
6 (a) The fact sheet contains information about oil sand.

Oil sand contains a mixture of approximately 90% clay, sand and water and about 10% bitumen.

Each grain of sand in the oil sand is surrounded by a layer of water and a layer of bitumen.



Key

- bitumen
- sand grain
- water

Bitumen is one component of petroleum (crude oil). Unlike crude oil, which can be extracted directly from the ground, the bitumen in oil sand has to be separated from the sand before it can be used. Once separated from the oil sand, the bitumen is processed in oil refineries to produce fuels such as gasoline (petrol) and diesel.

Deposits of oil sand are found all over the world, including Canada, United States of America, Russia and Venezuela. Many of the oil sand deposits are buried deep underground.

(i) Name the useful component in oil sand that can be processed to form a fuel.

....[1]

(ii) Suggest why extracting the useful component from oil sand is **not** economically viable.

....

....

....

....[2]

(b) Crude oil is a fossil fuel.

Describe the formation of crude oil.

....

....

....

....

....

....[3]

(c) The oil sand deposits in Canada are covered by forest.

The forest must be cleared before the oil sand can be mined.

(i) Describe the negative impacts of deforestation on an area of land.

....

....

....

....

....

....[3]

(ii) The wood from forests is used for timber.

State **one** way timber can be used more efficiently so that forests do not need to be cut down to provide the timber.

....[1]

(iii) A worker at a local forest said,

'Forests can be managed sustainably.'

To what extent do you agree with this statement? Give reasons for your answer.

....

....

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....[4]

(d) The information is about an environmental disaster in Canada.

In 2016, a wildfire destroyed a huge area of forest near the oil sand deposits in Alberta, Canada. The wildfire covered an area of more than 590 000 hectares and burned for two months.

The intense heat destroyed the roots of many trees. Clouds of ash were in the air for days and the smoke from the fires turned the sky black during the daytime. Large volumes of sulfur dioxide were emitted into the air during the fires.

The local government declared a state of emergency and forced nearly 90 000 people to evacuate the area during the fires. More than 1000 fire fighters, 45 helicopters, 138 fire engines and 22 aeroplane water tankers fought the fires.

About 2500 homes were completely destroyed and many more were damaged. Power and water supplies were disrupted. Despite this, only a very small number of people were killed during the wildfires.

(i) Suggest how people living in the area of the wildfires were affected by this environmental disaster.

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....[4]

(ii) The wildfire damaged the ecosystem of the area.

State the meaning of the term *ecosystem*.

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....[2]

(iii) Suggest the problems that this wildfire caused for the environment.

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....[5]

(iv) Evaluate the success of the local government's response to the wildfires.

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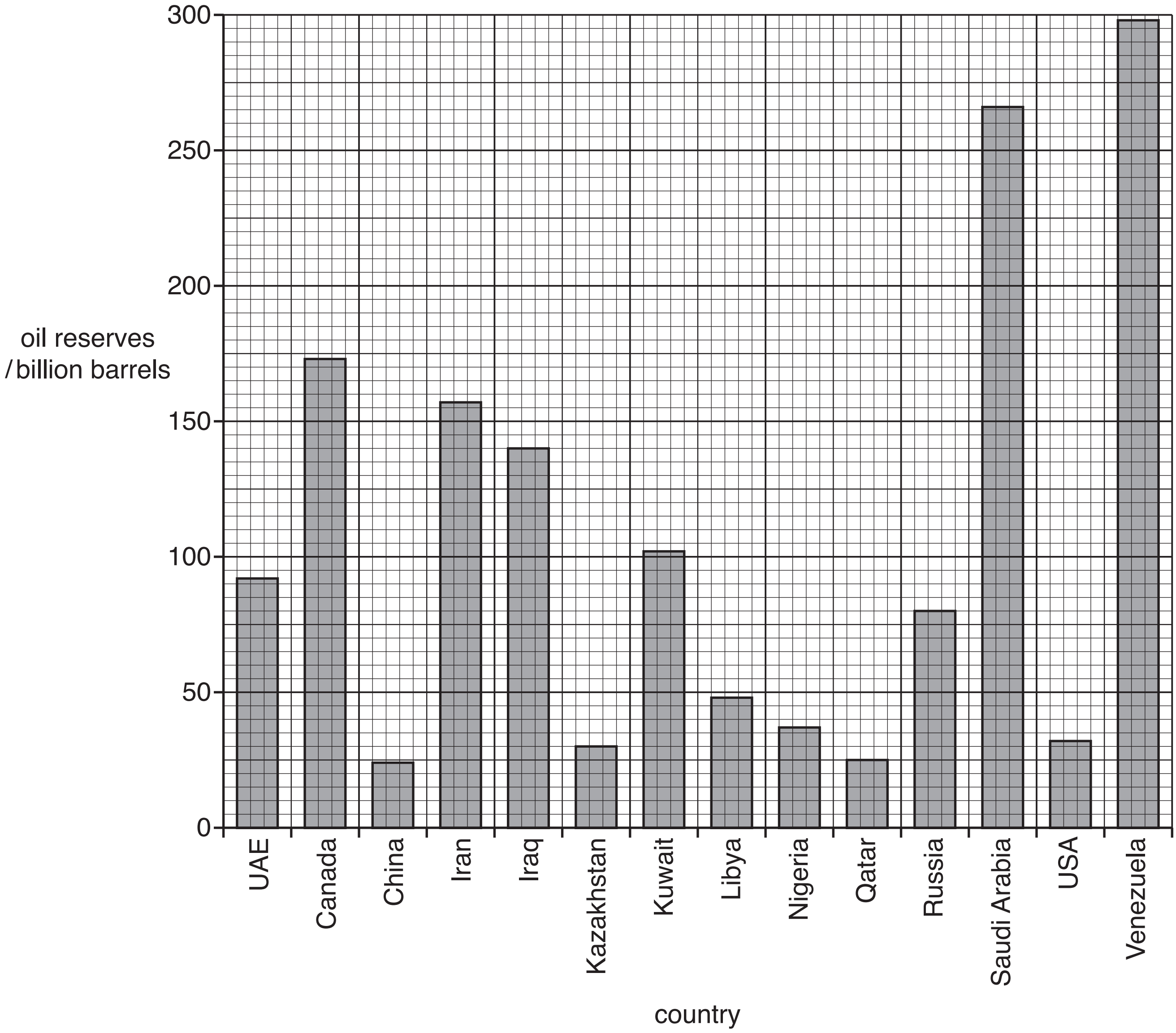
....[4]

(e) The graph shows the global reserves of crude oil in 2013 for some countries.



(i) State the amount of oil reserves in Canada.

.... billion barrels [1]

(ii) Name **one** country with more oil reserves than Canada.

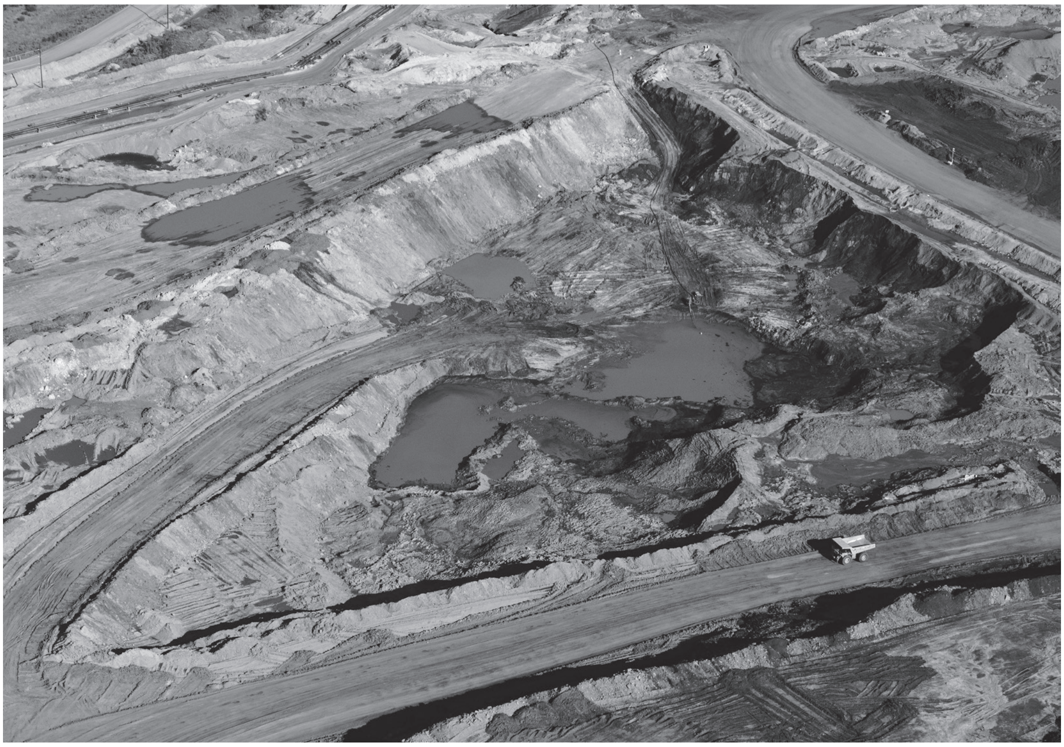
.... [1]

(iii) The value for oil reserves in Canada includes 168 billion barrels from oil sand deposits.

Calculate the percentage of Canada's oil reserves that come from oil sand deposits.

.... % [1]

(f) The photograph shows surface mining at an oil sand deposit.



(i) Describe how the mining company can restore the land after mining has finished at this location.

....

....

....

....[2]

(ii) The mining company wants to expand the oil sand mine. Some local people are against this plan.

The local government has decided to allow the expansion of the mine.

To what extent do you agree with this decision? Give reasons for your answer.

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....[6]

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Cambridge O Level

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **October/November 2016**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

CAMBRIDGE
International Examinations

Question	Answer	Marks
1(a)(i)	7.7;	**1**
1(a)(ii)	9.058 / 9.06 / 9.1 (%); allow ecf from part **(a)(i)**	**1**
1(b)(i)	July, August, September;	**1**
1(b)(ii)	return to villages / work on land / loss of money / cannot buy food / cannot pay for, education / rent / medical treatment / eq;	**1**
1(b)(iii)	*any 3 of:* mosquitoes breed in water; so (more) people bitten; so (more) malaria cases; yellow fever / dengue fever; drinking water more polluted; so cholera / dysentery / diarrhoea; no money to treat diseases; max 1 for named disease	**3**
1(b)(iv)	noise drives away animals / stop them breeding / other effect on domesticated animals; reference to stress / annoyance / scared of noise; dust reduced plant growth / leaves covered / eq; less photosynthesis; causes breathing problems; causes, eye / ear problems / eq; reduced biodiversity (of plants / animals); max 2 for noise and max 2 for dust	**4**

Question	Answer	Marks
1(c)(i)	*any 3 of:* safety masks; goggles / eq; protective clothes / eq; ear defenders / eq; helmets; boots; shelter from the sun; toilets;	**3**
1(c)(ii)	*any 2 of:* employers do not want to pay (unless they have to) / eq; business not profitable with these extra costs; reference to workers not organised in a union / eq; laws not enforced / no law exists / not employers problem;	**2**
1(d)	*any 4 of:* making a lake; landfill / infilling; create wildlife habitat / nature reserve; used for farming; further details such as adding topsoil / adding N fixing plants; planting, trees / plants;	**4**
1(e)	newest = 1; oldest = 3;	**2**
1(f)(i)	appropriate scale (plots to cover at least half of the grid); axes labelled (distance from top of waste pile / m and number of plants); one plotting error = two marks, two plotting errors = one mark;; allow bar or line graph	**4**
1(f)(ii)	increase in plants with distance (and then a plateau) / eq;	**1**

Question	Answer	Marks
1(f)(iii)	*any 4 of:* number of plants increase; less dust so more, growth / photosynthesis; number of species increase; more suitable pH / eq; pH goes down / eq; more minerals / minerals more soluble / minerals can be absorbed; temperature does not change / eq; so does not limit plant growth / eq; AVP;; allow ORA for all statements allow max 2 marks for description and max 2 marks for explanation	**4**
1(f)(iv)	*any 2 of:* only one transect / more transects needed; only one pile / more piles need to be sampled / eq; the transect / pile, could have been anomalous / eq;	**2**
1(f)(v)	*any 3 of:* same variety of wheat plants grown / seeds from same source / eq; split into two groups / experimental and control group; dust applied to one group / eq; all other conditions the same for both groups / eq; measure rate of, photosynthesis / growth / seed production / mass of seed; method repeated / replicated;	**3**
1(g)(i)	272 and 315 used; = 86 / 86.3 / 86.34 / 86.35 / 86.4(%) (not 86.0)	**2**

Question	Answer	Marks
1(g)(ii)	block makers: slurry cheap / free; no shortage of supply; make more blocks; so make more money / eq.; mine owners: can sell waste; no waste to dispose of; so not breaking the law; max 2 marks for block makers and max 2 marks for mine owners	**3**

Question	Answer	Marks
2(a)(i)	suitable key: livestock should be shown in the uplands but also allow livestock below the dam; crops shown below the dam or on either side;	**3**
2(a)(ii)	*any 2 of:* valid reasons for the areas shaded such as: plants need water for long growing season / eq; flat land for growing crops; animals only need some access to drinking water; land above dam has some food for animals; land not suitable for crops / eq;	**2**
2(a)(iii)	*any 2 of:* reference to, flow into ground / infiltration; water table stays high / eq; wells draw water from water table; reference to methods used to remove water from wells;	**2**

Question	Answer	Marks
2(b)	*any 3 of:* have to provide aid; it will cost government money; government may need to ask for overseas aid; people will expect aid; government may not be able to provide correct support / eq; AVP; e.g. to save face / eq /to stop stockpiling / to prevent panic or anxiety;	**3**
2(c)(i)	systematic / random sample / description of any valid method;	**1**
2(c)(ii)	three more questions related to animal survival or productivity;;; layout (one question must have at least two answer options);	**4**
2(c)(iii)	8.6 / 8.57(%);	**1**
2(c)(iv)	*any 3 of:* some animals can extract more water from food; some animals are, more drought resistant / adapted / do not need much water; some animals can feed on anything; (buffalos) are valuable so villagers keep feeding them; AVP; accept converse statements	**3**

Cambridge O Level

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/11**

Paper 1 **May/June 2016**

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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CAMBRIDGE International Examinations

Section A

Question	Answer	Marks
1(a)(i)	A; C; D;	**3**
1(a)(ii)	magma is rising / volcano formed;	**1**
1(b)(i)	*any 2 of:* it is very hot / molten; can set fire to objects with which it comes into contact; destroys buildings / vegetation / etc.; it is mobile; it can be rapid; kills wildlife / livestock / people;	**2**
1(b)(ii)	earthquake / tsunami;	**1**
1(c)	*any 3 of:* roads have poor surfaces / routes are cut (by lava / ash) / access is difficult; communication by telephone / internet is difficult; the disaster is large-scale; no / few evacuated before the disaster / disaster not forecast; poor economies / LEDCs / less developed country which lack sufficient numbers of relief teams / shelters / medical staff; major eruption will have greater impact on relief efforts than minor eruption; type of eruption – explained;;; (e.g. if Mt St Helens type with large amounts of ash, lahars, mudflows, etc. will make access more difficult and will cause more damage / affect more people / affect wider area AVP; *accept the opposite emphasis on why relief can be supplied more easily*	**3**

Question	Answer	Marks
2(a)(i)	800 000 and 1967–1969;	1
2(a)(ii)	the catch / fish stocks had been declining rapidly;	1
2(a)(iii)	no and fishing ended (in 1992); OR partly / yes and fish caught increased for a time;	1
2(b)	*any 4 of:* bottom sea trawling – damaged food chain / caught deep stocks; use of radar – enabled shoals to be found; large nets – caught large(r) numbers of fish; large boats / factory ships – could deal with more / large numbers of fish; refrigerated vessels – could stay fishing / at sea longer; *credit each line with up to two marks: one for the method and one for the explanation of its effect.* *allow a maximum of 3 for methods.* *effects must be related to a method.*	4
2(c)(i)	*any 1 of:* (too) few fish of reproductive age were left; pollution; illegal fishing; climate change;	1
2(c)(ii)	*any 2 of:* because many would lose their way of life / incomes / employment; the whole community would be badly affected / economic downturn / services would lose income; possibility that voters would turn against them; short-term politics; AVP;	2

Question	Answer	Marks
3(a)(i)	*any 2 of:* tropical; 10 °S to 20 °N Central America; South America; one / the Philippines in Asia;	**2**
3(a)(ii)	*any 3 of:* northern countries have the purchasing power; aware of the healthy properties of bananas / need for fruit in diet; north too cold to grow bananas / climate further south suits the growth of bananas; countries growing bananas / further south are less developed and more likely to rely on farming / export produce; bananas are a cheap fruit;	**3**
3(b)	horizontal line at 2000;	**1**
3(c)(i)	*any 3 of:* provides employment / income for workers; helps to pay for imports; provides foreign exchange / increases economy / GDP; helps to pay for the development of infrastructure / example of; improves standing in the world;	**3**
3(c)(ii)	crop grown for farmer and family to eat / not for sale (unless surplus);	**1**

Question	Answer	Marks
4(a)(i)	*any 3 of:* conical / coniferous; short branches; sloping branches; all look alike; thin; dark foliage;	**3**
4(a)(ii)	taiga;	**1**
4(b)	*any 3 of:* rocky ground; short growing season; very cold winters; frozen ground / permafrost; infertile / few plant nutrients; thin soil;	**3**
4(c)	*economic* timber / construction / furniture; poles; pulp / paper; employment *environmental* habitat / food for birds / animals / insects; prevents soil erosion; helps maintain the oxygen / carbon dioxide balance of the atmosphere; reduces global warming; transpires water vapour into the atmosphere; *at least one from each group for maximum.*	**3**

Section B

Question	Answer	Marks
5(a)(i)	both divisions correct;; (one division correct;) key completed to match pie chart;	**3**
5(a)(ii)	*air* respiration for soil organisms: respiration for plants; needed for decomposition; oxygen / respiration for roots; *water* needed for photosynthesis; dissolves minerals / nutrients for take up by plants; prevents wilting / hydrates plants; *max 2 on either air or water*	**3**
5(a)(iii)	*any 4 of:* decomposition; of (dead) plant material; of (dead) fauna); and turn them into nutrients / humus / minerals; worms aerate the soil; move down / mix organic material through the soil; role of organisms in: carbon cycle, nitrogen cycle, sulfur cycle;	**4**
5(b)(i)	2.5–3.0 times;	**1**
5(b)(ii)	correct scale on x and y axes; correct labelling of *y*-axis; 5 or 6 points plotted correctly;; (3 or 4 points plotted correctly;) *± half small square tolerance on plotting*	**4**

Question	Answer	Marks
5(b)(iii)	*any 1 of:* cheaper meat; from agricultural intensification / more meat available; increased wealth (in developing countries) – however indicated; growth of fast food chains;	**1**
5(c)(i)	Africa;	**1**
5(c)(ii)	North America and Oceania;	**1**
5(c)(iii)	*any 2 of:* tradition / beliefs; wealth explained / less developed explained / too expensive; infrastructure such as transport, refrigeration; suitability of land / climate for crop growing / animal rearing; much meat exported; other sources of protein;	**2**

Question	Answer	Marks
5(c)(iv)	*soil erosion* trampling destroys vegetation; trampling compacts soil; overgrazing; animals pull roots from soil; soil becomes exposed to wind / water; gets blown / washed away; *water pollution* produce (a lot of) waste / faeces; can add pathogens / harmful bacteria / viruses; fertilisers added (to fodder crops); gets washed into groundwater / streams / rivers; (or) animals defecate in water; nitrates increase in water; leading to eutrophication; *allow development marks, for example on impact of compacting soil, nutrients in water, etc.* *max 3 on either soil erosion or water pollution*	**5**
5(d)(i)	6.5;	**1**
5(d)(ii)	1000 times;	**1**
5(d)(iii)	*any 3 of:* cattle produce (large amounts of) methane; methane is a greenhouse gas; which has much greater effect per molecule than carbon dioxide; forests cleared to create cattle farms; burning forests give off carbon dioxide / reduces carbon dioxide uptake; carbon dioxide is a greenhouse gas; so could lead to (enhanced) global warming; explanation of how greenhouse gases trap heat to max 1;	**3**

Question	Answer	Marks
5(e)(i)	1;	**1**
5(e)(ii)	*any 3 of:* new pesticides less toxic; smaller amounts applied; laws to restrict / ban use of pesticides; restrictions on most hazardous / dieldrin and lindane;	**3**
5(f)	*Indicative content* excess fertilisers washed into water courses leading to algal blooms and eutrophication, with effects of plant and animal life in the water; controlled application to limit excess fertilisers and possibly use of organic fertilisers; clearance of algal blooms, oxygenation of water courses affected; Level 3 5–6 marks answers the question and provides detailed explanation of how fertilisers can damage the environment and then how those impacts can be reduced. Level 2 3–4 marks some detail of how fertilisers can damage the environment with some explanation of reduction of damage. OR provides detailed explanation of how fertilisers can damage the environment, but nothing worthy of credit on reduction or vice versa. Would need the one good part to be very good to achieve top of the level. Level 1 1–2 marks basic descriptive points with little or no explanation. May just be a list with little or no answer to the part on reduction. no response or no creditable response, 0.	**6**

Question	Answer	Marks
6(a)(i)	*any 2 of:* same amount of heat / energy in both rays; smaller area to heat up at Equator / larger area to heat at Arctic Circle / high angle of incidence at Equator, low angle at Arctic Circle; lower albedo at Equator / owtte; less energy loss at Equator as shorter distance through atmosphere;	**2**
6(a)(ii)	*any 2 of:* snow and ice have higher albedo / forests have lower albedo; so ice and snow reflect back most of solar energy; forests darker so absorb more solar energy; photosynthesis will absorb energy in forests / energy needed for growth;	**2**
6(b)	November; Bruce ; Christine; 3; April; 204;	**6**
6(c)(i)	16;	**1**
6(c)(ii)	5;	**1**
6(c)(iii)	6–7;	**1**
6(c)(iv)	*onto land* need warm water (to provide energy); mention of role of latent heat; so energy source cut off; *south* oceans become cooler; so less energy to power the cyclone; *max 2 on land or south*	**3**

Question	Answer	Marks
6(c)(v)	*any 4 of:* cyclones at their strongest at the coast; many coastal areas densely populated; low-lying areas liable to flooding from heavy rainfall; low-lying areas liable to flooding from storm surges; as strong winds create large waves; and low pressure raises sea level; allow flooding for 1 mark even if no cause given;	**4**
6(c)(vi)	*any 3 of:* less educated about coping with a cyclone; houses less strong; technology for advance warning less advanced; warnings may not get through to people; fewer (if any) cyclone shelters; search, rescue, medical facilities less good; poor communication system; evacuation less likely; medium and/or long-term effects to max. 2;;	**3**
6(d)(i)	*any 3 of:* easterly winds weaken; warm water current moves eastward / towards Peru; rather than the usual westward direction / towards Australia; cold Peruvian current blocked / replaced by warm water;	**3**
6(d)(ii)	*any 2 of:* warm water off the coast; so more evaporation; warm air (by ocean) likely to rise; leading to condensation / cloud / rain;	**2**
6(d)(iii)	heavy rain improves agriculture (or similar);	**1**

Question	Answer	Marks
6(e)(i)	phytoplankton; zooplankton; fish; all correct = [2] 1 or 2 correct = [1]	**2**
6(e)(ii)	*any 3 of:* less food for fish / whales so their numbers reduce; less fish for seals and penguins so their numbers reduce; less seals and penguins so shark numbers reduce; if no specifics can award 1 mark for general comment on reduction of all species in higher trophic levels; phytoplankton will start to increase;	**3**

Question	Answer	Marks
6(f)	*Indicative content* droughts tend to last longer and affect larger areas than floods or cyclones; floods and cyclones cause more damage to property than droughts; all can lead to soil erosion and loss of crops / animals; cyclones and floods are short-term and may require emergency rescue, shelter, food, etc.; effects of drought, being longer term, can be planned for, but can cause far more deaths than the others if no food aid, etc.; most environments recover from such disasters; it will depend on the severity of each disaster as to environmental effects, though flooding and cyclones more likely to result in pollutants being washed into water courses or the sea; Level 3 5–6 marks must communicate the features of at least one other climatic hazard besides droughts. Answers the question with detailed consideration of climatic hazards. Must look at both sides of the argument, so must be some discussion of floods and/or cyclones as well as droughts and their relative impacts. Level 2 3–4 marks Considers both sides (i.e. other climatic hazards besides droughts) with some details OR detailed consideration of one climatic hazard (probably droughts) Level 1 1–2 marks basic descriptive points with little or no reasoning. May just be a list of impacts of cyclones and an agreement without mention of other climatic hazards. no response or no creditable response, 0.	**6**

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **May/June 2017**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Question	Answer	Marks
1(a)(i)	(900 / 25.7 =) 35;	**1**
1(a)(ii)	25;; (*if answer incorrect, allow one mark for 35 × 2 or = 70 [1]*)	**2**
1(a)(iii)	*any one of:* saves buying them; to make more profit; AVP, e.g. recycling / environmental issue / small bags are scarce / expensive to buy;	**1**
1(a)(iv)	*any two of:* sand stolen; sand needs to be guarded at night; danger to life; AVP;	**2**
1(b)	*any two of:* vegetables supply; vitamins or any named vitamin; minerals or any named mineral; roughage / fibre; essential calories / supplies carbohydrate or protein or fat; prevent malnutrition; balanced diet;	**2**

Question	Answer	Marks
2(a)(i)	(**A**) anemometer; (**B**) rain gauge;	**2**
2(a)(ii)	thermometer / sunshine recorder / barometer / wind vane;	**1**
2(a)(iii)	*any two of:* grass surface; flat / stable ground; clear ground / away from trees or buildings / eq;	**2**
2(a)(iv)	7;	**1**
2(a)(v)	orientation and linear scale; *x*-axis fully labelled; *y*-axis fully labelled; all plots correct;	**4**
2(a)(vi)	April OR May OR June to September OR October;	**1**
2(a)(vii)	*any two of:* cheap or easy to construct; does not use too much labour / can be built by local people or farmers; more water bodies to feed nearby fields / eq; less relocation of people; *idea of* less flooding; AVP, e.g. easier to irrigate crops / greater access to water (to grow crops) / more sustainable / easier to repair / less pollutants likely to accumulate;;	**2**
2(a)(viii)	*any one of:* warm enough for plant growth; only a short dry season / eq;	**1**

Question	Answer	Marks
2(b)(i)	(solar / heat) energy absorbed at the surface / eq;	**1**
2(b)(ii)	0.0–0.5 m / 0–50 cm;	**1**
2(b)(iii)	(reversed shape due to) radiative or longwave cooling / cools most from the surface / it's the surface where the longwave radiation comes from / ORA / eq;	**1**
2(b)(iv)	*any three of:* *reference to* photosynthesis; further detail related to photosynthesis, e.g. optimal temperature for photosynthesis / photosynthetic rate increases / eq; *reference to* enzymes; further detail related to enzymes, e.g. discusses optimal temperature for enzymes; faster growth possible during day or when warm; some growth at night or cell elongation;	**3**
2(c)(i)	*after first year of cultivation* (rapid) drop from centre to edge of garden or from **P** to **T**; *after three years of cultivation* same availability across garden / eq;	**2**
2(c)(ii)	*any two of*: more people / population increases; more food needed; no more land (can be cultivated); AVP;	**2**
2(c)(iii)	*any two of:* not enough or loss of, nutrients / nitrates or other named minerals / eq; *reference to* damaging soil structure / soil degradation; poor plant growth; plant prone to disease;	**2**

Question	Answer	Marks
2(d)(i)	table drawn; suitable headings; minimum of 4 cells to collect data;	**3**
2(d)(ii)	*any three of:* not all paths surveyed; only for, 3 hours / limited time; only, some / two days a week; do not know quantity or mass or weight of wood / only people counted; AVP;	**3**
2(d)(iii)	*any three of:* use more people to do the survey / eq; survey all or more paths; more days; more hours / more times, a day; weigh wood / other measurement; AVP;	**3**
2(e)	*any two of:* being burnt (by the fire); being scalding (by the contents); damage to lungs / respiratory or breathing difficulties / named lung condition / eq; damage to eyes; toxic or poisonous gases / carbon monoxide; food poisoning / undercooked food / named illness;	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(f)(i)</td><td>6 correct [3]
4 to 5 correct [2]
2 to 3 correct [1]
<table><tr><th>open fire</th><th>new type of stove</th></tr><tr><td>5.75 (allow 5.5 to 6.0)</td><td>3.75 (allow 3.5 to 4.0)</td></tr><tr><td>15</td><td>10</td></tr><tr><td>5</td><td>13</td></tr></table>;;;</td><td>3</td></tr>
<tr><td>2(f)(ii)</td><td>256;</td><td>1</td></tr>
<tr><td>2(f)(iii)</td><td>any two of:
same volume of water;
same quantity or mass of hay;
same quantity or mass of beans;
same size of box;
same size pot;
type or species of wood;
AVP;</td><td>2</td></tr>
<tr><td>2(f)(iv)</td><td>14 / 14.1 / 14.09;;

(if answer incorrect, allow one mark for correct working or 42, e.g. 298-256 = 42 [1])</td><td>2</td></tr>
<tr><td>2(g)(i)</td><td>any three of:
less wood used / wood saved;
fewer trees cut down;
trees can be replanted;
wood is a renewable resource / eq;
more food can be cooked (to supply an increase in population);
less carbon dioxide emissions / reduced impact on climate / it's carbon neutral;
save time for cooking or gathering wood / cooks faster;
AVP, e.g. appropriate (sustainable) technology;</td><td>3</td></tr>
</table>

Question	Answer	Marks
2(g)(ii)	*any two of:* (limited) supply **and** (high) demand idea / ORA; (more) transport costs; AVP;	**2**
2(g)(iii)	*any two of:* not possible to cook on electricity due to cost; power cuts; supply of other fuel is unreliable; they know how to do it; reduce (running) costs / other fuel is more expensive; cools more slowly / keeps food hot for longer / cooks faster; safe to use;	**2**

Cambridge O Level

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **May/June 2017**

MARK SCHEME

Maximum Mark: 60

Published

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This document consists of **6** printed pages.

CAMBRIDGE International Examinations

Question	Answer	Marks
1(a)(i)	*any two of:* damage / interfere, with farmland / eq; so less food available / eq; *reference to* risk of pollution of Nile; risk of flooding homes; difficult to travel into the city; easier / more space, to the east; AVP, e.g. unsuitable ground for building;	**2**
1(a)(ii)	estimate between 100–200 (km^2);; (*if answer incorrect, allow one mark for* use of scale *[1]*);	**2**
1(a)(iii)	*any two of:* only desert; not used for farming; unused land; more space available for building; connects to existing housing / old city; connects to services / named services; low transport costs; no risk of flooding; AVP;	**2**
1(a)(iv)	*any three of:* roads / transport; electricity; telecoms; water supply; sewage removal; hospitals / healthcare / clinics; schools / education; *reference to* street lights / waste collections / other example; law enforcement / other named emergency service;	**3**

Question	Answer	Marks
1(a)(v)	*any two of:* cooling effect; *reference to* transpiration; shade; windbreak; absorbs some air pollutants; scenic value / pleasant environment / eq;	**2**
1(b)	*any three of:* for irrigation; so crops can grow all year; cheap / easy access to, water supply; easy to keep, animals / livestock; close to, population / farm workers; transport, crops / goods, to people; rich / fertile soil;	**3**
1(c)(i)	*any two of:* bacteria fix nitrogen gas from atmosphere / eq; pea plants do not need fertiliser; *reference to* amino acids / proteins / DNA / enzyme;	**2**
1(c)(ii)	5 / 4.6 / 4.58 / 4.583; g;	**2**
1(c)(iii)	5 OR 3–8;	**1**
1(c)(iv)	2400;; (*if answer incorrect, allow one mark for correct method, e.g.* 60 × 40 *[1]*)	**2**
1(d)(i)	*next two pairs circled* 4+6 AND 9+5; correctly drawn on grid;;	**3**

Question	Answer	Marks
1(d)(ii)	more plants / samples (3 not 2); more areas of field (10 areas rather than 1); random sampling; many samples remove bias / eq; anomalies / outliers, can be identified; and removed from average or have less effect; *reference to* validity;	**3**
1(e)(i)	*4 correct [2]* *2 to 3 correct [1]* 5, 2, 3, 1;;	**2**
1(e)(ii)	**R** <u>and</u> *any two of:* **highest** values for, BOD; bacteria; chromium; iron; allow a reason, e.g. not enough oxygen means organisms will die / chromium is toxic / etc.;	**2**
1(e)(iii)	**T** <u>and</u> low / no, BOD / bacteria count / no metal content;	**1**
1(e)(iv)	**R** <u>and</u> highest concentration of, chromium / iron;	**1**
1(e)(v)	*any two of:* organic matter / sewage / fertiliser, present; more bacteria / bacteria feed on this / multiply / eq; bacterial respiration increases / respire <u>more</u> oxygen; so oxygen used up / eq;	**2**
1(f)(i)	129 / 128.6 / 128.57 / 128.571(%);; (*if answer incorrect, allow one mark for* 4.5 e.g. 8.0–3.5 = 4.5, 4.5 / 3.5 × 100 *[1]*)	**2**
1(f)(ii)	to check there were no changes in temperature / make sure it was not an important factor in this canal;	**1**

Question	Answer	Marks
1(f)(iii)	*any two of:* dissolved oxygen decreased at site 3; dropped by a half / by 4.4 (between site 2–3); organic matter / sewage / fertiliser / other valid effluent, e.g. tannery waste; so bacteria increase in numbers and use up oxygen / BOD increase;	**2**
1(g)	*allow any three valid suggestions, such as:* more monitoring of water sources; strict laws about water quality; enforcing the laws; take polluters to court; polluter pays for clean up; AVP;	**3**
2(a)(i)	add all the responses for each question; express as a % of total number of questionnaire or out of 100;	**2**
2(a)(ii)	so decision makers can compare results (of different living areas); to find out if only people near the canal were in favour / ORA;	**2**
2(a)(iii)	*any two of:* reason why (government) need the money / eq; government can find out if people, approved of the project / value the project; give people / nation, a sense of ownership of project / eq; AVP;	**2**
2(a)(iv)	*any two of:* close to, canal / new roads, for easy transport; low cost of transport; encourages, import / export / trade; creates new jobs / canal already has skilled workers, so readily available / eq; provides source of water for industries;	**2**
2(a)(v)	tax breaks / investment grants / interest free loans / subsidies / eq or AVP;	**1**

Question	Answer	Marks
2(a)(vi)	*any three of:* *for*: saves, money / time; so more ships will use canal; more canal fees earned; AVP; *against:* canals already save days / weeks anyway (not going around Africa); new route only saves 8 hours; may not be demand for these improvements; AVP;	**3**
2(b)(i)	*any two of:* swim; flow / drift, in current; in ships bilges; cargo; stuck on hull; smuggled / tourists; AVP;	**2**
2(b)(ii)	*any three of:* disrupt food chain / web; *reference to* competition; e.g. effect of invasive producer or consumer; lack of natural predators; so population explosion; or population crash; habitat destruction; decrease in biodiversity; credit valid <u>aquatic</u> example; can carry, disease / pathogen / toxin;	**3**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **October/November 2018**

MARK SCHEME

Maximum Mark: 60

Published

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Generic Marking Principles

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GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

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- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	5.3;	**1**
1(a)(ii)	6 (°C);	**1**
1(a)(iii)	(19.5 + 3 =) 22.5 / 23 ;; (*if answer incorrect, allow one mark for 0.6 × 5 = 3 (°C) [1]*);	**2**
1(b)(i)	photosynthesis;	**1**
1(b)(ii)	*any two from:* Farm **A** warmer at a lower altitude / Farm **B** cooler at a higher altitude; faster growth (in warmer climate) / faster development; further details, e.g. rate of metabolism / flower earlier / ripen faster;	**2**
1(b)(iii)	*any two from:* low average rainfall; so fertiliser not washed off fields; so fertiliser not wasted; so fertiliser can be absorbed (as tree grows); needed to develop, flowers / fruit;	**2**
1(b)(iv)	*any three from:* fertilisers, leached / washed off field due to rainfall; (enters rivers / streams) causes eutrophication; algal bloom; which blocks light; death of plants due to lack of light; increase in bacteria; bacterial respiration; decomposition uses up oxygen; death of fish; disrupts, food chain / food web / reduction in biodiversity;	**3**

Question	Answer	Marks
1(b)(v)	*any three from:* not enough light for growth of crops; too difficult to harvest between trees; cultivation, disturbs / damages trees; competition for water; competition for nutrients; difficult to control pests; economic comment such as, value of avocado crop enough to support farmer / increased cost of, labour / seeds / fertiliser if crops grown as well;	**3**
1(c)	*both benefits for farmers and benefits for the government must be covered for maximum credit:* *max two from, benefits for farmers:* farmers, can always sell all the crop / have access to bigger market; make more money; invest in more, equipment / fertiliser / education / healthcare; *max two from, benefits for the government:* government collect taxes; improved, economic growth / GDP; so can spend money on infrastructure;	**3**
1(d)(i)	(0.17 / 1.47 × 100 =) 11.56 / 11.6 (%) ;; (*if answer incorrect, allow one mark for 1.64 – 1.47 = 0.17 [1]*);	**2**
1(d)(ii)	*any four from:* reduced interception; reduced infiltration; (increased) surface run-off; soil no longer held in place by roots; soil washed away; trees act as wind breaks; wind blows soil away;	**4**

Question	Answer	Marks
1(d)(iii)	produce hybrids / by cross pollination / selective breeding; genetic manipulation / genetic modification / gene transfer / genetic engineering;	**2**
1(e)(i)	192;	**1**
1(e)(ii)	62.54 / 62.5 / 63;	**1**
1(e)(iii)	be careful, with knife / cutting the fruit;	**1**
1(e)(iv)	*any two from:* use seeds to grow more trees; use to make compost / manure; use as animal feed;	**2**

Question	Answer	Marks
2(a)(i)	suitable linear scale; y axis fully labelled including units; x axis fully labelled; correct plots;	**4**
2(a)(ii)	the price (of copper) decreases (with time);	**1**
2(a)(iii)	*any two from:* too many copper mines operating / (world) oversupply of copper / recycling copper; less demand for copper; description of reduced demand, e.g. using alternative materials; local protests stop copper mining;	**2**
2(b)(i)	*any two from:* deposit not as big as first thought; not going to be profitable; AVP, e.g. unforeseen geological / environmental issue, such as underground flooding / dispute over mining rights / land ownership;	**2**

Question	Answer	Marks
2(b)(ii)	to make sure that mining can take place without doing too much damage to the environment / OWTTE;	1
2(b)(iii)	*any three from:* holes can be filled with, waste rock / overburden; landfill; landscape / spoil heaps, can be shaped; covered with soil; and soil planted; use plants that absorb toxins; reference to, safe drainage through soil heaps; AVP, e.g. National Park / wildlife reserve;	3
2(c)(i)	*any two from:* they want employment; all year round; better paid; named example of raising their standard of living, e.g. better housing / infrastructure; job security for 12 years; risk of being caught illegal logging;	2
2(c)(ii)	*any two from:* stable income / mining better paid (than logging); so reduced logging; people do not need to do illegal logging to earn money;	2
2(c)(iii)	*any three from:* mining produces toxic waste; their drinking water may become toxic; toxic materials kill the, fish / organisms in the river; could poison, humans / stock animals; reduce crop growth;	3
2(d)(i)	select at random / stratified sampling / AVP; further description on how it is done, e.g. volunteer sampling;	2

Question	Answer	Marks
2(d)(ii)	*any two from:* they earn money / it provides jobs (from working in hotels / tour guides / souvenir selling, etc.); to pay for, school / medical care / other named example; better infrastructure for everyone; help people learn another language;	**2**
2(d)(iii)	*any two from:* (they think) the butterfly is being protected; and tourists cannot visit other sites / can only visit two, sites / zones / controlled areas; tourists only visit in winter; it is ecotourism;	**2**
2(d)(iv)	*question about tourism such as,* Do you sell butterflies? / Do you earn money from tourists? / Do you work in tourism? / Do tourists want to see other species? ;	**1**
2(e)	*any two from:* raise awareness of the butterfly within Mexico; and in other parts of the world the tourists come from; AVP, e.g. generates money for conservation;	**2**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 **October/November 2018**

MARK SCHEME

Maximum Mark: 120

Published

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GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	**Z**; **Y**;	**2**
1(a)(ii)	circles / semi circles / arcs, with the highest intensity in the centre / with lower intensities away from the centre;	**1**
1(a)(iii)	*any four from:* large-scale destruction / widespread destruction / very disastrous; buildings destroyed; bridges destroyed / roads destroyed; broken / leaking, (gas / water) pipes; electricity / communication, damage; fires; deaths / injury; workplaces destroyed; large cost to the economy; loss / damage, of water supplies / sewers ; disease outbreaks / cholera;	**4**
1(b)(i)	*any one from:* mineral wealth / mining / quarrying; tourism; developed before danger was known; (may have) geothermal energy; fertile soils; availability of water; high population density;	**1**
1(b)(ii)	*any two from:* buildings / bridges, built to earthquake proof standard / strengthened; example of design feature; land use zoning / example of; education / earthquake drills;	**2**

Question	Answer	Marks
2(a)(i)	18 <u>million</u>;	**1**
2(a)(ii)	14;	**1**
2(a)(iii)	(18 000 000 / 100 × 14 =) 2 520 000;	**1**
2(b)	*any three from:* heavy / exhausting, to carry; takes time / time-consuming; water source may be far away; may need several journeys to fetch enough for a family; water-related disease / bacteria / example of; family members cannot work when affected by a water-related disease; (water contaminated by) industrial chemicals / fertiliser run-off; which may be toxic / example of; wells / water sources, dry up in hot weather / summer;	**3**
2(c)	*any four from:* treated to remove impurities; chlorination / filtration / desalination; piped to homes; wealth / able to afford the service; technology available; (efficient) sewage system prevents water contamination; government / authorities, monitor, water quality / safe levels;	**4**

Question	Answer	Marks
3(a)(i)	temperature;	**1**
3(a)(ii)	23(°C);	**1**
3(a)(iii)	32;	**1**
3(b)(i)	*any three from:* hot; dry / low rainfall; less than 250 mm p.a.; (hot and dry) all year; large diurnal range of temperature / hot days and cold nights; cloudless / high sunshine levels; low humidity; wind gusts / sandstorms;	**3**
3(b)(ii)	30° (circled);	**1**
3(c)(i)	*any two from:* vegetation would, wilt / die / be less healthy / less dense / stunted / less able to produce, seeds / flowers; increased, evapotranspiration / transpiration; less rainfall for plants to use; species may adapt over time / change, in mix of vegetation / biodiversity;	**2**
3(c)(ii)	remedial action by a few countries only would have very little effect / action of many countries would be required to slow the problem; OR pollutants from one country impact others / country borders do not exist in the atmosphere / whole planet affected;	**1**

Question	Answer	Marks
4(a)	*stage in model of demographic transition:* **A** – 3; **B** – 5; **C** – 2;	**3**
4(b)	*any four from:* economic development, slows / stops; rise in imported goods / impact on balance of payments; pressure on infrastructure / example of; food shortage / hunger / malnutrition / water shortage; insufficient funds to provide services needed; insufficient, schools / teachers; insufficient, hospitals / medical services / doctors / nurses; housing shortage; increased levels of, poverty / unemployment;	**4**
4(c)	*any three from:* educating women as well as men; later marriages; delay having children until later / able to have careers; understand the benefits of having fewer children / cost of children / understand the impacts of a large population; education on family planning; education on contraception;	**3**

Question	Answer	Marks
5(a)(i)	*any two from:* in between the tropics; mainly in southern hemisphere / most between Equator and Tropic of Capricorn; specific continents mentioned; specific countries outside tropics mentioned / location identified;	**2**
5(a)(ii)	*Coffea canephora* (circled);	**1**
5(a)(iii)	*any one from:* inability to take up (certain) nutrients; do not grow well in soils that are too acid or alkaline;	**1**
5(a)(iv)	*any two from:* (the type of) fertiliser added; pH of (parent) rock; pH of rainwater / acid rain; (local) pollution;	**2**
5(a)(v)	*species: (Coffea) canephora;* *explanation (any two from):* has good cold resistance; will cope with the pH of the soil; has good disease resistance; good / high, yield;	**3**
5(b)(i)	commercial croplands (circled);	**1**
5(b)(ii)	(a farming system with) high level of inputs (fertilisers / pesticides / water / machinery); to produce optimal production / resulting in high yields from an area of land;	**2**

Question	Answer	Marks
5(b)(iii)	*advantage:* work done faster / more efficiently / less labour needed / able to cultivate previously difficult sites; *disadvantage:* loss of other vegetation (windbreaks) to allow machines to operate / damage to soil structure / increased risk of soil erosion / loss of jobs locally / damage to plants / expensive to purchase / expensive to maintain or fix;	**2**
5(c)(i)	*any three from:* use of pest resistant varieties; use of natural predators; close management of crops to remove pests as they appear (by hand) / cultural techniques; crop rotation; physical barrier / example of; traps;	**3**
5(c)(ii)	1.32;	**1**
5(c)(iii)	*any four from:* risk of residues (left on the beans) harmful to humans; spray drift may impact local, ecosystem / wildlife; run-off into soil water affecting ecosystem; bioaccumulation in the food chain; reduction of beneficial insects (pollinators) in the crop; resistant species will impact on the natural balance; ethical preference;	**4**
5(c)(iv)	*any two from:* allows the growing of a rust disease resistant crop; identification of rust resistant genes in another species; insertion of gene into (high yielding) coffee species / change to DNA; faster than 'traditional' plant breeding;	**2**

Question	Answer	Marks
5(c)(v)	*any three from:* unknown long term effect on, ecosystem / humans; concerns of genes getting into the, natural / wild population; expensive technology / unaffordable for many farmers; religious / ethical, objection;	**3**
5(d)(i)	*any four from:* removal of tress reduces, interception / infiltration; increasing soil erosion; roots bind soil together; roots reduce risk of erosion; loss of habitats / loss of food sources / disruption of food chains / loss of shade for other vegetation; reduction in biodiversity; impact on, water table / waterlogging / water cycle; loss of soil, nutrients / fertility; desertification;	**4**
5(d)(ii)	*any three from:* legislation / preservation orders / fines; education / help people to understand their importance; subsidy payments to farmers to retain the trees; reserves to sustain the habitat; provide other areas to grow crop; provide alternative employment;	**3**

Question	Answer	Marks
5(e)	*Level of response marked question:* Level 3 [5–6 marks] Answers will be detailed and well-rounded. The reasons given will be developed. The best answers will include appropriate examples. The response will show a balance of both viewpoints and may also reflect a global viewpoint. Level 2 [3–4 marks] Answers will include ideas and / or reasons that have been developed or explained. They may include a specific example but lack some essential details within the answer. Answer may lack balance within the response and may focus on one viewpoint. Level 1 [1–2 marks] Answers may include a reason(s) presented in a list with little development. There may be repetition or comments not relevant to the question asked. No response or no creditable response [0]. *Level of response marking indicative content:* Sustainable agricultural techniques include, plant breeding, integrated pest control, mixed cropping, gene banks, new crop strains, trickle drip irrigation and organic alternatives to inorganic fertilisers. Candidates might include examples or descriptions of techniques that could be used. Many candidates will consider the cost of processes and the fact that farmers need income to invest in new technologies, so it only works if it is profitable. Farmers may not want to change current practices and may lack skills or need to learn new techniques. External / governmental pressure may be needed to change behaviours. Candidates could also include other details of reasons techniques would not be used. The best answers will include the global viewpoint, e.g. the increase in world population and the link to increase in demand for food and increasing production, and considerations such as, the difference between 'small farm' and 'large farm' situations needing different solutions.	**6**

Question	Answer	Marks
6(a)(i)	100% (circled);	**1**
6(a)(ii)	*material:* construction materials; *reasons (any two from):* increase in population; increase in urbanisation; improvements to infrastructure (e.g. roads); more buildings constructed; increased industry; (construction materials) harder to recycle;	**3**
6(a)(iii)	*any three from:* changes to technologies; improvements to recycling; scarcity of resources; use of other materials / demand for materials; cost (qualified); improved mining technology;	**3**
6(b)(i)	*any three from:* recycling; low waste technology / developing cleaner processes and products (at point of production); example of; processing waste and industrial products; example of; more efficient use of raw materials; example of; use of other technologies that do not produce waste; example of (e.g. switch to renewable resources); legislation to ban (certain) waste producing processes;	**3**
6(b)(ii)	28(%);	**1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>6(b)(iii)</td><td>
<table>
<tr><th rowspan="2">rank</th><th colspan="2">type of waste</th></tr>
<tr><th>low income country</th><th>high income country</th></tr>
<tr><td>1st</td><td>organic / biomass</td><td>paper and cardboard</td></tr>
<tr><td>2nd</td><td>paper and cardboard</td><td>organic / biomass</td></tr>
<tr><td>3rd</td><td>other</td><td>plastics</td></tr>
</table>
;;</td><td>2</td></tr>
<tr><td>6(b)(iv)</td><td>organic waste: decreases as the income of a country increases;

paper and cardboard waste: increases with increase in a country’s income;</td><td>2</td></tr>
<tr><td>6(b)(v)</td><td>organic waste:
low income countries have a large proportion of households involved in agriculture;

paper and cardboard waste:
high income countries have an increase in packaging / purchasing more from supermarkets, which means an increase in paper and cardboard / high use of paper in the home;</td><td>2</td></tr>
<tr><td>6(b)(vi)</td><td>any three from:
more employment;
health insurance for workers;
free meals (for workers);
child care;
cheap supply of compost as a fertiliser;
reduced need for irrigation;</td><td>3</td></tr>
<tr><td>6(b)(vii)</td><td>(addition of a mulch) reduces evaporation from the soil / suppresses weeds;</td><td>1</td></tr>
<tr><td>6(b)(viii)</td><td>methane / biogas / biofuel;</td><td>1</td></tr>
</table>

Question	Answer	Marks
6(c)(i)	*any four from:* better insulation in walls; double glazing of windows; improved ventilation; solar panels on roof (to supply electricity); rainwater harvesting; more glazing to reduce need for artificial light; power from waste; use of technologies to improve efficiency of use (qualified); heat recovery;	**4**
6(c)(ii)	*any three from:* governments might not see it as a priority; lack of suitable resources to implement; abundant supply of energy; a need to build large quantities of housing is the priority; economic situation / lack of capital;	**3**
6(d)(i)	*benefits (any two from):* cost effective source (compared to alternatives); does not damage own environment; lack of own resources; (good for) international relations; *risks (any two from):* continuity of supply; risk of price rises; dependant on other countries (political issues); risk of, explosion / accident;	**4**
6(d)(ii)	*any one from:* cleaner burning / less pollutants; less waste products to dispose of; less bulky to transport; can be produced from organic matter;	**1**

Question	Answer	Marks
6(e)	*Level of response marked question:* Level 3 [5–6 marks] A full response to the question, giving a balanced approach, utilising a range of different information to support their views. Answers will include different types of waste. Level 2 [3–4 marks] Shows understanding of the issues but lacking in detail or accuracy and often one-sided in approach. Level 1 [1–2 marks] Basic understanding shown. Descriptions and comments are very limited and superficial. Response may be in the form of a list and lacking expansion of the points made. No response or no creditable response [0]. *Level of response marking indicative content:* Burying waste in the ground may be supported by candidates because it is a cheap way of disposal, the waste can be transported in bulk and you don't need to sort the waste. Some candidates might consider other benefits, for example, it provides opportunities for collection of valuable items by scavengers (which might be lost if burnt), it's a way of restoring old mines, it reduces the dumping of waste into water courses, and there are income opportunities for taking waste for landfill. Some candidates might also consider the opportunities for methane production and collection and that the land may be used for other purposes afterwards, such as agriculture. Candidates might not agree with the statement because large areas of land are impacted, rubbish can blow into surrounding areas there is a risk of pollution to ground water supplies and an impact on local organisms. Some may consider the smell and gas produced. Others may consider the fact that there is a shortage of landfill sites and that some materials do not decompose or will take many years to do so. The best candidates will consider alternatives, for example burning of the waste could be used to generate electricity and as much as possible should be recycled.	**6**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/11**

Paper 1 **October/November 2018**

MARK SCHEME

Maximum Mark: 120

Published

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Cambridge Assessment International Education

Generic Marking Principles

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GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	all three segments plotted correctly (living 3%, dead 9%, decomposing 88%) ;; (*if plotting incorrect, allow one mark for one correctly plotted segment*) correct use of key;	**3**
1(a)(ii)	earthworms / beetles / woodlouse / AVP;	**1**
1(a)(iii)	fungi / bacteria;	**1**
1(a)(iv)	*any two from:* minerals / rock particles / weathered rock; air; water;	**2**
1(b)	*any three from:* when ploughed in it adds organic matter; decomposes to add, nutrients / plant foods; improves soil structure / helps soil particles stick together; adds / retains, moisture; raises pH / reduces acidity; animals enrich soil with manure while eating the grass;	**3**

Question	Answer	Marks
2(a)(i)	it is driven by winds that move in the same direction / wind blows west / wind blows from east / blown by SE Trade winds;	**1**
2(a)(ii)	*any two from:* warmer; more cloud; wetter / higher rainfall; less / no, mist / fog;	**2**

Question	Answer	Marks
2(a)(iii)	*any three from:* upwelling of cold water brings nutrients; cold Peruvian current brings nutrients; waters are rich in nitrates; waters are rich in plankton; plankton provides abundant food supply for fish (anchovies);	**3**
2(a)(iv)	warm water has, little oxygen / few nutrients / little food supply;	**1**
2(b)	*any three from:* need to feed increasing populations; disputes over fishing grounds; fishing stocks used before are depleted; may have mineral wealth; oil / natural gas, in demand for energy; government / military, policy; difficulty of, defining / establishing / policing boundaries;	**3**

Question	Answer	Marks
3(a)(i)	75;	**1**
3(a)(ii)	lower percentage of cities with increasing levels of fine smoke particles in higher income countries / ORA; 28% (higher income countries) and 52% (lower income countries);	**2**
3(a)(iii)	get into the lungs / lung cancer / COPD / respiratory problems / heart disease / strokes / asthma;	**1**
3(b)	*any three from:* exhaust emissions from many vehicles / high traffic levels; especially diesel; frequent traffic jams; gases from power plants; gases / smoke, from industries; dust from construction sites;	**3**

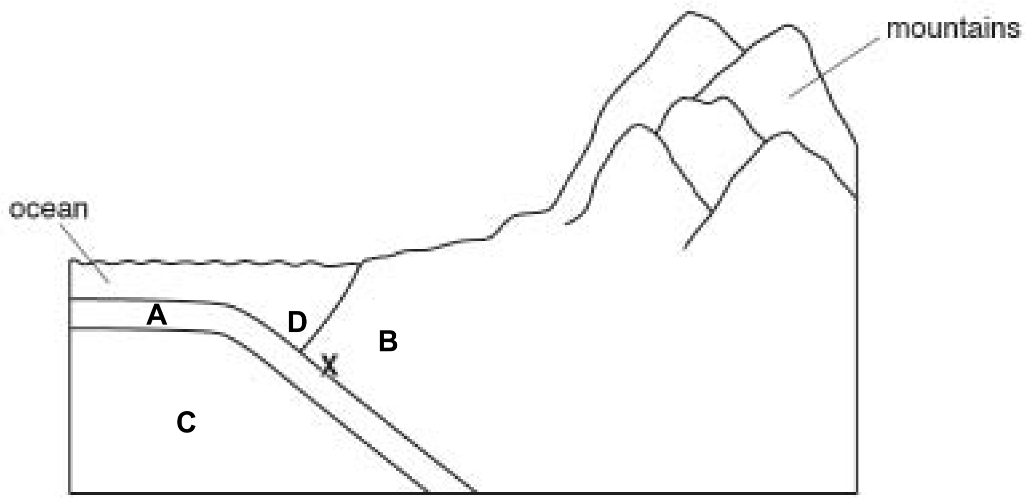
Question	Answer	Marks
3(c)	*any three from:* expense; need to gain revenue; importance of industry to the economy; high use of polluting, fossil fuels / coal / oil / wood; less environmental, awareness / education; little legislation to control atmospheric pollution; many open fires for cooking; burning of vegetation to clear for farming;	**3**

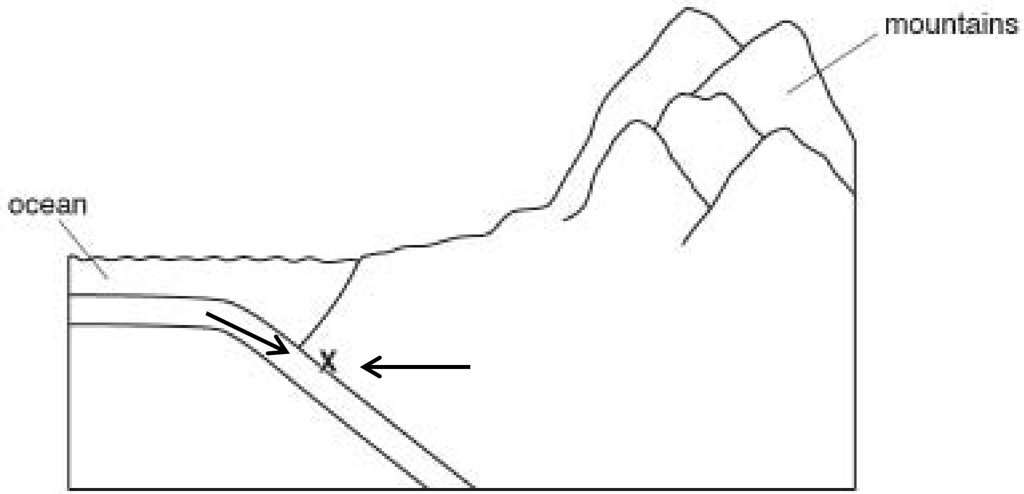
Question	Answer	Marks
4(a)(i)	**C** – *vegetational succession is being stopped*; **A** – *vegetational succession has been happening for a long time*; **B** – *vegetational succession has been happening for a shorter time*; *3 correct [2]* *1–2 correct [1]*	**2**
4(a)(ii)	pastoral;	**1**
4(b)(i)	*any three from:* different species; taller / tallest, that can grow in the climate; denser; fewer species than the stage before; reduced, ground / field layer / herbaceous plants / shrubs;	**3**
4(b)(ii)	new species are brought in (by wind / animals);	**1**

Question	Answer	Marks
4(c)	*any three from:* more varied / increased number of, habitats; animals move in; increased biodiversity; reduced soil erosion; deeper soil; (usually) more fertile soil; more moisture in the soil; more shade / cooler at lower levels; more moisture in the air;	**3**
5(a)(i)	*both benefits of tourism and negative impacts of tourism must be covered for maximum credit:* *max three from, benefits:* jobs for local people; income for local economy / foreign exchange / foreign currency; preserves rural services; increased demand for local produce or crafts; encourages conservation, of the habitats / awareness; improvement to infrastructure; *max three from, negative impacts:* litter / erosion (qualified) / fires / vandalism; more people / increased noise levels, disruption to wildlife / scares animals; traffic congestion / air pollution; local goods become expensive; shops stock products for tourists; housing prices increase; jobs are seasonal; increased sewage (to remove) / water pollution; deforestation / clearing land – qualified;	**4**

Question	Answer	Marks
5(a)(ii)	*any two from:* idea of education of tourists – qualified; encourage visitors to leave cars outside of parks; no 'idling' of car engines within parks; encourage visitors to buy local; create walkways / maintain paths (to prevent erosion); planning polices to control buildings of hotels etc.; restrict access to parks / quotas; charge visitors a fee to enter the park; ensure, facilities / attractions are spread out (to reduce concentration of tourists in an area); laws / fines – qualified; ecotourism; provide bins;	**2**
5(b)	*any four from:* (family) has always lived in the area; no (large) earthquakes in living memory / perceived threat is low; cheaper land / expensive to live elsewhere; work in the area / tourism; (confidence in country's) strategies in place for managing the impacts of earthquakes; earthquake resistant buildings; minerals / natural resources, often near plate boundaries; fertile soil; source of geothermal energy;	**4**
5(c)(i)	data correctly plotted;; *4 correct [2]* *2–3 correct [1]*	**2**
5(c)(ii)	29;	**1**
5(c)(iii)	167;	**1**

Question	Answer	Marks
5(c)(iv)	*any four from:* temperatures increase then decrease; hottest in, June, July and August / July / increase to July then decrease; maximum 23 °C / minimum –6 °C; precipitation fluctuates; wettest in June / May–September / summer; highest rainfall 60 mm / lowest 20 mm; correlation between high temperature and high precipitation; (some) precipitation falls as snow / freezing in winter;	**4**
5(c)(v)	**A** = rain / precipitation AND (rain) gauge; **B** = (wind) speed / force AND anemometer; **C** = sunshine; **D** = (wind) vane;	**4**
5(c)(vi)	*any four from:* loss of life / injury; crops / livestock, destroyed; lack of food / malnutrition / starvation; water supplies contaminated; may lead to, cholera / typhoid / water-related disease / water-borne disease; stagnant water for breeding mosquitoes; increasing risk of malaria; houses and contents, damaged / lost; homelessness / evacuation; infrastructure / transport / communication, disrupted; cost to government; businesses damaged; people out of work; loss of income; food costs increase; time / effort / cost, of restoration when floods recede;	**4**

Question	Answer	Marks
5(d)	*any two from:* it has melted; increased (emissions of), greenhouse gases / CO_2 / methane; climate change / global warming / (Earth's) temperature has increased;	**2**
5(e)(i)	*oceanic crust correctly labelled on diagram as* ***A***; *continental crust correctly labelled on diagram as* ***B***; *mantle correctly labelled on diagram as* ***C***; *trench correctly labelled on diagram as* ***D***; e.g.  	**4**
5(e)(ii)	subduction (zone);	**1**

Question	Answer	Marks
5(e)(iii)	*arrows show the direction of movement is towards each other*; e.g.  	**1**

Question	Answer	Marks
5(f)	*Level of response marked question:* Level 3 [5–6 marks] Answers will be detailed and well-rounded. The reasons given will be developed. The best answers may include appropriate examples and will present reasons to support both viewpoints. Level 2 [3–4 marks] Answers will include ideas and / or reasons that have been developed or explained. They may include a specific example but lack some essential details or breadth within the answer. Responses may focus on one viewpoint. Level 1 [1–2 marks] Answers may include a viewpoint presented in a list with little development. Alternatively the response may focus on one or two reasons. There may be repetition or comments not relevant to the question asked. No response or no creditable response [0]. *Level of response marking indicative content:* Many learners will tend to agree with the statement citing issues such as: It is a renewable energy resource and there are no ongoing fuel costs. Geothermic energy does not produce harmful polluting gases. Responses will typically identify that it is a constant source of energy or a limitless supply. More complex responses might include detail of the small land footprint compared with other power sources, and that the process is simple and reliable. Viewpoints that disagree with the statement will typically cite that: Most parts of the world do not have suitable areas where geothermal energy can be exploited or it is location specific. More detailed responses might include a reference to the high installation costs, the difficulty in transporting hot water from the source without heat loss – limiting the distance it may supply and the costs required by businesses / homes to retro-fit. Other responses might include the low public awareness. Longer term effects might include the risk that unstable ground may induce seismic activity when water is withdrawn and the unknown impact on agricultural irrigation. Other concerns could include the difficulty in drilling into the heated rocks and the difficulty in working with water at such high temperatures.	**6**

Question	Answer	Marks
6(a)(i)	bitumen;	**1**
6(a)(ii)	*any two from:* deposits deep underground; (oil sand bitumen mixture) only contains 10% bitumen; (oil sand bitumen) has to be separated from the sand before it can be used; expensive process compared to the return;	**2**
6(b)	*any three from:* formed over millions of years; (from remains of dead marine) organisms; (remains of dead organisms) fell to bottom of sea covered in, mud / sand / sediment; mud / sand / sediment, buried by more sediment; (changed to oil as) temperature / pressure increased;	**3**
6(c)(i)	*any three from:* loss of habitat / disruption to food chains / animals might move away / extinction / reduction in biodiversity; soil erosion; increased (surface) run-off / flooding / siltation of rivers; desertification; (local) climate change / (local) reduced rainfall;	**3**
6(c)(ii)	recycling / use more parts of the tree, e.g. chipboard or laminate;	**1**

Question	Answer	Marks
6(c)(iii)	*any four from:* *sustainable forestry activities;* agro-forestry; example or description of the above; community forestry; example or description of the above; reforestation; description of the above; sustainable harvesting of hardwoods; example or description of the above; fuel wood planting; example or description of the above; *potential problem of sustainable management of forests;* world population growing fast, so more demand for timber; example of the above; more demand for, cleared land / agricultural land; example of the above;	**4**
6(d)(i)	*any four from:* damage to homes / homes destroyed / evacuated / homeless; loss of life / injury / burns; danger to health from the (chemical or particulates in) smoke / breathing difficulties / eye irritation / asthma; loss of income; power / water, supply disrupted; lack of sanitation;	**4**
6(d)(ii)	living organisms and the physical factors in an area or habitat; reference to, interaction between them;	**2**

Question	Answer	Marks
6(d)(iii)	*any five from:* (soot / particulates / smoke) reduces photosynthesis; increased carbon dioxide emissions; describes an effect of increased CO_2 / climate change or greenhouse effect; (SO_2) acid rain formation; describes an effect of acid rain / damages crops or acidifies lakes; air pollutants / ash / soot deposited, damage trees / water sources; damage to animal lungs; habitat destroyed; loss of trees / roots burnt, causing soil erosion; food chains disrupted / death of organisms / loss of biodiversity;	**5**
6(d)(iv)	*both positive and negative response must be covered for maximum credit:* *max three from, positive response:* only a few people killed / people saved; state of emergency declared / evacuation; lots of support given to fight the fires / numbers of firefighters; *max three from, negative response:* many homes destroyed or damaged; water and power outages; fires burned for a long time; fires covered a large area; some people killed;	**4**
6(e)(i)	173;	**1**
6(e)(ii)	Saudi Arabia / Venezuela;	**1**
6(e)(iii)	(168 / 173 × 100 =) 97 / 97.1;	**1**

Question	Answer	Marks
6(f)(i)	*any two from:* filling in – qualified; landscaping; example of reclamation; waste management;	**2**

Question	Answer	Marks
6(f)(ii)	*Level of response marked question:* Level 3 [5–6 marks] Answers will be detailed and well-rounded. The reasons given will be developed. The best answers may include appropriate examples and will present arguments to support both viewpoints. Level 2 [3–4 marks] Answers will include ideas and / or reasons that have been developed or explained. They may include some reasons but lack some depth or breadth within the answer. Responses may focus on one viewpoint. Level 1 [1–2 marks] Answers may include a viewpoint presented in a list with little development, may focus on one or two reasons. There may be repetition or comments not relevant to the question asked. No response or no creditable response [0]. *Level of response marking indicative content:* Reasons to allow the expansion of the mine could include justification that more jobs will be created which will bring increased income into the area and improvements to the local economy. Responses might also identify the improvements to infrastructure in the area which will be needed to serve the enlarged mine. Some responses may focus on the impact nationally, namely the availability of oil reserves and the improvement in economic growth. Some might also identify the security to energy supplies that might result. Reasons against the mine are likely to include factors such as an increase in noise pollution, atmospheric pollution and the impact on greenhouse gas emissions. More locally, there may be reference made to the impact on local rivers due to run-off, the destruction of habitats as well as the impact of increased traffic. Broader and more detailed responses might also mention the fact that mining / extraction is carbon intensive and the country should be moving towards renewable resources, and oil sands are non-renewable;	**6**

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 **May/June 2018**

MARK SCHEME

Maximum Mark: 120

Published

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Question	Answer	Marks
1(a)(i)	52;	**1**
1(a)(ii)	*any two from:* use in industry increased whereas use for electricity, decreased / remained same; industry large(r) change / electricity small(er) change; industry more fluctuating / electricity more stable; correct use of data to express a comparison;	**2**
1(a)(iii)	*any two from:* concern about global warming; because carbon dioxide is emitted when oil is combusted; increase in the use of alternative fuels / other methods of electricity generation; expense;	**2**
1(a)(iv)	*any one from:* more people own cars; more flights; increase in, shipping / trade;	**1**
1(b)(i)	*any two from:* millions of years ago; from, dead / decomposed, plant / animal / plankton / sea creatures / organic matter; (fell to bottom of sea) and were covered by mud / sediment; increased the temperature and pressure;	**2**
1(b)(ii)	*any two from:* geological survey; use of sonar / seismic tests, to find out about rocks below the ground surface; geologist identifies, rock structures / upfold / anticline / salt dome / fault where oil is likely to be found; geologist identifies cap rock; examination of rocks from core samples (drilled); well is drilled;	**2**

Question	Answer	Marks
2(a)(i)	two lines in correct positions (e.g. at 61 and 71 or at 61 and 90); segments correct to key;	**2**
2(a)(ii)	90(%);	**1**
2(a)(iii)	*any four from:* increased demand (from growing population); increased size of fishing vessels / vessels go further from shore; development of factory ships; more effective fishing methods / new technology used; e.g. sonar to find fish shoals / satellite technology; increased size of nets / drift nets (as large as one km wide); fishers get a good price for the fish so catch as many as possible; illegal nets / fishing illegally / overfishing / ignoring quotas; no enforcement of rules / no rules / fishing during breeding season; lack of education over the consequences of overfishing;	**4**
2(b)(i)	*any two from:* a limit is set for the number of a type of fish that can be caught (within a time period); fishing then has to stop; leaves enough fish for stocks to increase / prevents overfishing;	**2**
2(b)(ii)	*any one from:* some fish caught that are not allowed / bycatch, often die before they are returned to the sea; quotas may be, incorrectly set / too high;	**1**

Question	Answer	Marks
3(a)(i)	*any three from:* gentle slopes / fairly flat / no steep slopes; no, field boundaries / gateways; large, fields / area; no rivers; no (other) obstructions; appropriate named machinery e.g. tractor / combine harvester;	**3**
3(a)(ii)	*any one from:* little fertiliser used; wheat takes, a lot of / same nutrients out of the soil / robber crop / impoverishes the soil / soil loses fertility; many weeds compete with the crop; pest / disease, build up;	**1**
3(a)(iii)	wind break / shelter from the wind / for timber / firewood / animal habitat / shade for house;	**1**
3(a)(iv)	*all three circled for **one** mark:* commercial, cropland, extensive (circled);	**1**
3(a)(v)	*any one from:* market demand; capital input; <u>can afford</u> large land area / the land <u>is cheap</u>; market value / profit (from crop); transport to market;	**1**

Question	Answer	Marks
3(b)	*any three from:* maintain soil fertility; crop rotation / in each field one crop is followed by a different crop the next year; intercropping / plant crops in between rows of other crops; strip farming / plant crops in strips / leave grass strip between crop strips; different crops have different nutrient requirements; some crops / alfalfa / legumes, have nitrogen fixing bacteria in root nodules; some crops / cereals, have high nutrient demands so are followed by plants that return a lot of nutrients to the soil / a grass ley; reduction of soil erosion; by improved soil structure; by, covering / protecting / trapping the soil; less chance of all crops being destroyed by pests and diseases;	**3**

Question	Answer	Marks
4(a)(i)	plot at 24% with correct application of key; plot at 100% with correct application of key;	**2**
4(a)(ii)	39;	**1**
4(b)	*any three from:* land reserved for wildlife takes up land of possible use, to people / for development; wildlife, destroy farmer's crops / eat pasture; wildlife can spread disease to farmers' stock; people with such a view may lack, education / awareness about biodiversity; wish to continue a, (traditional) activity / hunting / food source / income; danger to, humans / livestock;	**3**
4(c)	*any four from:* policies to keep the population of, wildlife / plants / animals to sustainable levels; wildlife reserves / national parks; biosphere reserves; gene banks; create / conserve, habitats of species / reduce pesticide use / reafforestation; legislation to control, hunting / collecting of endangered species / deforestation; educate for, conservation / ecotourism / campaigning; international agreements; establishing charities; captive breeding / zoos / re-introduction;	**4**

Question	Answer	Marks
5(a)(i)	16 (km); –74 (°C);	**2**
5(a)(ii)	*any three from:* temperature increases, to the stratopause / in the stratosphere / to 50 km / –74°C to –19°C; temperature decreases, to the mesopause / in the mesosphere / to 85 or 86 km / –19°C to –95°C; temperature increases, in thermosphere / –95°C to +80°C; (initial) further (small) decrease in temperature, in thermosphere / to 91 km;	**3**
5(a)(iii)	any indication across diagram between 20 and 30 km;	**1**
5(a)(iv)	*any two from:* ozone layer absorbs / protects, from UV radiation / light; which is damaging / harmful, to organisms; which causes, <u>skin</u> cancer / mutations / cataracts;	**2**
5(a)(v)	*any four from:* CFCs / halons (released into atmosphere); from, spray cans / aerosols / fridges / air conditioning / fire extinguishers; travel to / winds carry CFCs, to Antarctica; where they accumulate; reacts / act as catalysts destroying ozone; reaction of O_3 to O_2; exist in atmosphere for a long time;	**4**
5(b)(i)	sun heats Earth's surface, <u>which heats air above it</u>;	**1**
5(b)(ii)	*any two from:* normally temperatures decrease with height; in temperature inversion, temperature increases with height; to create a warm layer at height / warm layer trapped between cold layers;	**2**

Question	Answer	Marks
5(b)(iii)	*any three from:* pollutants from city released into atmosphere; inversion / warm layer, stops air rising; so pollutants, trapped / can't escape / reflected back; while more pollutants, are added over time / accumulate;	**3**
5(c)(i)	bar drawn to correct length (21.0%);	**1**
5(c)(ii)	15.6(%);	**1**
5(c)(iii)	*both electricity generation and agriculture must be covered for maximum credit:* carbon dioxide; methane; is a / are, greenhouse gas(es); *electricity generation:* <u>burn</u>, carbon-based fuels / fossil fuels / coal / oil / gas; *agriculture:* rice growing / decaying vegetation, gives off methane; cattle / other livestock, give off methane; tractors / trucks for transport of farm produce, burn petrol / diesel / oil; burning of, forest / crop residues;	**5**
5(c)(iv)	*any three from:* increased use of, renewables / alternative sources; solar / wind / HEP / geothermal / other example; increased use of nuclear power; improving energy efficiency / example of; reducing energy wastage; legislation; education;	**3**
5(d)(i)	(pH) 5.0–5.6;	**1**

Question	Answer	Marks
5(d)(ii)	*any three from:* in 1990 most of the country had rainfall of pH greater than 7.0; by 2005, general decrease (in pH) / <u>rain is more acidic</u>; larger area affected by acid rain; SE / NE / NW pH has fallen to less than 5.0; largest area less than 5.0 on SE; no change in, West / North;	**3**
5(d)(iii)	*any two from:* increase in industry; increase in vehicles; increase in electricity generation; increase in release of SO_2; increase in release of nitrogen oxides;	**2**

Question	Answer	Marks
5(e)	*Level of response marked question:* Level 3 [5–6 marks] The response will reach a conclusion that will agree with the statement. The answer will provide details of actions that may be taken internationally, supported with relevant examples or demonstrate an understanding of the difficulties in developing international implementation. Level 2 [3–4 marks] The response will agree with the statement and describe how atmospheric pollution is an international problem, identify examples of air pollution sources or impacts, but lack additional detail. Level 1 [1–2 marks] Basic descriptive points with little or no reasoning. May just be a list of for and / or against. Answers at this level may include discussion of some of the causes of air pollution and / or may suggest basic information on resolving the problems. No response or no creditable response [0]. *Level of response marking indicative content:* Candidates will most likely discuss sources of atmospheric pollution. Better answers will look at how atmospheric pollution spreads from source areas through winds and so reach a conclusion that it is necessary to have international cooperation to overcome the problems. Examples of the problems are likely to include global warming, CFCs and the ozone layer, and acid rain. The best answers will include ideas on actions that can be taken at an international level and include examples.	**6**

Question	Answer	Marks
6(a)(i)	PCBs **and** flame retardants;	**1**
6(a)(ii)	accumulation of toxins within the body of an animal; when predator in the food chain feeds on prey the toxin is amplified;	**2**
6(a)(iii)	*any two from:* from, factories / domestic / agriculture; washed into oceans via, rivers / groundwater / drains; from airborne toxins washed out of atmosphere by rain; from, ships / oil rigs;	**2**
6(b)(i)	Peru (current);	**1**
6(b)(ii)	*any three from:* circular motion in, North Atlantic / South Atlantic; circulate clockwise (N); anticlockwise (S); further details such as name <u>and</u> direction of currents; Equatorial currents run East-West / parallel to Equator; warm currents flow towards poles; cold currents flow toward Equator;	**3**
6(b)(iii)	*any two from:* via Somali current; N. Equatorial current; and Agulhas current; from ships transporting oil from Middle East;	**2**
6(b)(iv)	*any two from:* lowers temperature; low rainfall; coastal fog / mist;	**2**
6(c)(i)	10 (m);	**1**

Question	Answer	Marks
6(c)(ii)	*any three from:* water flows (through turbines / floodgates), into lagoon / from sea (after low tide); water flows (through turbines / floodgates), from lagoon / into sea (after high tide); turns the turbines; which turn generators;	**3**
6(c)(iii)	tides / water always present; can be used over and over again / tides always move;	**2**
6(c)(iv)	155 000 (homes);	**1**
6(c)(v)	*any three from:* rivers carry silt; which will fill the lagoon; and clog turbines; reducing power generated; require lots of dredging; shorten life of scheme; would stop boats moving, up / down stream; impact on fish, movement / breeding; flooding of lower parts of river valleys;	**3**
6(c)(vi)	*any four from:* fish may get killed in turbines; species trapped in lagoon; leading to inbreeding; damage to habitats during construction; loss of habitats; changes to currents / tides, may damage habitats; possible erosion of beaches; food chain disrupted; specific pollution during construction – qualified;	**4**
6(d)(i)	both plotted points and line correct;	**1**

Question	Answer	Marks
6(d)(ii)	*any three from:* increased 1960 to 1968; from 1.1 to 1.9 (million tonnes); decreased until 1977; to 0.45 (million tonnes); up and down (a little) until 1990; decreased to 0.03 (million tonnes) by 1994 / 5;	**3**
6(d)(iii)	*any one from:* overfishing; fish stocks so depleted they cannot recover;	**1**
6(d)(iv)	the interaction of; living organisms (biotic) in conjunction with the physical environment (abiotic) components;	**2**

Question	Answer	Marks
6(e)	*Level of response marked question:* Level 3 [5–6 marks] A detailed and balanced response covering plastics, raw sewage and heavy metals, resulting in a valid conclusion. Level 2 [3–4 marks] May discuss at least two of the pollutants, with some detail. The response might focus on plastics or another pollutant providing justification and detail. Level 1 [1–2 marks] Concentrates on one of the pollutants (not necessarily plastics) giving some detail of the dangers posed. Alternatively answers may include some descriptive points from a range of pollutants with little detail. May just be a list of risks. No response or no creditable response [0]. *Level of response marking indicative content:* The better answers will compare the impact of plastics, raw sewage and heavy metals. Fully comprehensive answers are not expected. Many candidates will agree with the statement and then try to justify this. Credit damage to marine ecosystems. There is no correct answer, but assess on the quality of the response. The following give examples of damage caused. *Plastics:* swallowed by many creatures – build up can block digestive system killing the creature. Organisms can get entangled. Microplastics may clog gills. Ingestion by filter feeders and scavengers damages gut and plastics accumulate in stomachs of predators reducing food intake. *Raw sewage:* can cause eutrophication. Shellfish and filter feeders consuming poisons – bioaccumulation up the food chain. *Heavy metals:* these are poisonous above low levels and while some creatures seem to be unaffected, predators are more likely to suffer effects through bioaccumulation – possibly linked to marine mammals decreasing in number, and beaching themselves.	**6**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/11**

Paper 1 **May/June 2018**

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Cambridge Assessment International Education

Generic Marking Principles

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GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	open-pit / opencast / open-cut / strip / outcropping / quarrying;	**1**
1(a)(ii)	*any three from:* uneven ground; lake / depression; waste heaps; of loose material / overburden; steep slopes; bare ground; removal of vegetation; loss of habitats / new habitat (lake); water pollution;	**3**
1(b)	*any three from:* (possible) use of lake for, recreation / swimming / boating / fishing / water supply for industry; lack of population in the area to complain about the eyesore; vegetation too poor for grazing; land of, no / little economic value; expense of, restoration / landscaping; climate too cold for agriculture; too little time has passed since mining ceased; mining company bankrupt / cannot afford the costs;	**3**
1(c)	*any three from:* recycle; example or detail of recycling; use more efficiently; example of more efficient use; substitute other materials; example of substitute material; use a renewable energy source; educate about conservation;	**3**

Question	Answer	Marks
2(a)(i)	North America is mainly without malaria whereas most of South America has it / less area with malaria in North America / more area in South America with malaria;	**1**
2(a)(ii)	Chile / Uruguay; Venezuela;	**2**
2(b)	*any three from:* differences in the ability to afford to spend on, remedial measures / drugs / insecticides / nets; differences in availability of the remedial measures / example of; differences in organisational ability; differences in education about how to solve the problem; differences in the availability of, health care / hospitals / doctors / nurses; differences in the, amount of suitable land area / area with a suitable climate for mosquitoes to breed / measures to eliminate breeding areas; differences in aid received from foreign countries;	**3**
2(c)	*any four from:* all people need water; deaths; long lasting illness; debilitating illness / causes weakness; reduces ability to work; reduces (personal) incomes; reduces economy of, region / country; reduces, cost / need of medical care;	**4**

Question	Answer	Marks
3(a)(i)	plot of days at 140 and correct use of key; plot of yield at 4.5 and correct use of key;	**2**
3(a)(ii)	to develop a quick(er) growing variety / variety with a high(er) yield;	**1**
3(b)	*any three from:* the poor(est) did not gain; could not afford to buy what would help increase productivity; one example, e.g. fertiliser / irrigation / hybrid-seed / machinery; to compete had to borrow; borrowing left the farmers in debt;	**3**
3(c)(i)	*any two from:* use of, natural pest control / biological pest control / predators / competitors; use of pest-resistant crop varieties; use, nets / screens to protect crops; use chemicals only when essential; change the environmental conditions that favour pests;	**2**
3(c)(ii)	*any two from:* it minimises the use of pesticides; it reduces, risks to human health / poisoning; to reduce risk to the, environment / ecosystem; long-term solution / future generations will not need to cope with pesticide resistance;	**2**

Question	Answer	Marks
4(a)	East Asia and the Pacific's share decreased greatly / from 52% to 17%; South Asia's share increased / from 29% to 35%; Sub-Saharan Africa's share increased greatly / from 14% to 43%;	**3**
4(b)	*any four from:* per capita incomes; inadequacy / shortage of, housing; levels of disease; levels of, nutrition / malnutrition; AVP;	**4**
4(c)	*any three from:* improve trade; fair trade; financial aid; to build, power projects / roads; to build dams / provide irrigation; improve education; training in new / appropriate, farming practices; introduce new strains of crops / crop strains more suitable for the area; aid for medical improvements / to improve the health of the people; organise cooperatives / marketing schemes; improve domestic water supplies;	**3**

Question	Answer	Marks
5(a)	The wettest month is **Dec / December**, which has a mean monthly rainfall of **306** mm. Temperatures are highest in **October** and the annual range of temperature is **7.2** °C. Ndola has a savanna climate and is located in the **southern** hemisphere. ;;;;;	**5**
5(b)(i)	*any two from:* there are trees in the tree savanna, not in the shrub savanna / greater variety of species in the tree savanna; vegetation is, higher / taller in the tree savanna than in the shrub savanna; trees different height / shapes, whereas shrubs same height / shape; lower density of trees in tree savanna / higher density of shrubs in shrub savanna;	**2**
5(b)(ii)	grass;	**1**
5(b)(iii)	*any four from:* low / decrease in, rainfall / ground dries out; increased grazing / overgrazing; so vegetation dies; trees removed / fires destroy trees; no roots to bind soil; less transpiration so rainfall reduced still more; soil open to erosion / winds carry soil away / soil bare; encroachment of dunes from desert;	**4**
5(c)	dispersal **D** competition **E** food web **A** habitat **B** photosynthesis **C** ;;;; *4–5 correct [4]* *3 correct [3]* *2 correct [2]* *1 correct [1]*	**4**

Question	Answer	Marks
5(d)(i)	*any three from:* sand / beach / sunbathing; sea / ocean / coast / swimming / watersports; clear (blue) skies / no or little cloud / dry / no rain / hot / warm; attractive scenery / mountains / exploring / walking; flat land; bay;	**3**
5(d)(ii)	*any four from:* quarrying for, stone / resources to build; clearance of vegetation; loss of habitats; loss of, flora / fauna / biodiversity / animals move away; noise pollution during construction; waste from digging foundations; air pollution from machinery / dust; increase in sediment washed into the sea / soil erosion; effects of increased sediment; sand stripped from, beach / seabed for building; soil compaction; soil pollution; visual pollution / spoils view;	**4**

Question	Answer	Marks
5(d)(iii)	*both shortage of water and waste and sewage to process must be covered for maximum credit:* encouraging tourists to, save water / use water carefully / restrict use / reduce waste; *shortage of water:* recycling of water; tax / raise price of, water; require swimming pools to use sea water / legislation on building or reducing pools; desalination of seawater / storage / aquifers / importing water; *waste and sewage to process:* idea of improvement of sewage treatment works; recycling of, grey water from sewage / wastes (e.g. plastic, cardboard); waste for bioenergy; sewage for manure / waste for compost;	**3**
5(d)(iv)	*any four from:* (large numbers of) planes / ships / ferries; (large numbers of) vehicles, e.g. coaches / taxis / buses; traffic congestion; large amounts of electricity needed; for, air conditioning / other uses; barbecues / other recreational activities, e.g. fireworks; result in, sulfur / nitrogen / carbon, oxides / greenhouse gases, being released; result in particulates being released;	**4**
5(e)	scale of x axis and orientation; scale of y axis; x and y axes correctly labelled; 4 / 5 <u>points</u> plotted correctly;	**4**

Question	Answer	Marks
5(f)	*Level of response marked question:* Level 3 [5–6 marks] Balanced coverage of both environment and economy and reaches a conclusion based on the detailed explanation. Level 2 [3–4 marks] Covers environment far more than economy or vice versa. Reaches a conclusion, but explanation is limited. Level 1 [1–2 marks] Basic descriptive points with little or no reasoning. May only cover one side of the environment / economy debate. No response or no creditable response [0]. *Level of response marking indicative content:* The best answers will look at both parts of the statement, i.e. why tourism is good for a country's economy, but harmful to the planet. Candidates can use ideas from previous part-questions on environmental damage and also include examples, such as building of airports. Answers should also include how tourism benefits the economy through for example, jobs, investment, foreign exchange and tax revenue. Expect most candidates to agree with the statement, though some may argue that sustainable / eco-tourism is good and that money from tourism can pay for protection, as happens in Kenya for example.	**6**
6(a)(i)	*any two from:* trawl dragged along sea-floor; chain / frame, holds trawl open / allows it to sink; fish, enter net / are trapped (at narrow end); fish loaded onto trawler;	**2**
6(a)(ii)	*any two from:* heavy chains dig into the sea-floor / skids cut grooves in the sea-floor / sediment disturbed; (both) destroy, sea-floor habitats / food sources / plants / animals / e.g. corals; parts of the trawl left on sea-floor causing, damage / pollution; lack of time for sea-floor to recover;	**2**
6(a)(iii)	when more fish are caught than can be replaced (by breeding) / fishing that is not sustainable;	**1**

Question	Answer	Marks
6(a)(iv)	*any three from:* large / increasing, human population; large / increasing, demand / need for fish; better, technology / fish detection; example of improved technology, e.g. sonar, freezer trawler; big nets / large number of fishing boats / bigger boats; small mesh size; lack of, regulation / policing;	**3**
6(a)(v)	*any three from:* fish population decreases; loss of food for, predators / consumers / species higher up the food chain; so their numbers decrease / death of predators; predators will feed on other prey species / other prey species decrease; increase in, species lower down the food chain / plankton;	**3**
6(b)(i)	20;	**1**
6(b)(ii)	*any three from:* overall increase / increases from 2005 or 2007 to 2014; no increase 2005–2007; small increase 2007–2009; large increase since 2009; greatest yearly increase between 2011 and 2012; data to illustrate change, including at least two figures or calculation of change;	**3**
6(c)(i)	152;	**1**
6(c)(ii)	56;; *(if answer incorrect, allow one mark for 152–96 [1]);*	**2**
6(c)(iii)	fish stocks are, increasing / improving / recovering;	**1**

Question	Answer	Marks
6(c)(iv)	*any three from:* increase mesh size; ban fishing in certain areas; ban fishing for particular species; allow fishing for only part of the year / avoid fishing in breeding seasons; licences or permits or sensible legislation; authorities should eliminate hidden subsidies, e.g. subsidising the cost of fuel for travelling to fishing grounds;	**3**
6(d)(i)	2006;	**1**
6(d)(ii)	2003–2005;	**1**
6(d)(iii)	7.4;	**1**
6(d)(iv)	*any three from:* some cyclones more powerful; those that reach land cause more damage; some hit, populated areas / high density of buildings; some hit low-lying areas; damage more costly to repair in developed nations; areas less prepared / areas without defences suffer more damage / houses not as strong;	**3**
6(d)(v)	correctly marked at 8,42;	**1**
6(d)(vi)	*any two from:* no relationship; no best fit line / damage does not increase with frequency; data to justify answer;	**2**
6(d)(vii)	*any three from:* warm water / ocean temp 27 °C or greater; energy from evaporation; warm air rises (quickly); causing low pressure; draws in more air at ocean surface; the air spins due to the, Coriolis effect / Earth's rotation; cold air sinks in centre; increase in atmospheric energy due to global warming;	**3**

Question	Answer	Marks
6(e)	*Level of response marked question:* Level 3 [5–6 marks] Detailed explanations of pollutants, reduction and challenges. Needs to reach a conclusion based on the evidence presented. Level 2 [3–4 marks] Some explanation of a source of pollution, how it can be reduced and an idea about the challenges faced. Level 1 [1–2 marks] Limited reference to pollution sources and their reduction. Little or no explanation. No response or no creditable response [0]. *Level of response marking indicative content:* Expect candidates to deal with major pollution sources in oceans, such as oil spills, sewage, plastics and possibly fertilisers / heavy metals. It is not necessary to cover all of these. The main focus should be on whether the sea pollutants can be reduced. Very difficult to remove, though oil spills are 'scooped up' to some degree. So expect details on making oil tankers safer, government initiatives to reduce waste being dumped into the oceans and the role of environmental groups in terms of education and action. Answers may also include difficulties, as international cooperation is needed and currents spread pollution from source areas.	**6**

Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **May/June 2016**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Question	Answer	Marks
1(a)(i)	*any 3 of:* more jobs / reduces unemployment; people earn more money; learning new skills; more exports; more money, qualified; maintain standard of living / reduce poverty; more taxes; to pay for more infrastructure; AVP e.g. an example of a business / diversity of business;	**3**
1(a)(ii)	*any 2 of:* coastal location; most / all producers are near the plant / eq; low transport costs / eq; use of scale on map to provide figures;	**2**
1(a)(iii)	to see how much the processing will affect the environment / eq; any consequence; e.g. budget constraints / change site of processing plant;	**2**
1(a)(iv)	*any 3 of:* *supply of water* supply-seawater / desalinated water; method-piped / pumped to plant; *supply of electricity* supply-solar / tidal / wind / nuclear / coal / oil / gas / geothermal power; method-power station / generator / turbine / power lines;	**3**
1(a)(v)	4400; 25;	**2**

Question	Answer	Marks
1(b)	*any 2 of:* collected in large tanks / eq; processed so they do not harm environment / underground / eq; export / sell wastes to other processing plants / for other uses; put in landfill sites / eq; AVP;	**2**
1(c)	*any 3 of:* more employment; more trade; so more taxes; taxes pay for other infrastructure;	**3**

Question	Answer	Marks
2(a)	*any 3 of:* ref to low total rainfall / eq; no rainfall for four months / eq; quoted data; too hot for plants to survive; so evaporation / transpiration fast; wilting described;	**3**
2(b)(i)	higher temperatures / heat energy evaporates more water; so cooler air cools plants;	**2**
2(b)(ii)	*YES* the yield will remain high; after construction house should last for many years; solar powered so no pollution; AVP e.g. saves space / soils; *NO* energy input required; minerals must be supplied; water supply qualified; high skills needed; AVP e.g. cost qualified;	**3**
2(b)(iii)	*any 2 of:* provides vitamins / ref to vitamin deficiency e.g. scurvy; minerals / named mineral; fibre / roughage / prevents constipation / digestive disorder; AVP e.g. ref to any other health benefit e.g. low in fats;	**2**

Question	Answer	Marks
3(a)(i)	mass and length decrease (with increase in salinity);; diameter no real change / eq;	**3**
3(a)(ii)	*any 1 of:* number of fruits per tree; sugar content of fruits; acidity / pH taste: colour; storage time; AVP e.g. volume;	**1**
3(a)(iii)	*any 3 of:* same soil type; same nutrients; soil pH; soil structure / drainage; one aspect of same climate described; to compare in context / fair comparison/test;	**3**
3(a)(iv)	21.8;	**1**
3(a)(v)	11–19.6 / 8.6; 19.4–24.5 / 5.1;	**2**
3(b)(i)	axes fully labelled; plots;; scale;	**4**
3(b)(ii)	A D; give low yields;	**2**
3(c)(i)	*plan three* has as a control dish / for comparison / eq; with and without salt / eq;	**2**
3(c)(ii)	table drawn one column for days and three columns for salt concentrations; headings with units for salt; enough space for 10 days;	**3**

Question	Answer	Marks
3(c)(iii)	find the mass; measuring volume; dissolving / eq; count the seeds; use of two pieces of equipment; the sequence works;	**6**
3(d)(i)	*any 2 of:* nitrate increases yield; no need for fertiliser; so more profit / less expense; some N left in soil for next crop;	**2**
3(d)(ii)	*any 2 of:* to keep DNA / genes / alleles from wild plants; in case they become extinct; can be reintroduced to wild; used for research / described; protect biodiversity; AVP e.g. to develop drugs;	**2**
3(d)(iii)	*any 2 of:* grow some into adult plants to make more seed; selective / cross breeding by pollination; to produce hybrids; with desirable qualities/described; gene transfer; AVP e.g. other method of transferring genetic material;	**2**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **October/November 2018**

MARK SCHEME

Maximum Mark: 60

Published

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GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	*any two from:* overpopulation in urban areas; increase in disease related to overpopulation; unemployment; low wages; no / few, schools / hospitals; AVP, e.g. housing / infrastructure, cannot keep up with population growth / depopulation of rural areas / increase in a named pollutant;	**2**
1(b)(i)	23.4 / 23 ;; (*if answer incorrect, allow one mark for 539 000 / 23 000 [1]*);	**2**
1(b)(ii)	*any one from, farm workers:* make more money; more job opportunities; *any one from, the government:* government collect taxes; revenue spent on, infrastructure / services; goods for export; (increased) economic growth;	**2**
1(b)(iii)	*any two from:* low wages / seasonal work, in rural areas; poor standard of living in rural areas; converse in city; better (named) services in city;	**2**
1(c)(i)	27;	**1**
1(c)(ii)	185 ;; (*if answer incorrect, allow one mark for 5000 / 27 [1]*);	**2**
1(c)(iii)	sample, is not representative / is biased;	**1**

Question	Answer	Marks
1(c)(iv)	*any two from:* no student bias as (selection is) random; several rows sampled; in different parts of the field; sample is representative (of whole field); AVP, e.g. largest fruits not chosen;	**2**
1(c)(v)	(*correct position indicated on diagram*) row 7, plant 3; (*correct position indicated on diagram*) row 8, plant 5;	**2**
1(c)(vi)	table drawn that can record all the data; correct headings including units; order of data entered;	**3**
1(d)(i)	6.6;	**1**
1(d)(ii)	*any two from:* warm enough for growth every month; and so flowering; and ripening; no frosts to kill plants; AVP related to temperature;	**2**
1(d)(iii)	November to April;	**1**
1(d)(iv)	trickle drip irrigation / AVP;	**1**
1(d)(v)	*any three from:* leaching; of mineral nutrients; increased evaporation; draw salts to the surface; salinisation; AVP, e.g. waterlogged soils lack oxygen for root growth;	**3**

Question	Answer	Marks
1(d)(vi)	*any two from:* widespread crop failure; spread of insect pests and other diseases; reference to, need to use expensive treatments; nutrients used by crop need to be replaced; soil becomes, less fertile / infertile;	**2**
1(e)(i)	*any two from:* asthma; lung disease; breathing problems; burns; eye problems (from smoke); fire risk / described; AVP, e.g. particulates;	**2**
1(e)(ii)	22.76 / 22.8 / 23 (%) ;; (*if answer incorrect, allow one mark for 28 million / 123 million [1]*);	**2**
1(e)(iii)	*any two from:* cost; do not know how to use it; not traditional / do not want to change;	**2**
1(e)(iv)	*any two from:* promotion activities to show advantages and word of mouth; make the new stoves cheaper; lots of people now know how to use it; train people to use it; provide people with materials;	**2**

Question	Answer	Marks
1(e)(v)	*any three from:* reduces deforestation / habitat destruction; reduces air pollution; less carbon dioxide released; maintains species diversity; (reduces impact on) climate change / less global warming; AVP, e.g. less particulates;	**3**

Question	Answer	Marks
2(a)(i)	*any two from:* only, small / single (hand-held) net; limit to how many fish can be caught by one person; lake has provided fish for many years;	**2**
2(a)(ii)	*valid collection method described keeping details the same at each sample point, e.g.* length of fishing / number of fish; size of net; mesh size; time of day; time of year; AVP, e.g. fish at the surface / at a specified depth;	**4**
2(a)(iii)	suitable linear scale; y axis fully labelled; x axis fully labelled; correct plots;	**4**
2(a)(iv)	**F** AND as it has lowest, number of different fish species / species diversity / number of charal fish;	**1**
2(a)(v)	*any five from:* eutrophication; increase in bacteria; bacterial respiration; use up oxygen; algal bloom; block light; death of plants; death of fish; disrupts, food chain / food web;	**5**

Question	Answer	Marks
2(b)	*any four from:* do not allow development too close to the shoreline; to prevent / control, (land / water / air) pollution, e.g. litter; build sewage treatment plants; efficient rubbish collections; restrict visitors to some areas; restrict holiday activities such as, speed boats / tour boats; maintain local fishing activity; restrict vehicles near lake; advertise rules; implement fines / punishment, for abuse of rules; employ rangers to monitor, area / lake;	**4**

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

5014 ENVIRONMENTAL MANAGEMENT

5014/11 Paper 1, maximum raw mark 120

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CAMBRIDGE
International Examinations

Section A

1 (a) (i) name: open cast / open pit / quarrying;
method: remove soil / overburden;
blasting / explosives;
remove rock in layers;
broken rock removed by bulldozers / diggers;
= 2 [3]

(ii) noise of blasting / bulldozers;
heavy lorries in roads;
dust / air pollution;
ugly / visual pollution / scar on the landscape / eyesore;
vibrations from blasting;
loss of land for other purposes;

One mark for basic statement and allow second mark for development. [4]

(iii) employment / income improves standard of living;
shops / services develop (to serve the workers);
roads improved; [1]

(b) plant trees around it;
remove quarry buildings;
fill it in and put the soil back;
so vegetation regrows / grass it over (development);
make it into a lake;
use it as a nature reserve / recreation; [2]

2 **(a)** **(i)** A: water evaporates (into the air) / water changes to water vapour;

B: water vapour condenses / changes to water droplets / precipitation / rainfall;

C: the rain runs off (down slope) back to the sea; [3]

(ii) infiltration; [1]

(iii) leaves intercept rain and it evaporates;
roots extract water from the soil;
space between roots and soil particles allow infiltration;
leaves provide shade so less evaporation occurs; [2]

(iv) extraction for irrigation;
extraction for industrial water;
extraction for domestic water;
water taken into canals for navigation;
dammed rivers / reservoirs reduce flow downstream;
climate change (impact explained); [3]

(b) too far inland / heart of continent / too far from a sea / ocean;
flat / no relief to make the air rise / other causes of uplift and rainfall;
rain shadow regions,
cold so precipitation falls as snow / rivers / lakes freeze;
areas of impermeable rock have no infiltration / percolation;
areas of bare rock have no infiltration;
deforestation reduces infiltration / increases run-off / decreases interception;
not a closed system; [1]

3 **(a)** **(i)** Wilma Megi Yasi;;

Three correct for two marks. Two correct for one mark. [2]

(ii) August to October; [1]

(iii) the sea is at its warmest / hot / 27 °C / above 27 °C; [1]

(b) strong winds uproot trees;
strong winds destroy crops;
strong winds damage / destroy buildings / homes / industries;
strong winds destroy power lines / telephone communications / other;
strong winds / floods result in deaths of people / animals;
strong winds cause storm surges;
heavy rain causes flooding;
flooding pollutes water supplies / disrupts sewage treatment;
flooding disrupts transport;
etc. [3]

(c) differences in advance warning times given;
differences in the strength of the buildings;
different population densities;
differences in the provision of shelters stocked with water / food;
differences in the ability to pay for preventative measures / post-cyclone help;
differences in the amount of provision of emergency relief / aid;
differences in the numbers evacuated;
differences in the relief of the coastline / delta / low / cliffs; [3]

4 (a) (i) two correct lines plotted for two marks;;
correct use of key; [3]

(ii) any sensible land use, such as:
urban / transport / water storage / H.E.P. / mining; [1]

(b) (i) displaced from their homes;
way of life changed / loss of livelihoods;
loss of source of fuel;
lack of forest products for food;
lack of traditional medicines from the forest;
conflicts with the logging companies; [2]

(ii) replant after felling;
selective logging / clear only the individual tree species needed;
select only the mature trees;

Allow development marks. [2]

(c) (i) P / core area to conserve the ecosystem / species / landscape; [1]

(ii) eco-tourists / small groups / people interested in nature / always with a ranger / official guide; [1]

Section B

5 (a) (i) temperatures increase then decrease;
precipitation increases then decreases;
cold winters / below freezing in winter;
warm / hot summers;
low rainfall / approx. 120 mm per year;
max. rainfall is 20 mm per month / max. temp is 23 °C;
less than 5 mm per month in winter;
rainfall lowest in winter / highest in summer;
allow one mark for average temperature or precipitation;
data from graph to a max. of two marks;; [4]

(ii) insufficient rainfall (for crops or pasture);
high evaporation in summer;
cold winters – no pasture for animals; [2]

(b) decreased in size;
split into two or more 'seas';
mainly disappeared from the east / now mainly on the west;
more disappeared from south than north;
most rapid decrease since 1984;
c.25% loss by 1984;
c.60–70% loss by 2007;
allow width measurements to show reduction; [3]

(c) (i) correct scale on *y*-axis;
correct labelling of axes;
all three bars correct;;
one or two correct;

If line graph then max. three marks.
If cumulative bar graph, max. two marks. [4]

(ii) 12 (km^3); [1]

(iii) evaporation;
more water evaporated than flowed into the Aral Sea;;
some lost by infiltration; [2]

(d) (i) fertilisers will lead to growth of algae;
eutrophication;
reduced oxygen in water causes fish to die;
algal blooms block sunlight;
plants die due to lack of photosynthesis;
pesticides will poison fish;
increase in salt will cause fish / plants to die;
fewer fish means fewer birds that feed on fish;
disruption to food chain;
less water surface so less evaporation and so less rainfall;
desertification;
vegetation around lake changes to salt-tolerant vegetation; [5]

(ii) less fish means loss of work / income for fishermen / farmers;
less rain so more difficult to grow crops;
reduction in food supply (either fish or crops);
health problems from poor diet;
health problems from chemicals in the water / inhaled;
birth defects;
desalination plants cannot cope with increased salinity; [3]

(e) (i) trees (deciduous) / shrubs / bushes; [1]

(ii) little vegetation;
animals have grazed it bare;
rain washes soil away / soil erosion (by wind or water);
deep gullies (where soil washed away);
desertification; [2]

(iii) *Basic explanation such as 'terracing' for one mark; and extra detail for second mark.*

terracing, to stop water and soil flowing down the slope;
contour ploughing, to stop water and soil flowing down the slope;
mixed cropping, so soil never left completely bare;
plant tree crops with grass underneath, so soil always covered;
prevent overgrazing, by reducing stock levels; [4]

(iv) *Allow development marks for detailed responses.*

with no soil crop growth reduced;
soil loss containing nutrients leads to infertile land / soil;
leads to farmer going out of business / malnutrition;
once soil lost cannot be replaced / irreversible;
soil may be washed / blown into rivers;
  leading to flooding;
  and decline in water quality; [3]

(f) wide range of possibilities concerned with:

soil conservation
integrated pest control
improved irrigation
organic fertilisers
genetic plant breeding
reducing meat / dairy consumption

however, world population growing fast so more food needed
and demand for meat / dairy increasing
desertification, salination, soil erosion, etc. decreasing agricultural land

Do not expect every aspect to be covered, even for answers in the top level.

Level 3 5–6 marks

Answers the question giving at least two ways sustainable agriculture can meet <u>growing</u> demand and one reason why it cannot or other way round (one that it can and two that it cannot). But can achieve five marks on one view only if done very well.

Level 2 3–4 marks

Some detail of at least two ways agriculture can be sustainable or two ways that it cannot or one each way. With a basic explanation will lift it to the top of the level. Max. three marks if limited detail.

Level 1 1–2 marks

Basic descriptive points with little or no explanation. May just be a list or just one view of sustainable agriculture or problems due to population growth.

No response or no creditable response scores zero marks. [6]

6 **(a)** term letter

term	letter
food chain	W
habitat	Y
population	Z
producer	X

All four correct for three marks.
Three correct for two marks.
Two correct for one mark. [3]

(b) **(i)** *All four correctly plotted for two marks.*
Two or three correctly plotted for one mark.
One or none correctly plotted scores zero marks. [2]

(ii) 11 (°C); [1]

(iii) 18 (mm); [1]

(iv) winter / cool season; [1]

(v) March:
leaves on trees;
tall / green grass;

August:
some trees have lost leaves;
sparse brown / yellow grass;

Max. two marks on either.
One mark for general comment more / less grass / leaves. [3]

(vi) March – wet season; August – dry season;
rainfall / water availability; [1]

(c) **(i)** for water / mud bath; [1]

(ii) (large area) destroyed / vegetation trampled / eaten / debarking trees / overgrazed; [1]

(iii) more vegetation destroyed / less vegetation;
loss of biodiversity;
loss of habitats for other animal species;
less food for other herbivores;
therefore less food for predators;
soil erosion as no interception, no plants to absorb soil moisture;
increased compaction of the soil;
increase in dung so more dung beetles; [4]

(d) WWF is a nature conservancy / environmental organisation / wildlife protection;
elephant population already under threat / endangered / to prevent extinction;
elephants are part of the ecosystem;
so reducing their numbers upsets the balance of the ecosystem; [2]

(e) **(i)** fewer than 700 000; [1]

(ii) scattered;
none in extreme north / North Africa / north of Tropic of Cancer;
few in West Africa;
spread through other regions;
largest areas in: East Africa, northern parts of Southern Africa, around Equator in West Africa;
most between Equator and Tropic of Capricorn;
savanna regions;
etc.;
Credit any three valid points. [3]

(iii) *Increasing because:*
hunting is banned;
creation of national parks / reserves;
trade in ivory banned;
stable government to enforce ban;
more severe penalties for poaching;

Decreasing because:
poaching still goes on;
loss of natural habitat;
farmers killing elephants to stop them eating / destroying crops [3]

(f) **(i)** 2.5 (billion); [1]

(ii) 8.9 (billion); [1]

(iii) Asia and Oceania; [1]

(iv) 12–18 (years); [1]

(v) vaccinations / eradication / reduction in diseases (such as smallpox);
improved medical care;
improved sanitation / water supply;
so people living longer;
fewer die as infants / more infants survive;
so grow up to reproduce;
better nutrition;
role of education explained;
etc. [3]

(vi) wide range of possibilities concerned with:

loss of biodiversity and habitats on land and / or in oceans
soil exhaustion, erosion
increasing pollution leading to global warming, eutrophication, etc.

Do not expect every aspect to be covered, even for answers in the top level.

Level 3 5–6 marks

At least two environmental problems described and explained in detail, or three in less detail, but still needing explanation.

Level 2 3–4 marks

Some detail of at least a couple of environmental problems. With a basic explanation will lift it to the top of the level.

Level 1 1–2 marks

Basic descriptive points with little or no explanation. May just be a list or just one environmental problem.

No response or no creditable response scores zero marks. [6]

[Total: 120]

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

5014 ENVIRONMENTAL MANAGEMENT

5014/12 Paper 1, maximum raw mark 120

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CAMBRIDGE
International Examinations

SECTION A

1 (a) (i) it is dragged along the sea bed;
it is very heavy;
the skids scrape along the bed / creates ridges / OWTTE;
heavy chains rip into the sea bed;
destroys seabed habitats; [2]

(ii) the net is very large;
other creatures also live on / near / just above / just below the sea bed; [1]

(b) *Accept any sensible suggestions, such as:*

(depletes fish population so) reproduction does not replace the lost fish;
loss of food for organisms higher up the food chain;
species becoming endangered / extinct;
biodiversity effects explained;

unemployment / loss of income;
shops in fishing ports do less trade / close;
poverty (area or individual);
shortage of protein / food in the diet;
catch is insufficient to feed the (growing) population;
cost of fish increases;
less fish for future generations;
international conflicts;

Max. three for either group. [4]

(c) quotas exceeded;
use of net sizes smaller than allowed;
fishing in exclusion zones;
fish in by-catch die;
no laws out of territorial waters;
difficult to make / keep to international agreements;
difficult to police such large areas / high cost of policing;
short-term profit motives;
corruption; [3]

2 (a) (i) it increases (with increased depth) / decreases nearer the surface; [1]

(ii) water heated by contact with hot rocks;
water changed to steam;
steam turns the / turbines / generators; [2]

(iii) geothermal; [1]

(iv) renewable / sustainable;
constant generation unlike solar / wind power;
clean / little / no pollution; [2]

(v) igneous / hot rocks not near the surface;
lack the finance / expensive to construct;
lack the technology; [1]

(iv) earthquake; [1]

(b) magma / lava / molten rock;
cools / cooling / solidifies; [2]

3 (a) (i) increase to 15 / 16 km;
decrease above 15 / 16 km; [2]

(ii) 13 / 14 / 15 to 19 / 20 / 21; [1]

(b) use of CFCs / halons / sulfur dioxide (much smaller impact than chlorine);
in air conditioning / refrigerators / aerosols / hairspray / insect sprays / detergent sprays etc.;
compounds containing chlorine / chlorine gas released into the atmosphere;
moves up to the ozone;
chlorine atoms catalyse / cause break down of ozone (molecules); [3]

(c) (i) pollutants are spread round the Earth by its rotation;
carried by wind / owtte; [1]

(ii) the gases that destroy are stable / remain for a long time;
old equipment / appliances / example of are still being used;
such a large problem will take a long time to repair itself; [1]

(iii) skin cancer;
cataracts; [2]

4 **(a)** **(i)** completion of bar at 60 000; [1]

(ii) any from 45–49 to 90–94; [1]

(iii) 225 000 to 235 000; [1]

(iv) it reduced; [1]

(v) *Accept current problems of population structure such as:*

small working population to support large under 20 group;

unemployed young workers;
pressure on maternity services / lack of midwives;
pressure on high schools / polytechnics / universities (any aspect, e.g. shortage of buildings / places / lecturers);
unemployment of primary teachers / unused primary school buildings;

future problems, such as:
shortage of workers in future;
unemployed teachers / lecturers / unused higher education buildings in future;
large proportion of population of reproductive age so future population will grow quickly; [2]

(b) *Accept any sensible suggestion such as:*

birth control / contraception;
education / campaigns about family planning;
education about benefits of reducing family size;
more education for women;
improve the standing in society of women;
provide more jobs for women;
encourage later marriage;
laws against polygamy;
reduce the desire for more children by better health care / food supplies;
any special policy, e.g. China's one child policy / sterilisation;
paying allowance for the first one / two children only;
educating against traditional / cultural beliefs for large families;
reduce immigration; [4]

SECTION B

5 (a) (i) up then down;
very low Jan to May;
starts increasing (rapidly) in June;
reaches peak in August;
then declines (rapidly);
max. two marks for quoting data from the graph: peak ca.36 000, lowest <1000, below 2500 Jan to May etc.;; [4]

(ii) discharge increase with rainfall / positive relationship;
peak a month later than peak rainfall;
because water takes time to reach / flow down the river; [2]

(iii) end of / mid July; to start of / mid October;
August and September; [2]

(iv) *Allow one mark per point plus one mark for development where appropriate.*

loss of life;
crops / livestock destroyed;
lack of food / malnutrition / starvation;
water supplies contaminated;
may lead to cholera / typhoid;
stagnant water for breeding mosquitoes;
increasing risk of malaria;
houses and contents damaged / lost;
homelessness;
infrastructure / transport disrupted;
cost to government;
businesses damaged;
people out of work;
loss of income;
food costs increase;
time / effort / cost of restoration when floods recede; [5]

(b) (i) 3% [1]

(ii) $1390 \times \frac{3}{100}$ (million)
41.7 million km^3

*Accept ECF from **(b)(i)**.* [2]

(iii) most water not available for use;
as it is in glaciers and ice caps;
or deep underground;
only small amount in rivers;
water not evenly spread over the Earth / wide variations in rainfall;
rapid population growth;
increasing demand for industry / irrigation / etc.;
rivers / lakes polluted; [4]

(c) (i) 2; [1]

(ii) *Max. one mark for a list of where. Requires some description of clustering, so two clusters described and a list of others would give max. three marks.*

not evenly spread;
mainly close to coasts;
some clusters such as south of N America (Gulf of Mexico); W Europe;
a few on W coast of Africa;
most in N hemisphere;
many / most in Atlantic (includes Gulf of Mexico); [3]

(iii) mainly on shipping routes;
more likely near to where oil produced;
or where it is to be used; [3]

(iv) oil poisons marine life;
reduces ability of gills to absorb oxygen;
and causes them to die;
oil penetrates feathers of sea birds;
so they no longer insulate birds / cannot fly;
can blind birds / fish / sea mammals; so they cannot find their prey;
affects ability to smell; so cannot find prey;
blocks filter feeding apparatus, e.g. corals;
layer on surface blocks sunlight;
so reducing photosynthesis;
and reducing gaseous exchange / oxygenation;
disrupts food chain; [4]

(d) Two marks for accurate pie graph.
One mark if one or two sectors inaccurate.
One mark for correctly completed key. [3]

(e) Indicative content:

Wide variety of ideas to be explored, such as:

Ocean currents and winds cause pollution to spread (required for top level).
Pollution caused by all nations bordering oceans and by landlocked countries via rivers and wind.
Outside territorial waters there is no individual country ownership of or responsibility for the ocean / sea.
Ships often under 'flags of convenience' so little control by that nation.
Often hard to identify polluter so all need to take responsibility.
Many nations need the oceans resources of fish so all need to work together to preserve environment.

Do not expect Level 3 answers to cover all aspects. Mark on quality of response.

Level 3 5–6 marks

Answers the question and provides at least two reasons explained well or three in less detail. Must cover the fact that pollution is spread to all parts of the oceans no matter where it originates.

Level 2 3–4 marks

Some detail of at least two reasons with some explanation. If both done well the answer will reach the top of the level.

Level 1 1–2 marks

Basic descriptive points with little or no explanation. May just be a list or one good point.

No response or no creditable response scores zero marks. [6]

6 **(a)** **(i)** 380–390 ppm [1]

(ii) temperature: 20 °C;
how long ago it occurred: 125 000 to 130 000 years ago; [2]

(iii) similar shapes / when one goes up or down so does the other;
recent increase in CO_2 not matched by such a large temperature increase; [2]

(b) **(i)** Two marks for all four correct.
One mark for two or three correct. [2]

(ii) largely dependent on fossil fuels / example of a fossil fuel;
which contain a large amount of carbon;
which when burnt produces CO_2,

May answer in terms of why renewables / nuclear unable to provide sufficient electricity, or are costly for max. two marks. [3]

(c) **(i)** doubled / 4 billion tonnes / from 4 to 8 billion tonnes; [1]

(ii) all increased;
largest / massive increase in Asian emissions;
small increase in Europe and / or in N America;
others more than doubled;
S America and Oceania and Africa still small emissions but much larger than in 1970;

Max. one mark on data.
Max. one mark if just a list with no comparisons. [4]

(iii) *Must be able to be seen as possible strategies. Allow development marks.*

change to renewables / nuclear for electricity generation;
increased efficiency of engines;
insulation to reduce heat loss and energy demand;
recycling materials uses less energy than primary processing;
public transport policy;
banning of old vehicles;
policy for reducing deforestation;
etc.

Two points well explained can achieve all four marks. [4]

(d) **(i)** fuel derived from prehistoric / ancient plants and animals;
formed by anaerobic decay / heat / pressure;
carbon (hydrocarbon) based; [2]

(ii) two from oil (petroleum), coal, gas; [1]

(e) (i) soil / overburden stripped away;
loosened by explosives;
mineral extracted using diggers / mechanical shovels / etc.;
taken away by trucks; [2]

(ii) visual pollution;
waste heaps;
noise pollution from blasting / machinery;
loss of habitat / wildlife;
atmospheric pollution / dust;
transport of mineral causing air pollution;
pollutants seeping into groundwater / lake / rivers; [4]

(f) (i) does not produce carbon dioxide / contribute to enhanced greenhouse effect;
less transport of fuel needed; [2]

(ii) in favour because:
provides employment;
and social / economic benefits as a result;
provides power supply;

against because:
risk of leaks / accidents;
so releasing radioactive materials / radiation;
radioactive waste problem;
which may cause cancer / illness / death;

Max. three marks on either side of the debate. [4]

(g) Indicative content for:
Long lasting supplies.
Fossil fuels diminishing rapidly.
Renewables such as solar or wind not reliable.
Access to resources not a limiting factor for siting.
Biofuels cannot meet demand.

Indicative content against:
No solution to waste problems.
Risk of leak, accidents, terrorism.
Expensive to build and decommission.

Do not expect Level 3 answers to cover all aspects. Mark on quality of response.

Level 3 5–6 marks

Answers the question and provides at least two reasons explained well or three in less detail. Must look at both sides of the argument.

Level 2 3–4 marks

Some detail of at least two reasons for and/or against. May answer the question but provide only reasons with limited development.

Level 1 1–2 marks

Basic descriptive points with little or no reasoning. May just be a list of for and / or against

No response or no creditable response scores zero marks. [6]

[Total: 120]

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **May/June 2016**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)	10.2; 34.0 / 3.9 / 3.99;	**2**
1(b)(i)	three correct squares shaded;;;	**3**
1(b)(ii)	1115 – 675 / 440;	**1**
1(b)(iii)	correct use of scale to give answer in the range 330–390 m^2;	**1**
1(b)(iv)	922 × answer from **(ii)** so answers in the range 304 260 to 359 580 (g);	**1**
1(b)(v)	the root system of plants; is also biomass;	**2**
1(b)(vi)	repeat (with more squares) / to control variables / to use a standardised procedure;	**1**
1(c)(i)	*any 2 of:* the biomass is increasing; increased by 430 g between year 1 and 3; greater increase between year 1 and 2 that 2 and 3; further comment using figures e.g. grows about 200 g per year; biomass nearly doubles between year one and two / eq; less recovery between year 2 and 3; AVP;	**2**
1(c)(ii)	(plan to) protect damaged / unprotected saltmarsh / eq for more than three years / up to five years for complete recovery / know how long for the saltmarsh to recover / needs to be protected;	**1**
1(d)(i)	*any 3 of:* identify species (with a key/eq); count the number of different species; use of suitable scale e.g. ACFOR; found in each quadrat; process data e.g. find average; find diversity index;	**3**
1(d)(ii)	table drawn; headings; 10 spaces for recording;	**3**

Question	Answer	Marks
1(e)(i)	adds mineral nutrients; named nutrients such as N ,P ,K ; speeds up growth;	**2**
1(e)(ii)	*any 2 of:* eutrophication / described / eq; some species die out/numbers reduced; may cause disease; disrupts food chains; adds toxic substances;	**2**
1(f)(i)	*any 2 of:* produce energy / electricity; for air conditioners; and refrigerators; to make water from desalination plants; AVP e.g. long hours of darkness;	**2**
1(f)(ii)	photosynthesis; (do not accept if respiration also stated)	**1**
1(f)(iii)	prevent damage(in future from) rising sea levels; reduce possible climate change / global warming; conserves fossil fuels for longer; preserves habitats; maintains biodiversity;	**3**

Question	Answer	Marks
2(a)	*any 4 of:* it reproduces quickly / many / 220–300 eggs; quick hatching; larvae damage trees feeding for 90 days; larvae quickly change to adults; flying adults move to new trees; 3 life cycles a year possible / one female can lay 600–900 eggs a year / eq; AVP;	**4**
2(b)(i)	*any 2 of:* concentration of chemical; temperature; food; water; same source of larvae; dosage; same species of weevil;	**2**
2(b)(ii)	*any 2 of:* as a control; to compare / eq; to find out how many would die anyway;	**2**
2(b)(iii)	5, 70, 40;; 2 correct = [1]	**2**
2(b)(iv)	*any 3 of:* all survive in water after 5 days; none survive chemicals after 5 days / eq; more males paralysed by both chemicals; water does not paralyse; females all dead after 5 days but some males alive for both chemicals; more are killed at 5 days than 2 days; use of comparative figures;;	**3**

Question	Answer	Marks
2(b)(v)	*carbosulfan or dimethoate* *any 3 of:* kills both males and females; kills all the females; kills males quickly; kills more females after 2 days; stops reproduction / laying eggs;	**3**
2(c)	both axes labelled;; sensible scale; plots; allow ± ½ square for plots	**4**
2(d)(i)	spray in Jan / Feb before the population increase / to kill larvae / eq;	**1**
2(d)(ii)	masks / gloves / other protective clothing;	**1**
2(d)(iii)	biological control method / described in outline;	**1**
2(e)(i)	*any 4 of:* to maintain traditional food / culture; keep people employed; improve food security; maintain / develop export market; to develop scientific knowledge in UAE; become world leader in producing weevil resistant palms; AVP e.g. government revenue/improve standard of living	**4**

Question	Answer	Marks
2(e)(ii)	*any 3 of:* find DNA / genes / alleles; resistant to weevil attack; reference to cloning / tissue culture produces uniform trees; genetic engineering / GM trees; drought resistant; salt tolerant; produce higher yields; top quality/highest value fruit; reference to healthy food; extra income for the country;	**3**

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **May/June 2018**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)	(Oahu island and Laysan island) 1600 (km); (Hawaii island and Kure island) 2800 (km);	**2**
1(b)(i)	1 481 200;	**1**
1(b)(ii)	67.8(%) ;; *(if answer incorrect, allow one mark for 1 003 700 ÷ 1 481 200 × 100 [1]);*	**2**
1(b)(iii)	*from highest to lowest:* Oahu, Hawaii, Maui, Kauai ;; *4 correct [2]* *2-3 correct [1]* *1 correct [0]*	**2**
1(c)(i)	table drawn; *headings:* (three) days; (number of) fish / (blue) marlin caught;	**3**
1(c)(ii)	(to get results they can) compare / unbiased / valid; (quantitative data so) means / averages, can be calculated;	**2**
1(c)(iii)	*any two from:* not enough females to breed; males cannot find female to mate with; less eggs / young fish, produced; so population, reduced / decreases / die out;	**2**
1(c)(iv)	large fish increase **and** small fish decrease; as less large fish eaten by marlin; more large fish to eat small fish;	**3**

Question	Answer	Marks
1(c)(v)	*any five from:* *control of,* number of boats / people; limited fishing days; quotas; landing checks; licences; enforce laws / fines; closed season; avoid breeding season; temporary fishing bans; no fishing zone; fish finding / attracting, devices not allowed; control of bait, type / size; AVP, e.g. education / making fisherman aware / catch more males;	**5**
1(d)(i)	*any two from:* far away from most diving sites; (long way to go so) not many divers visit site 20; long way / about 40 km, from state capital / Honolulu; the site may only have limited species to see;	**2**
1(d)(ii)	the number of different species;	**1**
1(d)(iii)	fish rot down / used as compost; to release, nutrients / mineral / named mineral; increase, organic matter / humus;	**2**

Question	Answer	Marks
1(d)(iv)	*any three from:* *bad strategy* no control over total number caught; no records kept; could easily cause other problems by lots of spear-fishing; *good strategy* it does give some control of the grouper; it only controls the target species; helps farmers / increases crop yield; allows young fish to, grow / mature / become future breeding stock / maintains population of other fish species; allows biodiversity to recover;	**3**

Question	Answer	Marks
2(a)(i)	37; *from* 53 *to* 101 **and** *from* 10 *to* 85;	**2**
2(a)(ii)	x axis labelled islands; y axis labelled average; linear scale and plots; key;	**4**
2(a)(iii)	more plastic pieces (at Kure island); more mass of plastic (at Kure island);	**2**
2(b)(i)	10.48(%) ;; *(if answer incorrect, allow one mark for (27.4–24.8) = 2.6 [1]);*	**2**

Question	Answer	Marks
2(b)(ii)	*any two from:* use more energy flying; use more energy finding food / harder to find food; so less energy / food reserve for, mating / laying eggs; AVP;	**2**
2(b)(iii)	*any one from:* *reference to* climate change; more energy distributed from equator to poles;	**1**
2(c)(i)	*any two from:* bottle tops made of thicker plastic; a different plastic that breaks down more slowly; more bottles are removed by people than bottle tops;	**2**
2(c)(ii)	systematic;	**1**
2(c)(iii)	*working to show,* (south beach) 400 g (in 1 m^2) / (north beach) 900 g (in 1 m^2); (south beach) 600 (kg); (north beach) 2025 (kg);	**3**
2(c)(iv)	the wind blows from, the north / north east / towards the south / south east (plastic blown onto north beach from sea);	**1**
2(c)(v)	*any one from:* plastic waste cleared from (south) beach / collects on north beach; more people visit north beach;	**1**
2(c)(vi)	landfill; burning;	**2**
2(d)	*any three from:* microplastics eaten; some absorbed into tissues; cannot be, broken down / excreted / (microplastics) remain in food chain; organisms at the top of the food chain keep eating doses of microplastics; more plastic is being disposed of;	**3**

Question	Answer	Marks
2(e)	*any four from:* increase in demand for plastic; difficult to get agreements between countries; difficult to enforce international agreements; difficult to identify the source of some plastics / plastics come from many different sources; different levels of development; poor waste management / lack of money to reduce the problem / too expensive to deal with; generations / decades, of accumulation in the ocean, will not break down / already there; no alternative to plastic / cost of product; difficult to remove;	**4**

Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **May/June 2018**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)	(Oahu island and Laysan island) 1600 (km); (Oahu island and Midway island) 2300 (km);	**2**
1(b)(i)	1 418 300;	**1**
1(b)(ii)	68.8(%) ;; *(if answer incorrect, allow one mark for 976 200 ÷ 1 418 300 × 100 [1]);*	**2**
1(c)(i)	3 (°C);	**1**
1(c)(ii)	(driest) June **and** (wettest) December;	**1**
1(d)	*any three from:* comfortable temperature every month; low rainfall / no wet season; humidity constant; use of figures ;;	**3**
1(e)	*any three from:* more damage to land (from building) / loss of habitat / deforestation; soil erosion; risk of flooding; increased traffic; more vehicle emissions; more, air / water / noise / visual pollution; loss of biodiversity / disruption of, food chains / food web / other specific effect on wildlife; more, sewage / domestic waste ; AVP, e.g. more turbidity / cloudiness / sediment in water / shortage of water / effect of this on wildlife, e.g. less photosynthesis / death of coral / animals scared away;	**3**
1(f)(i)	11–15 **and** 16–19;	**1**

Question	Answer	Marks
1(f)(ii)	closer to the capital; easy to access; so more people visit; *idea of,* more interesting sites;	**2**
1(f)(iii)	*any three from:* broken / damaged coral; water pollution (from boats); e.g. fuel / motor oil; named example of pollution, e.g. more garbage / plastic; damage from boat / boat anchors; more fish killed; AVP, such as noise can scare away wildlife / disturb or disrupt breeding;	**3**
1(g)(i)	*any three from:* no control of the total number of collectors; no control of the number of fish caught / overfishing; no control of the type / species, of fish caught ; so the populations will become depleted / large numbers die / 90% fish die; breeding / reproduction, reduced; alter the, food chains / food web; possibility of local extinction of species / species become endangered; AVP;	**3**

Question	Answer	Marks
1(g)(ii)	*any three from:* make the collector licence cost more; limit the number of, licences / diving sites; restrict number of collection sites; set a quota for each, collector / species of fish; create laws; police the collectors; enforce laws, e.g. heavy fines; ban collection of (rare) fish species; have a closed season; declare a marine reserve; AVP, e.g. making people aware / education / conservation projects / control net size / mesh size / control industrial / agricultural waste;	**3**
1(h)(i)	to get, results they can compare / unbiased results / valid results;	**1**
1(h)(ii)	more data collected / more, boats / boat captains / people asked; more days / specified days (rather than any 3 days from each month);	**2**
1(h)(iii)	table drawn; *headings:* Tues(day) and Thurs(day); (number of) fish / (blue) marlin caught;	**3**

Question	Answer	Marks
2(a)(i)	fourth line drawn in appropriate position **and** 5 sample points on each line; equally spaced / scale used;	**2**
2(a)(ii)	systematic;	**1**
2(a)(iii)	(distance / m) 10 **and** (direction) North (235);	**1**
2(a)(iv)	the concentration (of lead) decreases with increasing distance; same pattern in all directions;	**2**
2(a)(v)	as a control; to compare with test samples;	**2**
2(b)(i)	wind direction, blows from the north east / blows to south west; higher concentrations of lead / paint chips, found (at greater distances) to south and west of buildings;	**2**
2(b)(ii)	*any three from:* lead is, toxic / poisonous; cannot be broken down / excreted by animals; (bio)accumulates; *idea that,* small / young chicks / animals, only need a small dose to be fatal;	**3**
2(b)(iii)	*any two from:* high cost; uninhabited island / does not affect people; no new source of lead; problem only small scale; eventually the paint chips will disperse; disturbance might make the problem worse; AVP; e.g. bigger problems to tackle first / remote location / difficult or time consuming to get to (the island);	**2**
2(c)(i)	use of linear scale; x axis fully labelled; y axis fully labelled (must be average leaf area / cm^2); correct plots;	**4**

Question	Answer	Marks
2(c)(ii)	leaf area decreases with increasing mass of lead nitrate; decrease not linear / figures used to illustrate this;	**2**
2(c)(iii)	*instruction 3:* number / mass / size, of seeds **or** type / species, of bean stated; *instruction 4:* conditions described, e.g. specified temperature / hours of light / light intensity / soil mixture / pH / CO_2; *instruction 5:* volume of water used / frequency of watering;	**3**
2(c)(iv)	*any two from:* try experiment on another type / species, of bean / plant; grow to see change in, crop yield / height / (bio)mass / dry mass / number of leaves / flowers / seeds / fruits produced; use of other masses / concentrations, of lead nitrate; state a specific condition for the experiment, e.g. waterlogged; AVP, e.g. change in, respiration / photosynthesis rate;	**2**
2(d)	*any three from:* it is toxic to organisms / lead can poison / kill, plants and animals; lead absorbed into soils; enters groundwater; enters, food chains / food web; bioaccumulates; lead does not break down / lead pollution lasts a long time; damage to development of children / brain damage; needs to be controlled due to increased transport; AVP, e.g. adults get high blood pressure / kidney damage when exposed / to keep people healthy;	**3**

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 Management in Context **October/November 2020**

MARK SCHEME

Maximum Mark: 80

Published

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>‘List rule’ guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)(i)	3.6(21) (million);	**1**
1(a)(ii)	*any two from:* increased, cost / use of, (health) care; increased (cost of) pensions; younger people need to care for older people / people have more children to care for older people / more (older) dependants; shortage of (working age) workers; increased potential for selling, goods / services, to older people; increased knowledge base; effect on economy with reason e.g. increased economy as, more businesses / more people / more taxes, decreased economy as fewer taxpayers	**2**
1(b)(i)	132 (mm);	**1**
1(b)(ii)	*x*-axis fully labelled: month **AND** *y*-axis fully labelled with unit: (average) temperature / °C; linear scale such that plots occupy at least half of grid; 8 plots correct; 11 plots correct;	**4**
1(b)(iii)	increase / high / more **AND** more refrigeration / air conditioning units / fans, used / needed;	**1**
1(b)(iv)	(sun)<u>light</u> all year round; (light needed) to produce food (in plants) / for photosynthesis;	**2**

Question	Answer	Marks
1(c)(i)	*any four descriptive comments but:* *max 3 about category or direction of hurricane:* starts as category 5 (offshore); (with direction of travel) category decreased / category 5 to category 1; reference to distance travelled at stated category, e.g. cat 2 for, 200 km / 450 km over land; hurricane travelled from S / SE **OR** moved towards N / NW; AVP; *max 2 about storm surge size or location:* range 0.6–2.4 m; highest in north east / Atlantic Ocean coast; lower storm surge on north west / Gulf of Mexico coast; increased through Atlantic Ocean; AVP; *max 1 about wind speed:* (with direction of travel) wind speed decreases / wind speeds greater than 252 (km/h); AVP;	**4**
1(c)(ii)	*any four from:* *idea of short-term effect:* (extraction or export) reduced / stopped; due to, dust / debris (in air); flooding (of mine); no ships / planes / transport (for export) / roads damaged / blocked; damage to machinery; land damaged **OR** soil contaminated (after flooding); infrastructure damage, e.g. electricity / communication; *idea of long-term effect:* idea of, recovery time / reduced demand; water must be pumped away / wastewater must be treated; phosphate entered groundwater (so must be cleaned up) / contaminated water; workers, not safe to work / killed / injured / sick from disease; economic impact, e.g. customers go elsewhere / loss of jobs / less money made as less extracted / loss of profit;	

Question	Answer	Marks

1(c)(iii)	one mark for minimum of three stated strategies;
	
	any five developments from:
	radio announcement:
	non-English speakers might not understand message / other method could have been used;
	early warning system in place;
	    which, reduced loss of life / allowed evacuation;
	one road:
	could have caused delays in evacuation;
	    build other, road / means of escape route;
	road, surrounded by water / could have flooded;
	shelters:
	(some tourist / residents) did get to a shelter (so strategy effective);
	indicate prior disaster prevention planning;
	signs:
	(signs have a picture) so easy to understand / help non-English speaking people / ORA;
	food and water:
	people less likely to get diseases / don't eat contaminated food / drink contaminated water;
	doctor available:
	people do not need to leave the shelter if they are ill / reduced risk of, injuries / disease;
	phone signal:
	build, more / hurricane-proof, phone masts / find other way of getting messages;
	message home:
	prevent rescuers searching unnecessarily for people;
	families knew if people missing;
	temporary housing:
	reduces risk of death from, cold / heat;
	people had a place to live after the hurricane;
	reduces risk of diseases;
	AVP;
		6

Question	Answer	Marks
2(a)(i)	**A** sampling points, are only at one end / are close to each other / don't cover whole line; **C** sampling points, not at even distances / randomly spaced; **D** not enough sampling points;	**3**
2(a)(ii)	price of phosphate has decreased;	**1**
2(a)(iii)	*any two from:* questions, do not address the questionnaire purpose / are not about the mine; questions, leading / biased / only refer to benefits; not enough questions / not enough data; AVP, e.g. no information on who was questioned or selection method, e.g. age / sex;	**2**
2(a)(iv)	*any suitable question <u>about the mine or impact of the mine</u>, e.g.:* Are you in favour of a new phosphate mine?; Do you think there is a need for a new phosphate mine?; Would you object to a new phosphate mine?; Do you think the mine will cause pollution?; AVP;	**1**
2(a)(v)	2.7–3 cm distance measured on map; 1.3–1.5 km, so therefore can mine;	**2**
2(a)(vi)	*any two from:* removes, habitats / shelter / food source; (so) species, migrate / starve / die; (which leads to) loss of biodiversity / genetic depletion / extinction / disruption of food, chains / webs; correct <u>effect</u> of deforestation, e.g. soil erosion, desertification, reduced soil fertility; AVP;	**2**
2(a)(vii)	*any two from:* increased efficiency of (phosphate) extraction; recycling (of phosphate rock); education of customers (on how to efficiently use phosphate);	**2**

Question	Answer	Marks
2(a)(viii)	*any two from:* *land reclamation / ORA:* costs less / economic reason; quicker / easier; gives a named beneficial use (for the land), e.g. farming, tourism;	**2**
2(b)(i)	N = nitrogen; K = potassium (ions);	**2**
2(b)(ii)	*any two from:* adds, nutrients / minerals / ions (to soils); which are needed for plant growth; increases yield; leads to increased profit;	**2**
2(b)(iii)	*any two from:* concern about, excess / overuse of, fertiliser; leads to nutrient enrichment (of water) / eutrophication / <u>water</u> pollution; want to grow organically; economic reason, e.g. cost of fertiliser; crops are growing well without fertilisers / soil already has nutrients;	**2**
2(c)(i)	table with two columns / rows **AND** headings (distance, concentration); units: (distance) km **AND** (concentration of phosphate) mg / *l*; five sets of results recorded in order of distance;	**3**
2(c)(ii)	(north of mine) concentration of phosphate will be, lower / not contaminated with phosphate / owtte;	**1**
2(c)(iii)	*any three from:* formed over millions of years; from, dead marine organisms / dead organisms on seabed; (remains of dead organisms) covered in, mud / sand / sediment; temperature / pressure, increased (causes oil to form);	**3**

Question	Answer	Marks
2(c)(iv)	*any two from:* carbon dioxide is a greenhouse gas; leads to, climate change / enhanced greenhouse effect / global warming / global warming described; effect of climate change. e.g. flooding, drought, forest fires, loss of land, stated effect on crop yield or plant growth;	**2**
2(c)(v)	*any three from:* carbon capture / carbon storage; transport policies; (international or national or local) agreement / policies / laws / treaties; reduction in reliance on fossil fuels / use renewable resources; have fewer children; switch to a plant-based diet; reforestation / afforestation / decrease deforestation; use (more), energy-efficient devices / energy-efficient cars / electric cars; AVP, e.g. switch to renewable supplies, turn off appliances, improve building design, tax fossil fuels;	**3**
2(c)(vi)	no borders (in the atmosphere) / idea that if neighbouring country is polluting, then this will reach other countries / atmospheric circulation or wind patterns / idea of point of no return to act;	**1**

Question	Answer	Marks
3(a)(i)	5180 (km^2);	**1**
3(a)(ii)	*any two from:* building of canals; disruption of water-flow (due to human activities); drainage (of wetlands); urbanisation, e.g. road construction, water supply, infrastructure, homes; (intensive) agricultural (practices); climate change / global warming; mining / raw materials;	**2**

Question	Answer	Marks
3(a)(iii)	*any two from:* sustainable tourism / ecotourism / limit tourism; water conservation area; limit, storm / run-off; educate people / raise awareness, about the area; restrict activities, e.g. ban, hunting / cutting down trees / grazing / canal building; fines (for illegal activity); (make the non-national park area into) wildlife reserve / corridor / extractive reserve / (world) biosphere reserve / buffer zone; AVP;	**2**
3(b)	can't tell them apart / might kill a crocodile by mistake;	**1**
3(c)(i)	orchid (plant);	**1**
3(c)(ii)	*any three from:* vector control; (mosquitoes killed by spraying with) insecticide; sterilise male mosquito; removal of, stagnant water / breeding grounds; cover wells / use sloping roof buildings; place oil on water; biological control; antimalarial drugs to protect people; (prevent bites by using) repellent / nets;	**3**
3(d)(i)	*any two from:* out-competes native species / uses up resources; disrupts, food chains / food webs / biodiversity; no (native) consumer; needs to be cleared;	**2**
3(d)(ii)	*limitation:* can only record plants that are large enough to be seen (from the air) / can't fly if weather is poor / expensive / difficult to identify different species (from a distance); *advantage:* covers large areas / quick / covers all terrains;	**2**

Question	Answer	Marks
3(d)(iii)	*any five from:* use transect **OR** a grid (to determine the location of the quadrats); *transect*: systematic / regular intervals, e.g. at every 5 m, take 10–25 readings **OR** *grid*: random; size of quadrat 1 m × 1 m; estimate the percentage or coverage of pepper plants (in the quadrat); record results in a suitable format, e.g. table or tally or chart; repeat **AND** take an average; calculate the (average) percentage or coverage <u>per quadrat</u>; determine percentage coverage per m^2;	**5**
3(d)(iv)	*any two from:* native / other, plants and animals also burnt; kills endangered species; danger of causing a wildfire / idea of fire getting out of control; causes, atmospheric pollution / increase in carbon dioxide concentrations; leads to, global warming / named effect; damages, settlements / farming land / national park; (possibility of causing) respiratory illness / difficulty in breathing / named effect, e.g. asthma;	**2**

Cambridge Assessment International Education

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 Theory **October/November 2020**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1: Marks must be awarded in line with: • the specific content of the mark scheme or the generic level descriptors for the question • the specific skills defined in the mark scheme or in the generic level descriptors for the question • the standard of response required by a candidate as exemplified by the standardisation scripts.
GENERIC MARKING PRINCIPLE 2: Marks awarded are always **whole marks** (not half marks, or other fractions).
GENERIC MARKING PRINCIPLE 3: Marks must be awarded **positively**: • marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate • marks are awarded when candidates clearly demonstrate what they know and can do • marks are not deducted for errors • marks are not deducted for omissions • answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.
GENERIC MARKING PRINCIPLE 4: Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>‘List rule’ guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states ‘show your working’. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	*any three from:* idea of water, stored / contained / held, in reservoir / by dam wall; water moves (down) through, (sluice) gate / pipe; water, rotates / turns, turbine; turbine, rotates / turns / activates, generator (which generates electricity);	**3**
1(b)	*(list rule applied)* *any one from:* loss of land / loss of habitat / deforestation; additional localised flooding; shortage of water further down river; affects fish, migration / breeding; visual / noise, pollution; reduction in CO_2 emissions if switching from fossil fuels;	**1**
1(c)	*any two from:* no water source; climate is not suitable / little rainfall / water frozen; unable to fund; have abundance of other fuel sources, e.g. fossil fuels / solar power; geology / terrain, not suitable;	**2**

Question	Answer	Marks
2(a)	*any one from:* to keep fish in; to keep, predators / other fish, out; to control diet; so fish, live / breed, in their natural ecosystem; to, harvest / monitor, fish easily;	**1**

Question	Answer	Marks
2(b)	*any two from:* regulation of, mesh size / net design; other species-specific methods, e.g. pole and line; fishing quotas ; closed seasons; protected areas / reserves; conservation laws; international agreements / implementation / monitoring;	**2**
2(c)	*any two from:* parasites / sea lice; disease; habitat destruction; pollution from waste; invasive species; use of antibiotics;	**2**

Question	Answer	Marks
3(a)	exponential / log (phase) ;	**1**
3(b)	*any three from:* increasing / introducing, availability of, family planning / contraception; providing / improving, education on family planning; increasing, career / job / higher education, opportunities for women; encouraging, marriage / children, later in life; improving accessibility to childcare; improving provision of health care; introducing / using, pronatalist / antinatalist, policies; AVP;	**3**

Question	Answer	Marks
4(a)	respiration; combustion / burning;	**2**
4(b)	(the) Sun / (sun)light;	**1**
4(c)	methane; water vapour;	**2**

Question	Answer	Marks
5(a)(i)	*any two from:* Africa has (ora): more LEDCs / less money, to invest in safe water; fewer safe water sources; fewer water-treatment facilities, e.g. chlorination / desalination ; less water-delivery infrastructure, e.g. pipes, bottling; lower levels of sanitation; more areas of water shortage; more, water pollution / contamination / water-borne diseases; more conflict (restricting access to water);	**2**
5(a)(ii)	*any three from:* aquifers; rivers; lakes; (natural) springs; reservoirs / dams; desalination / treatment / filtration, plants; (stand)pipes; rain(water) (harvesting);	**3**
5(b)(i)	2011;	**1**
5(b)(ii)	Africa;	**1**

Question	Answer	Marks
5(b)(iii)	fluctuates / no clear pattern / overall decrease (with some outbreaks);	**1**
5(c)	*any two from:* damage to / contamination of / shortage of, fresh water sources; damage to, sewerage systems / infrastructure; damage to / lack of, medical, provision / supplies; lots of people living closely together in temporary accommodation; poor sanitation;	**2**

Question	Answer	Marks
6(a)(i)	(direction of) energy transfer / flow;	**1**
6(a)(ii)	hawk;	**1**
6(a)(iii)	*any one from:* increases as no predators; stays the same because, other predators / hawks, eat them;	**1**
6(b)(i)	*any two from*: provide a, habitat / shelter, for a variety of different, plants / animals; prevent toxins from harming the environment; provide, water / food, source for local organisms; prevent flooding of other habitats nearby;	**2**
6(b)(ii)	provide, income from tourism / ecotourism / incentive to maintain the environment to attract visitors;	**2**

Question	Answer	Marks
6(b)(iii)	*max five marks for points made from comment bubbles:* *the wetland should be drained because:* provides people with, homes / shops / roads; breeding ground for mosquitoes that spread malaria; *the wetland should not be drained because:* provides a habitat for several rare species of birds and frogs; which may become extinct; reduces the risk of flooding; *plus max five marks for candidate's own valid points:* *additional examples of reasons for draining:* people of the town would benefit / supports the growth of the town; provides other forms of employment; provides better transport links, e.g. for commerce; reduces associated health risks, e.g. malaria; *additional examples of reasons for not draining:* provides, income / employment, for people working there; provides area to grow, rice / food; source of fish to eat; acts as migratory network (for birds);	**7**

Question	Answer	Marks
7(a)	*any two from:* idea of existing rock; (is subjected to) high temperature **AND** pressure; to change (physical / chemical) structure / rock crystals;	**2**
7(b)(i)	40 100 bar line;	**1**
7(b)(ii)	(54 500 ÷ 89 900 × 100 =) 60.6 (%);	**1**

Question	Answer	Marks
7(b)(iii)	*any two from:* more mechanisation; fewer mines / mines closed / low coal price / poor profitability; concern over danger of working in mines; less demand for coal; cheaper imports available; non-renewable resource / finite reserves / running out; switched to other forms of energy resources; named environmental concern, e.g. climate change, acid rain; government policy;	**2**

7(c)(i)	*any three from:*
	development of infrastructure;
	    improved, transport links / roads;
	    improved, internet / communications services;
	improved local economy;
	    increased commerce, e.g. services, shops, hotels;
	        increased tax revenue which could be spent locally;
	    improved, standard of living / quality of life / income;
	greater access to coal;
		3

7(c)(ii)	*any two from:* landfill site; nature reserve / public space / park; flood to make a, lake / reservoir; visitor attraction / museum / science centre; other recreational area, e.g. race track, dry ski slope, zip wire; land restoration;	**2**
7(d)	*any three from:* have large reserves (of coal); infrastructure / mines, already in place; industry employs a large number of people in the country; lack of finance for, investment / research in alternatives; cheap imports of coal available; lack of other energy resources, e.g. nuclear, wind, solar;	**3**

Question	Answer	Marks
8(a)(i)	491;	**1**
8(a)(ii)	axis labels and units; sensible linear scale so graph uses at least half plotting space; points correctly plotted; line joining points;	**4**
8(a)(iii)	yield will stay the same / no increase;	**1**
8(a)(iv)	*(list rule applied)* *any two from:* temperature; amount of rainfall; amount of (sun)light; pests / invasive species; disease;	**2**
8(b)	*any two from:* due to, rain / irrigation; fertilisers dissolve in water; (surface) run-off; infiltration / via ground water;	**2**
8(c)	*any two from:* larger yield per plant, e.g. bigger / heavier / more productive, plants; faster growth / multiple harvests per year; resistant to environmental conditions; drought-resistant; temperature-resistant, e.g. frost-resistant; resistant to, disease / pests; AVP;	**2**

Question	Answer	Marks
9(a)(i)	an area of land covered by water when rivers overflow / owtte;	**1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>9(a)(ii)</td><td>any one from:
have fertile soil;
have a layer of nutrient-rich sediment;
store floodwater;</td><td>1</td></tr>
<tr><td>9(b)</td><td>table drawn;
column / row, headings and unit;
4 sets of data recorded correctly;

<table><tr><td>year</td><td>area flooded / km^2</td></tr><tr><td>1987</td><td>57 300</td></tr><tr><td>1988</td><td>82 000</td></tr><tr><td>1998</td><td>77 700</td></tr><tr><td>2004</td><td>70 800</td></tr></table></td><td>3</td></tr>
</table>

Question	Answer	Marks
9(c)	*Level of response marked question:* Level 3 [5–6 marks] **A coherent response is given that develops and supports the candidate's conclusion using relevant details and examples.** Indicative content and subject-specific vocabulary are generally used precisely and accurately. Good responses are likely to present a balanced evaluation of the statement. Level 2 [3–4 marks] **Development and support of the conclusion is evident, though the response may lack some coherence and/or detail.** Irrelevant detail may be present. Indicative content and subject-specific vocabulary are used but may lack some precision and/or accuracy. Responses contain evaluation of the statement, but this may not be balanced. Level 1 [1–2 marks] **The response may be limited in development and/or support.** Contradictions and/or irrelevant detail may be present. Indicative content and subject-specific vocabulary may be limited. Responses may lack structure or be in the form of a list. Evaluation may be limited or absent. No response or no creditable response [0 marks]	**6**

Question	Answer	Marks

9(c)	*Indicative content for:*
	*‘**Small-scale** flooding is a good thing even though damage can occur.’*
	
	benefits of small-scale flooding include:
	layers of sediment deposited on land
	soil becomes fertile
	less / no, need for additional chemical fertiliser
	saves on cost of fertilisers
	reduces environmental impact (of chemical fertilisers)
	replaces need to irrigate
	replenishes underground water supplies
	salt deposited by high rates of evaporation removed by flood water
	
	drawbacks of small-scale flooding include:
	potential waterlogging of soil
	(minor) damage to infrastructure, e.g. potholes in roads, communications lines down
	    makes, travel / transportation / business / work, difficult
	cost of repairing damage
	restricts use of mechanisation for farming
	may delay agricultural operations, e.g. harvesting, sowing
	possible, overflowing of / damage to, sewerage systems
	possible pollution of drinking water sources
	    leading to potential disease outbreaks (e.g. cholera)
	may provide areas of standing water for mosquitoes to breed
	    leading to potential malaria outbreaks	

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

5014 ENVIRONMENTAL MANAGEMENT

5014/21 Paper 2, maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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CAMBRIDGE
International Examinations

1 (a) (i) damage to, crops; infrastructure; buildings; roads; communications; loss of products; unemployment; inability to trade e.g. tourism; AVP; such as too sick to work; [2]

(ii) money sent back / e.g.; high costs of repair / recovery; [1]

(b) (i) provides fertiliser / nutrients / minerals / named mineral; for (plant) growth; rapid root development; manure retains water / eq.; [2]

(ii) increased density increases yield; increased yield decreases yield per tree; only by a small amount; doubling the number of trees does not double yield / eq.; use of figures to support any point; [3]

(iii) high cost of labour; for digging holes; cost of manure / bone meal; cost of extra seeds / seedlings / trees; loss of soil quality; extra investment only gives small increase in yield; [3]

(iv) intercropping / agroforestry / intensive; [1]

(v) nitrogen fixed (from the air); so more (nitrogen for crop) growth; more crop to sell / eat; fodder for animals; increased fertility; [3]

(c) (i) 253; [1]

(ii) 253 / 385 = 65.7–66.0 (%); [1]

(iii) knife or spoon / scales / bowl / notebook and pen;
Four for two marks. Three or two for one mark. [2]

(iv) care with knife / gloves to handle seed / wash hands; [1]

(v) use seeds for new planting; composted (to make fertiliser); animal feed; [2]

(d) (i) foreign currency helps balance of payments / eq.; more tax revenue; reduces poverty / improves standard of living; creates jobs / eq.; [2]

(ii) high cost of fertiliser and / or insecticide; regular hurricanes could destroy crop; drop in world demand; risk of going bankrupt / eq.; [2]

(iii) cross-breeding two varieties; selecting the offspring with desired characters; identify the allele / gene for large fruits; genetically engineer (a native variety); further detail of genetic engineering; ref. to grafting; [2]

(e) (i) product lasts longer; can be exported all year round; exported when demand / prices high; lower transport costs; native plants need less care / eq.; makes use of native species; AVP;

First point for one mark, two or three points for two marks. [2]

(ii) large amount of raw material needed / eq.; high production cost; skilled labour needed; difficult to dry flesh in a tropical climate; cost of heating / eq.; [2]

(iii) give grants / loans / subsidies for building ovens / buying gas; government education campaigns aimed at farmers / product promotion; [1]

(iv) sustainable: less chemical inputs needed; low risk of pollution; lower costs of production; still part of the local ecology / eq.; AVP;
OR
not sustainable: too difficult to process / store dried fruit; need to produce more fruits; demand may drop; small fruits are easy to export when there is demand;
AVP; [4]

2 (a) (i) one line in correct orientation; correct size each side of power line; [2]

(ii) plan 3 is in the correct orientation but plan 1 is not;
plan 3 goes into the forest but plan 1 and 2 do not;
plan 3 is repeated; plan 3 can check the data; so can take an average; [3]

(iii) line graph; correct orientation and both axes labelled; plots; [4]

(iv) plant species increases; then decreases; [2]

(v) person B is right with a reason, e.g. species diversity similar under lines and in forest; maximum diversity at the boundary; further detail may include use of data; [2]

(vi) survey animals; more power lines; each year to measure changes; identify species; survey for named abiotic factors; [3]

(b) (i) H.E.P. does not generate greenhouse gasses / eq.; acid rain; water is a renewable source; abundant supply; use as a reservoir; only a small amount of forest lost; AVP; [3]

(ii) to pay for the building of the dam / turbines / eq.; [1]

(iii) macaw not saved: as power for people more important than the habitat of one species / eq.; some loss of species has to be accepted;
OR
macaw saved: as if it becomes (locally) extinct (where will destruction stop);
keeping biodiversity is important for the future / eq.; AVP; [2]

(iv) silt builds up behind the dam; so less water held / flow of water reduced;
turbines turn less / generate less electricity; [1]

AVP = Alternative Valid Point.

[Total: 60]

Cambridge Assessment International Education

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 Theory **May/June 2020**

MARK SCHEME

Maximum Mark: 80

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

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Marks must be awarded in line with:

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
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5 <u>'List rule' guidance</u>

For questions that require ***n*** responses (e.g. State **two** reasons …):

- The response should be read as continuous prose, even when numbered answer spaces are provided
- Any response marked *ignore* in the mark scheme should not count towards ***n***
- Incorrect responses should not be awarded credit but will still count towards ***n***
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 <u>Calculation specific guidance</u>

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 <u>Guidance for chemical equations</u>

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	*any three from:* causes the (enhanced) greenhouse effect; CO_2 is a greenhouse gas; increased concentrations prevent heat loss from atmosphere; global temperatures rise; causes increase in melting of glaciers / polar ice caps; causes expansion of water as water warms;	**3**
1(b)(i)	$23 - 12 = 11$; $(11 \div 23 \times 100 =)$ 47.8 / 48;	**2**
1(b)(ii)	*any three from:* increased flooding / flood risk; loss of industry / tourism; greater housing density; loss of farmland (to build houses); less food produced; families need to emigrate; salinisation of water sources / fields;	**3**

Question	Answer	Marks
2(a)	*any three from:* volatile organic compounds (from industrial processes); atmospheric pollution from vehicle emissions / from industry; temperature inversion layer / cold air, traps the air pollution; sea wind blows air pollution inland; mountains act as a barrier to prevents air movement;	**3**
2(b)	*any three from:* irritation to eyes; increase in asthma; particles increase risk of lung cancer; breathing difficulties / named condition; heart problems / stroke;	**3**

Question	Answer	Marks
2(c)	*any two from:* increase / introduce, legislation / taxation to control vehicle emissions; increase availability of public transport; encourage switch to electric / hybrid / hydrogen cell vehicles; restrict access of vehicles to city centre; car sharing incentives;	**2**

Question	Answer	Marks
3(a)	175 km;	**1**
3(b)	*any two from:* species are, invasive / not native to the Mediterranean Sea; impacts on food web / disrupts food chains / introduces predators; may cause competition with existing organisms, for habitat / for food; may introduce, disease / pathogens; possible extinction of native species;	**2**
3(c)	increased traffic means more sea water in canal / increased fertiliser run-off / sea water from Red Sea or Mediterranean Sea has entered canal;	**1**

Question	Answer	Marks
4(a)	*any three from:* reduced areas in 2016; reduced in southern and northern areas; named area / continent, e.g. southern part of South America; northern Africa; central Asia;	**3**
4(b)(i)	*any three from:* use of antimalarial drugs; mosquito nets; control of vector / mosquito with pesticides; removal of stagnant water; application of oil to water sources;	**3**

Question	Answer	Marks
4(b)(ii)	356 000;	**1**
4(b)(iii)	*any three from:* remoteness of population; lack of money for control methods; lack of health care; lack of awareness amongst population; resistance to antimalarial drugs; areas with lots of stagnant water present;	**3**

Question	Answer	Marks
5(a)(i)	opencast;	**1**
5(a)(ii)	*any four from:* over-burden removed; stored for mine restoration; cut away in, sections / benches; use of machinery; use of explosives; rock taken away for processing;	**4**
5(b)(i)	1500 million tonnes ÷ 20 million tonnes = 75 (years); (2008 + 75 years =) 2083;	**2**
5(b)(ii)	*any three from:* landfill; lake for fishing / water storage; parkland / recreation; nature reserve; afforestation / agriculture;	**3**

Question	Answer	Marks
5(c)	*any three from:* abundant coal reserves locally; lack of other resources locally; other sources / technologies, expensive / (coal) technology already in place; large economic benefit (exports / employment) from coal industry; (some countries) do not think the pollution issue is a priority;	**3**
5(d)	oil / gas / nuclear;	**1**

Question	Answer	Marks
6(a)(i)	3039 (million USD);	**1**
6(a)(ii)	*bar graph with:* appropriate use of scale; labelled axes with units; all plots correct;;	**4**
6(b)(i)	*any two from:* reduction in insect pollinators; impact on food web; water pollution;	**2**
6(b)(ii)	*any five from:* local people can use clean water; then grey water used on big farms; big farms could invest in, water storage systems / reservoirs; water conservation techniques could be used, e.g. rain water harvesting; investment in accessing aquifers / use of boreholes; charging big farms to extract water / water meters; agreements on levels of water use / quotas; choice of crops which do not use so much water; use of efficient irrigation techniques, e.g. trickle drip, clay pot; better education in, water use / conservation of water;	**5**

Question	Answer	Marks
7(a)(i)	penguin / leopard seal / squid / cod;	**1**
7(a)(ii)	phytoplankton; *(krill given)* penguin / cod **and** killer whale / leopard seal (in correct sequence);	**2**
7(a)(iii)	*any four from:* increase in phytoplankton; increase in krill (as more food available / less predation); increase in zooplankton (as more food available / less predation); increase in penguins (as more krill to eat); leopard seals will eat more penguins and squid (as no cod to eat);	**4**
7(a)(iv)	*energy lost by:* movement; respiration; reproduction; incomplete digestion of prey; excretion;	**2**
7(b)	*any four from:* restriction of net types; restriction on mesh sizes; fishing quotas; closed seasons; no-fishing zones; encourage fish farming;	**4**

Question	Answer	Marks
8(a)(i)	three points correctly plotted; completion of line to join;	**2**
8(a)(ii)	growth from 1890 to 1990; decline from 1990 to 2015;	**2**
8(b)	**A**;	**1**
8(c)	*Level of response marked question:* <u>Level 3</u> [5–6 marks] A coherent response is given that develops and supports the candidate's conclusion using relevant details and examples. Indicative content and subject-specific vocabulary are generally used precisely and accurately. Good responses are likely to present a balanced evaluation of the statement. <u>Level 2</u> [3–4 marks] Development and support of the conclusion is evident, though the response may lack some coherence and/or detail. Indicative content and subject-specific vocabulary are used but may lack some precision and/or accuracy. Irrelevant detail may be present. Responses contain evaluation of the statement, but this may not be balanced. <u>Level 1</u> [1–2 marks] The response may be limited in development and/or support. Contradictions and/or irrelevant detail may be present. Indicative content and subject-specific vocabulary may be limited or absent. Responses may lack structure or be in the form of a list. Evaluation may be limited or absent. No response or no creditable response [0 marks]	**6**

Question	Answer	Marks
8(c)	*Indicative content for:* 'Improved health and education are the best ways to manage the population size of a country' *agree:* improved healthcare reduces infant mortality therefore more children likely to survive so parents have fewer children better education provides information and options regarding family planning improved education for women means more job opportunities and children at an older age better jobs mean less need for children to provide income may take more than one generation for the effects of better healthcare to be seen *do not agree:* government can supply incentives to support population development, e.g. raise the age of marriage, antinatalist and pronatalist polices, sterilisation, immigration and migration etc. availability of family planning methods affects use cultural and religious views impact on family size increased immigration due to better healthcare and education (description of other ways as better)	

Cambridge Assessment International Education

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 Management in Context **May/June 2020**

MARK SCHEME

Maximum Mark: 80

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 <u>'List rule' guidance</u>

For questions that require ***n*** responses (e.g. State **two** reasons …):

- The response should be read as continuous prose, even when numbered answer spaces are provided
- Any response marked *ignore* in the mark scheme should not count towards ***n***
- Incorrect responses should not be awarded credit but will still count towards ***n***
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 <u>Calculation specific guidance</u>

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 <u>Guidance for chemical equations</u>

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)(i)	318 000;	**1**
1(a)(ii)	*any three from:* birth and death rates; migration; carrying capacity; improved health and education / medical access; access to birth control / family planning; national policies / pronatalist / antinatalist; famine / disease / natural disasters; war / conflict;	**3**
1(a)(iii)	*any three from:* the population already has a high % of young males and females; of reproductive age; so the birth rate is likely to be high / high number of children per woman; older people die / very few above 62 years / low life expectancy; AVP, e.g. no access to birth control / cultural influences ;	
1(a)(iv)	*any two from:* sold as a cash crop; exported; wasted / not good enough to sell / spoilt; fed to animals; used as fertiliser; non-food crops, e.g. biofuels, oil for products;	**2**
1(a)(v)	*any two from:* food could become too expensive to buy; may not have, much food / food reserve / food security; at risk of crop failure; leading to, famine / starvation / death;	**2**

Question	Answer	Marks
1(b)	*any two from:* space for, trees to grow / branches to spread out; easy access to pick fruit; allows farm machinery space to pass through; can grow other crops / graze animals between trees; AVP, e.g. helps to control / reduce plant diseases;	**2**
1(c)(i)	6.4 (tonnes);	**1**
1(c)(ii)	*any two from:* over 50 km from Dakar; too far from a port; too far to go around the Gambia; high transport costs make exporting unprofitable; so fruits will spoil / short shelf life of fruit; costs of crossing the Gambia border; AVP, e.g. bad roads damage fruit;	**2**
1(c)(iii)	*any two from:* fruits release nutrients into soil; taken up by tree next year; reduces chance of mineral deficiency;	**2**
1(c)(iv)	low value due to oversupply / lack of demand / cost too much too pick;	**1**
1(d)	2050; 410;	**2**
1(e)(i)	0.5; 5.8; 5;	**3**
1(e)(ii)	*any suitable question about mango farming such as:* How much does it cost to produce mango fruits? / How much profit do you make? / How much do mango fruits sell for? ;	**1**

Question	Answer	Marks
1(e)(iii)	results do not agree / false conclusion; evidence to support, e.g. average is 3 / only four farms out of 18 grow two varieties;	**2**
1(e)(iv)	*any one from:* grow well in local climate / better adapted to local environment; resistant to local pests; less expensive to buy; farmers know how to get the best fruits;	**1**
1(f)	*one from each section plus any other one:* *benefits to farmers:* faster transport by road; cheaper transport by road; may be able to export by air; more income; vehicles last longer / are not damaged; fruits not bruised / arrive in better condition; *benefit to government:* airport used more; more, revenue / airport taxes / eq; so more money to spend on, other projects / named example; more import from / export to, places further away;	**3**
1(g)(i)	*any two from:* farmer does not rely on (selling) one crop / has a variety of produce; unlikely that all crops would fail in the same year / more than one crop reduces chance of malnutrition / famine; source of protein / meat; sources of grain; source of vitamin C; varied diet covering all major food groups;	**2**

Question	Answer	Marks
1(g)(ii)	*any two from:* prolonged drought / floods / natural disaster; infestation of pests / plague of locusts; harvest failure; new disease of, crops / animals; illness / injury to, farmers / workers; conflict; too many farmers replace food crops with cash crops;	**2**
1(h)(i)	Jun / Jul to Sep / Oct; soil washed away; **or** Nov / Dec to Apr / May; dry soil blown away by wind;	**2**
1(h)(ii)	*any five from:* protect underlying soil from heavy rain; roots bind soil; roots take up water from soil; increase interception; so more infiltration; surface run off less likely; reduce wind speed / act as wind break; organic matter from trees helps bind soil;	**5**

Question	Answer	Marks
2(a)(i)	*any three from:* vegetation cleared; soil removed; overburden removed; explosives used; mining machinery used to dig out rocks / ore / phosphate rock; solid ore transported to processing plants (on trucks);	**3**
2(a)(ii)	*any one from:* only a small body of water; and level ground / no slope towards stream/river;	**1**
2(b)(i)	both axes fully labelled; linear scale such that plots occupy at least half the grid; line graph plotted correctly;;	**4**
2(b)(ii)	over production / reduced demand;	**1**
2(c)(i)	$880 - 93 / 787$; $(787 \div 93 \times 100 =)$ 846(%);	**2**
2(c)(ii)	*one from each section plus any other one:* *benefit to government:* increase value of exports; stronger economy; earns foreign exchange; *benefit to people:* more jobs available, in processing / transporting; earnings increase; improved standard of living, qualified / named example, e.g. more medical facilities;	**3**
2(c)(iii)	nitrate (ion) / NO_3^-; potassium (ion) / K^+;	**2**

Question	Answer	Marks
2(c)(iv)	*any five from:* (toxic / harmful) algal blooms; subsequent bacterial blooms / decomposers; respiration increases; oxygen depletion; fish / aquatic animals die; loss of / change in, biodiversity; clogging / drying up of, ponds / lagoons; makes water smell / decreases water transparency;	**5**

Question	Answer	Marks
3(a)	*any three from:* hot / liquid magma; from the mantle; moves into the Earth's crust; cools and solidifies; within or on surface of Earth; metallic crystals form; quick cooling forms small crystals / slow cooling forms large crystals;	**3**
3(b)(i)	*any two from:* washing / swirling; to separate sediment and gold / wash lighter sediment away; looking for heavy gold left / gold pieces left in pan;	**2**
3(b)(ii)	*any three from:* easy to extract gold / efficient method; know how to do it; mercury, very cheap compared to value of gold / readily available; want / need, to earn money (quickly); do not worry about long term toxicity; already invested in this method;	**3**

Question	Answer	Marks
3(c)(i)	*any three from:* absorbed by, plants / producers / first consumers; small fish eat plants and small amounts of (toxic) mercury; large fish eat small fish, accumulate large amounts of (toxic) mercury; passes up the food chain / bioaccumulates; high concentration of mercury in large fish causes death; mercury poisons drinking water supply;	**3**
3(c)(ii)	*any two from:* not a traditional method / did not know about them; could not afford to buy the machines; did not know how to use / maintain them / had no training; machines were not available;	**2**
3(c)(iii)	*any two from:* to find out if the miners are still using / maintaining the machines; to check that the claim they can recover more gold is correct; to update training / train more miners; to help with repairs / maintenance;	**2**
3(c)(iv)	*any two from:* many families would have to keep mining anyway; it is a traditional / well-established activity; difficult to enforce a ban / it might be very unpopular; helps to keep people on the land; reduces pull factor to urban areas; families will not require help from the government;	**2**

Cambridge Assessment International Education

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 Theory **October/November 2021**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Generic Marking Principles

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

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- marks are awarded when candidates clearly demonstrate what they know and can do
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- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

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Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>‘List rule’ guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	*clockwise, from top:* carbon dioxide, sunlight, oxygen, water 2 correct; 4 correct;	**2**
1(b)	chlorophyll;	**1**
1(c)	respiration;	**1**

1(d)	*one mark for named factor, one mark for linked explanation:*
	controlled environment;
	    so optimum conditions (can be provided);
	higher temperatures;
	    so the rate of photosynthesis will be higher;
	adding carbon dioxide;
	    for optimum concentration to favour photosynthesis;
	managed water (supply);
	    so the correct amount is available for plant processes;
	control of pests;
	    so energy is retained by the producer / not lost to the consumer / less competition;
	supply of nutrients;
	    so plants have access to optimum levels;
		2

Question	Answer	Marks
2(a)	commercial arable;	**1**
2(b)	*any two from:* crop rotation; fertilisers; insect control; (insecticide); weed control (herbicide); fungi control (fungicide); genetically modified organisms / better varieties / higher yielding varieties; mechanisation;	**2**
2(c)	*any two from:* waterlogging; dries out / compaction / capping; soil erosion; salinisation; leaching;	**2**

Question	Answer	Marks
3(a)	*any three from:* drink safe water; cholera is a water-borne disease; wash your hands with soap; spread by touch / contaminate food or people / prevent transmission / eq; have a vaccination; prevent infection / reduces spread / allows humans to fight the infection / lead to eradication / provides immunity; keep cooking areas clean; prevents (cross) contamination;	**3**
3(b)	*any two from:* LEDC / ora: less effective sanitation; fewer sources of clean water; less access to, medical care / vaccinations; different economic priorities; less awareness; larger number of informal settlements; AVP;	**2**

Question	Answer	Marks
4(a)	*any two from:* large population and limited land; economic; employment; safer; better access to medical facilities / education; better infrastructure; AVP;	**2**

Question	Answer	Marks
4(b)	*any two from:* availability of contraception; access to health care; improved education on family planning; education of women / careers; examples of pronatalist or antinatalist policies, e.g. taxation / legislation ;	**2**

Question	Answer	Marks
5(a)(i)	surface / opencast / open-pit / open-cut;	**1**
5(a)(ii)	*any three from:* air pollution due to transport; noise pollution due to transport / blasting; habitat loss due to land clearance; loss of biodiversity due to land clearance; dust pollution due to extraction; water pollution due to leaching / run-off;	**3**
5(a)(iii)	limestone is a sedimentary rock / formed by sedimentation; formed by the accumulation of shell (material); by compaction / weight of overlying sediment; over a long time period;	**2**
5(a)(iv)	*any three from:* availability / ease of extraction / cost of extraction; accessibility / ease of transportation; environmental impact assessment / away from settlements / protected areas / local opinion; supply and demand / profitability / economic factors; supply of labour; AVP;	**3**
5(b)(i)	2016;	**1**
5(b)(ii)	consumption is increasing; relevant quoted data, e.g. from 244 to 296 <u>million tonnes</u> / 21% increase;	**2**
5(b)(iii)	4.5% of 298 / 13.41; (298 + 13.41 =) 311.41 / 311 / 311.4;	**2**

Question	Answer	Marks
6(a)(i)	correct plotting of 3.5;	**1**
6(a)(ii)	(19 + 11 =) 30 (g);	**1**
6(b)(i)	*any three from:* along the equator / between tropics; mainly coastal; west coast of Africa; west coast of South America; east coast of Asia; east coast of North America; west coast of North America AVP;	**3**

6(b)(ii)	*one mark for strategy AND one mark for associated description:*
	any two strategies from:
	net types / mesh size;
	    reduces by-catch / allows juveniles to mature;
	quotas;
	    prevention of overfishing;
	closed seasons;
	    allows breeding;
	protected areas / reserves;
	    protects specific species / endangered species / allows population recovery;
	conservation laws;
	    forces people to comply with rules;
	international agreements;
	    aims to ensure fairness between countries;
		4

Question	Answer	Marks
7(a)(i)	*any two from*: lack of rainfall / high rate of evaporation; lack of rivers / lakes / water sources; overpopulation; water pollution; (geographically) inaccessible / frozen / permeable rock / lack of ability to extract; agriculture; climate change; poor management / water wastage;	**2**
7(a)(ii)	*any two from*: no research yet been done; very low population; inaccessible; not a priority;	**2**
7(b)	*any one from:* death of organisms; decline in crop yields; starvation; (increased) soil erosion; desertification; decrease in air quality; increased risk of wildfires;	**1**

Question	Answer	Marks
7(c)	*desalination:* *benefit:* unlimited supply; no further treatment needed / safe water supply; *limitation:* need to be by the sea / not suitable for landlocked areas; expensive (to build / run); large amount of electricity needed; disposal of waste / salt; *reservoirs:* *benefit:* can be large or small; low technology; low cost; rainfall can be stored; multipurpose uses; *limitation:* dry up / weather dependent; can be polluted; can cause earthquakes; displaces people / animals; loss of habitat / land lost; may silt up; *plastic bottle water:* *benefit:* plastic is light; easily transported; available in different sizes; source of potable water; can be used in times of disaster;	**5**

Question	Answer	Marks
7(c)	*limitation:* plastic pollution / waste disposal; not suitable for large scale use / farming / industry; short term solution; expensive method;	

Question	Answer	Marks
8(a)(i)	45;	**1**
8(a)(ii)	diesel;	**1**
8(a)(iii)	*y*-axis: CO_2 in arbitrary units; *x*-axis: engine type AND four labels; sensible linear scale for *y*-axis using at least half the graph paper; 4 bars plotted;	**4**
8(b)	*any three from:* transport policies e.g. congestion charge; taxation (of other vehicle types); subsidies / incentives to purchase electric vehicles; subsidies / incentives to use electric vehicles; preferential right of way or access; increase availability of electricity charging points; raise awareness; AVP;	**3**
8(c)	*any four from:* emissions cause climate change / global warming; which affects <u>all</u> countries / populations; idea that carbon dioxide crosses international boundaries; example of climate change effect e.g. melting of ice sheets, glaciers and permafrost, rise of sea level, flooding; causes loss of land / forced migration; requires international cooperation to address; climate change causing changes in biodiversity;	**4**

Question	Answer	Marks
9(a)	*any two comparisons from:* fuel: uranium vs coal / oil / gas; process: fission vs combustion; mass of fuel: small vs large(r);	**2**
9(b)	column or row headings: country, percentage and mass; unit for mass / tonnes in heading; 3 sets of data recorded correctly;	**4**
9(c)	*any two from:* demand increase / more power stations being built; resulting in limited supply; potential price fixing / cartel; political instability; AVP;	**2**

Question	Answer	Marks
9(d)	*Level of response marked question:* Level 3 [5–6 marks] **A coherent response is given that develops and supports the candidate's conclusion using relevant details and examples.** Indicative content and subject-specific vocabulary are generally used precisely and accurately. Good responses are likely to present a balanced evaluation of the statement. Level 2 [3–4 marks] **Development and support of the conclusion is evident, though the response may lack some coherence and/or detail.** Irrelevant detail may be present. Indicative content and subject-specific vocabulary are used but may lack some precision and / or accuracy. Responses contain evaluation of the statement, but this may not be balanced. Level 1 [1–2 marks] **The response may be limited in development and/or support.** Contradictions and / or irrelevant detail may be present. Indicative content and subject-specific vocabulary may be limited or absent. Responses may lack structure or be in the form of a list. Evaluation may be limited or absent. No response or no creditable response [0 marks]	**6**

Question	Answer	Marks
9(d)	*Indicative content for:* To meet world demand and combat climate change, all power stations built in the future should be nuclear power stations. *agree / advantages of nuclear power generation:* does not produce CO_2 or other greenhouse gases; does not produce smoke; a lot of energy from a small amount of fuel; reliable; small volume of waste; low maintenance; can meet increasing world demand; existing technology; *disagree / disadvantages of nuclear power generation:* expensive to build; expensive to decommission; danger of terrorism; danger of explosion; radioactive waste; limited availability of uranium; technically difficult to build / expertise needed; international regulation; other alternative sources available (solar, wind, hydroelectric, geothermal, etc.); variety of power stations more reliable; some countries have other resources naturally available; too dependent on Kazakhstan / Australia / Canada;	

Cambridge Assessment International Education

Cambridge O Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 Management in Context **October/November 2021**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)(i)	*any two from:* increasing; no fluctuations / increased gradually; increased by 11% / increased from 68% to 79%;	**2**
1(a)(ii)	6.8;	**1**

1(a)(iii)	*any two from:*
	better employment / economic opportunities;
	better access to services/infrastructure;
	    education;
	    healthcare;
	    clean water / sanitation;
	    electricity;
	    transport;
	improved security;
	less chance of natural disasters, e.g. volcanic activity;
	AVP;
		2

1(a)(iv)	*any two from:* fewer people to do jobs / loss of jobs; change the traditional way of life / loss of cultural identity; negative impact on economy; increased waste; loss of land / loss of homes; reduced production of food / less farms / less farming; AVP;	**2**

Question	Answer	Marks
1(a)(v)	*any two reasons and any two explanations that link to the reason:* education; people know how to lead a healthy life / are able to get better jobs / earn more money; healthcare; illnesses can be cured or prevented or treated / access to vaccines / access to medicines or doctors / access to family planning; clean water or sanitation; less chance of, drinking contaminated water / catching a water-related disease, e.g. cholera or typhoid; electricity; temperature control / refrigeration; AVP;	**4**
1(b)(i)	*from* May *to* Dec;	**1**
1(b)(ii)	400 – 85; 315 ;	**2**

Question	Answer	Marks
1(b)(iii)	*max two use of data from:* hot (all year); wet (all year); relevant quoted data, e.g. 26.2 max to 23.9 min °C; *max three reasons from:* *idea of photosynthesis:* higher temperatures lead to more photosynthesis / more crop growth; plants need water for, photosynthesis / growth; carbon dioxide + water → glucose + oxygen; *idea of high rainfall effect on soil:* risk of flooding / soil does not dry out / increased surface runoff; soil erosion; crops washed away; soil waterlogged; leaching of nutrients / decreased fertility; AVP;	**max 4**
1(c)(i)	*any three from:* loss of habitat / biodiversity; loss of arable or farm land / land for growing crops / plants can't grow / loss of vegetation; food shortages / famine / malnutrition; silting / sedimentation, of rivers; land less able to soak up water / increased runoff / leads to flooding; loss of nutrients / decrease soil fertility; decrease profits for farmers / economic impact; migration / displacement of people; desertification;	**3**

Question	Answer	Marks
1(c)(ii)	*any two from intercropping:* different <u>root</u> depths; <u>roots</u>, bind / hold the soil; improves or maintains soil fertility / nutrients / structure; ground is always covered by crops / greater percentage plant cover; (increases) interception; (increases) infiltration; *any two from bunds:* hold back or stores water or soil / soil eroded from higher up the slope is deposited behind the bund; slow down the flow of water / soil; allows greater time for infiltration;	**4**
1(c)(iii)	*any three from:* increases overall /decreases then increases / fluctuated; use of data on forest cover, e.g.: 1940–1980: decrease; 1980: lowest; 1980–1990: increase; 1990–2000 decreased / fluctuated; 2000–2019: increasing; 1940 and 2019: (slight) increased / very similar; 2019 had the largest cover; AVP;	**3**

Question	Answer	Marks
1(c)(iv)	*conclusion based on any four supported pieces of evidence from comments:* *any four from:* *YES* *payments:* reduce <u>deforestation</u> / prevents loss of habitat; maintain biodiversity / reduced genetic depletion / extinction; alternate way of making a living / financial incentive / economic development / generate foreign income; discourages cash crop growth / intensive cultivation or grazing; encourages, <u>ecotourism</u> / sustainable tourism; soil erosion / desertification is prevented; it is a sustainable strategy / or description of sustainability; reduces climate change; (trees) regulate the water cycle; (trees) provide food / medicine / raw materials; AVP; *NO* *forms to fill / not compulsory:* might prevent people from applying / forms are time consuming / need to be literate; may need a computer to complete forms / some people do not have access to electricity; not everyone will join / may not stay in the scheme (as not compulsory); land might be more valuable for other purposes / named purpose; reduced export (or crops); increased government spending; AVP;	**4**
1(c)(v)	*any four from:* trees capture carbon (from the atmosphere); trees act as carbon, sinks / stores; when tree dies / rots / burned, carbon is released; carbon dioxide is a greenhouse gas / increase in greenhouse gas (in atmosphere); leads to global warming / enhanced greenhouse effect; tree loss affects water cycle / named effect of water cycle; e.g. less infiltration / transpiration / evaporation; AVP;	**4**

Question	Answer	Marks
1(c)(vi)	*any one from:* conditions are no longer suitable for some animals (in Costa Rica); species not adapted (to new conditions); loss of habitat; lead to extreme weather / increase in wildfires; extinction (of species); migration of animals; change in hibernation patterns / lack of food; increased in competition; AVP;	**1**

Question	Answer	Marks
2(a)(i)	named producer AND arrow direction → ; rest correct with **four** trophic levels; grass or leaves → cricket → (blue-sided tree) frog → snake / bird;	**2**
2(a)(ii)	camouflaged / cannot be easily seen (by predators or prey);	**1**
2(a)(iii)	*any two from:* (ozone depletion has led to) increased UV / radiation reaching Earth; frogs captured for sale / part of illegal pet trade; disruption due to tourism; predation; less food; can't' lay eggs (on leaves) due to deforestation or loss of trees; AVP;	**2**
2(b)(i)	D;	**1**

Question	Answer	Marks
2(b)(ii)	axes labels; *x*-axis: (breeding) programme AND letters A–D AND *y*-axis: percentage of adult frogs (still living) after one year; sensible linear scale that covers half the plotted space; correct plotting ± half a small square tolerance; bars of equal width;	**4**
2(b)(iii)	*any one from:* no information on the number (of surviving eggs / tadpoles / frogs); only percentages given; no information on conditions or location of breeding programme e.g. temperature / time of year / when collected; AVP;	**1**
2(c)(i)	unbiased / representative;	**1**
2(c)(ii)	*any one from:* can take an average; data more representative; highlights anomalous results; improves reliability;	**1**
2(c)(iii)	*any two from:* student should not be disturbing an endangered species; difficult to count at night; hard to see / camouflaged; move around / active at night / nocturnal; dangerous to do fieldwork in the dark; AVP;	**2**

Question	Answer	Marks
2(c)(iv)	*any four from:* position of quadrat: at random / using a grid; quadrat size stated, e.g. 1 m^2, 25 cm^2, 50 cm^2; repeat; observe each quadrat for fixed period of time; observe at a distance / use a camera / take photograph to count; count (all the blue-sided treefrogs in the quadrat); take an average of results; record results in a, table / tally chart; scale up for whole area;	**4**

Question	Answer	Marks
3(a)(i)	*any four from:* damage to: buildings / loss of home; infrastructure e.g. road, hospitals; (farm)land / food / livestock destroyed; area unusable for many years; (cost of) relocation; rebuild / clean-up, costs; (cost of) emergency aid / medical treatment; export / import effect / decrease in foreign investment; tourism decline; loss of trading / job loss / people not being able to go to work / business destroyed; AVP;	**4**

Question	Answer	Marks
3(a)(ii)	*any three from:* drills; public awareness; early warning / monitoring system; emergency rescue; emergency supplies / stock piling (of food / water / medicines) / field hospital; avoid building in high-risk locations / exclusion zones; planned, evacuation / evacuation location / evacuation route; AVP;	**3**
3(b)(i)	*any four from:* idea of use of heat from, ground / rock cracks / volcanic areas / (decaying) radioactive elements; hot water turns to steam; steam, turns / drives / moves / runs a turbine; turbine, turns / drives / moves / runs a generator (which produces electricity); steam condensed / steam turns to water, at surface / cooling tower; cool water (forced) back into rocks / underground;	**4**
3(b)(ii)	*any two from:* money invested in other technologies / do not have the technology; lack of volcanoes; terrain or conditions not suitable / hot rocks are too deep; can be expensive (to install); other energy sources are available; AVP;	**2**
3(b)(iii)	*any two from:* fertile soil (for growing crops) / better crop growth; mining industry / extraction of minerals ; tourism; scientific research;	**2**
3(c)(i)	table drawn with headings: week AND pH; all data recorded; separate entry for week 6 and 7;	**3**

Question	Answer	Marks
3(c)(ii)	(yes); pH is, decreasing / becoming more acidic; this suggests, more emissions / more SO_2 / sulfuric acid is produced;	**2**
3(c)(iii)	*any two from:* temperature; humidity; water; oxygen; salinity; light, carbon dioxide;	**2**
3(d)(i)	combustion / burning / extraction of fossil fuels / coal / oil / natural gas;	**1**
3(d)(ii)	*any one from:* reduction in fish populations; damage to crops / vegetation; damage to buildings;	**1**